

Lecture note on MAS480A Matroid Theory

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1 Week01-1, 2019.09.02. Mon

Motivations of the matroid theory are graphs and vector spaces.

Example 1.1 (Matrices). Let $A = R \times E$ matrix over a field $\mathbb{F}$. Let $X \subseteq E$. What are the properties of $A[X]$, the submatrix of A by taking all columns in X ?

- Independence/dependency of column vectors
- (column) Basis: maximal independent set of (column) vectors
- Rank

It is fact that all bases have the same size.

Example 1.2 (Graphs). Let $G = (V, E)$ be a graph. It is fact that all maximal sets of edges inducing no cycles have the same size. Note that such sets are edge-sets of spanning forests of G . Moreover, the size is $(\# \text{vertices} - \# \text{components})$.

Example 1.3 (Matchings in a bipartite graph). Let G be a bipartite graph with a bipartition (A, B) . We say that $X \subset A$ is matchable to B if there is a matching (a set of edges pairwise vertex-disjoint) M covering X . It is fact that all maximal matchable subsets of A have the same size.

Proof. Let $X, Y \subseteq A$ be maximally matchable to B . Suppose $|X| < |Y|$. Let M, N be matchings covering X, Y , respectively. Note that by the maximality $|X| = |M|$ and $|Y| = |N|$. Let consider $H := G[M \cup N]$. Each component of H is a path or a cycle. Moreover, each path of length $(= \# \text{edges}) \geq 2$ or cycle is alternatively labeled by M and N . A component isomorphic to K_2 , i.e., just an edge, is possibly labeled by both M and N . If there is an edge $e \in N - M$, then we can directly add it into M . It contradicts to the maximality of M . If there is no such an edge, we can find a path P which is a component of H and its two end-edges are contained in N since $|N| > |M|$. Let $M' = M - (M \cap P) + (N \cap P)$. It is also a matching in G , and gives $X' \subseteq A$ which is matchable to B . Note that $X \subsetneq X'$, so it is a contradiction. $\square$

If we solve problems in matroid, we can think it (matroid) as one of them (vector spaces/graphs/matchings). Then we may get idea to solve.

Definition 1.1. A *matroid* is a pair $M = (E, \mathcal{I})$ of a finite set E and a set $\mathcal{I}$ of subsets of E satisfying the following:

- (I1) $\emptyset \in \mathcal{I}$,
- (I2) $X \in \mathcal{I}, Y \subseteq X \Rightarrow Y \in \mathcal{I}$, and
- (I3) $X, Y \in \mathcal{I}, |X| < |Y| \Rightarrow \exists e \in Y - X$ s.t. $X \cup \{e\} \in \mathcal{I}$.

Let us say that

- X is independent if $X \in \mathcal{I}$,
- X is dependent if $X \notin \mathcal{I}$,
- X is a base if X is a maximal independent set, and
- X is a circuit if X is a minimal dependent set.

The definition of matroid is possibly extended to infinite E . However, there are no settled one. The extensions are messy and hard to handle.

See Example 1.2, an independent set is a forest of G . A base is a spanning forest. A dependent set is an edge-set containing a cycle, and a circuit is a cycle.

Theorem 1.1. For $X \subseteq E$, all maximal independent subset of X have the same size.

Proof. Let $A, B \in \mathcal{I}$ with $A, B \subseteq X$. Suppose $|A| < |B|$. By the axiom (I3), there is $e \in B - A$ such that $A \cup \{e\} \in \mathcal{I}$, which is a subset of X . It contradicts to a maximality of A . $\square$

Definition 1.2. The *rank function* of a matroid M is

$$r_M(X) := (\text{a size of a maximal independent subset of } X).$$

It is well-defined by the previous theorem. The *rank* of a matroid is

$$r(M) := r_M(E).$$

$E(M) = E$ is the *ground set* of the matroid M .

Example 1.4 (Basic classes of matroids). Now we will introduce some basic classes of matroids.

1. *Uniform matroid*, $U_{r,n}$ ($0 \leq r \leq n$). $|E| = n$. X is independent iff $|X| \leq r$. It satisfies all axioms of the matroid.
2. *Vector matroid* (or *linear matroid*). $A := R \times E$ matrix over $\mathbb{F}$. X is independent iff the column vectors of $A[X]$ are linearly independent. Here, we denote $M = M(A)$.

Theorem 1.2. *A vector matroid is a matroid.*

Proof. (I1) and (I2) are trivial. Also, (I3) is obvious by an argument of dimensions, but we will explain it more precisely. By a permutation of E , WMA first $|X \cap Y|$ columns of A correspond to $X \cap Y$, next $|X - Y|$ columns correspond to $X - Y$, next $|Y - X|$ columns correspond to $Y - X$, and remainings corresponds to $E - (X \cup Y)$. Note that elementary row operation does not affect on dependency of column vectors. So we can change first $|X|$ columns to $I_{|X|}$. By the independency of Y , next $|Y - X| > 0$ columns cannot be a zero matrix. This implies that we can find such e from these column vectors. $\square$

In this case, we say that M is *representable* over $\mathbb{F}$ or $\mathbb{F}$ -*representable*. A is a *representation* of M .

Similarly, we can define matroids from a graph G (Example 1.2 and 1.3). Setting $E = E(G)$ and $X \in \mathcal{I}$ iff X is a forest, then $M = (E, \mathcal{I})$ is a matroid. We call it as a *graphic matroid* or a *cycle matroid*. Assume G is bipartite with a bipartition (A, B) . Setting $E = A$ and $X \in \mathcal{I}$ iff X is matchable (to B), then M is also a matroid. We call it as a *transversal matroid* or a *matching matroid*.

1.1 Whitney, 1935

Definition 1.3. Let M be a finite set. Let r be a function from 2^M to $\mathbb{Z}_{\geq 0}$. Then we call a system (M, r) a *matroid* if satisfies below three axioms:

- (R1) $r(\emptyset) = 0$,
- (R2) for any $N \subseteq M$ and $e \in M - N$, $r(N + e) = r(N)$ or $r(N) + 1$, and
- (R3) for any $N \subseteq M$ and $e_1, e_2 \in M - N$, if $r(N + e_1) = r(N + e_2) = r(N)$ then $r(N + e_1 + e_2) = r(N)$.

We call r a *rank function*, and $r(N)$ a *rank* of N .

Without doubt, it is fact that for any $N \subseteq M$ a system $(N, r|_{2^N})$ is a matroid. We call it a *submatroid* of (M, r) . When we write a matroid, we can omit a rank function if it is apparent in a context.

Definition 1.4. Let define a two functions $\rho, n : 2^M \rightarrow \mathbb{Z}_{\geq 0}$ as

$$\begin{aligned}\rho(N) &= |N|, \\ n(N) &= \rho(N) - r(N).\end{aligned}$$

$\rho(N)$ is just a cardinality of N . We call n a *nullity function*, and $n(N)$ a *nullity* of N .

- N is *independent* if $n(N) = 0$,
- N is *dependent* if $n(N) > 0$,
- N is a *base* if it is a maximal independent set ($M - N$ is called a *base complement*), and

- N is a *circuit* if it is a minimal dependent set.

By the axiom (R1) and (R2), $0 \leq r(N) \leq \rho(N)$. So $0 \leq n(N) \leq \rho(N)$.

Lemma 1.3. *For $N \subseteq N' (\subseteq M)$, $r(N) \leq r(N')$ and $n(N) \leq n(N')$.*

Proof. By (R2), $r(N) \leq r(N')$. Also the axiom implies that $r(N') \leq r(N) + \rho(N') - \rho(N)$. It is equivalent with $n(N) \leq n(N')$. $\square$

Lemma 1.4. *Any subset of an independent set is independent.*

Proof. Let $N \subseteq M$ be an independent set, i.e., $n(N) = 0$. Let $N' \subseteq N$. By the previous theorem, $(0 \leq) n(N') \leq n(N) = 0$. Therefore, $n(N') = 0$. $\square$

Theorem 1.5. *N is independent iff N is contained in a base iff N does not contain any circuit.*

Proof. (1 $\Rightarrow$ 2) Let N be an independent set. If N is a base, done. If not, there is $e \in M - N$ such that $N + e$ is independent. Repeat this work until get a base. It actually finishes since M is finite.

(2 $\Rightarrow$ 1) By the previous lemma, done.

(1 $\Rightarrow$ 3) Suppose N contains a circuit C . Then $0 < n(C) \leq n(N)$, which is a contradiction.

(3 $\Rightarrow$ 1) We will show a contraposition. Let N be a dependent set. If N is a circuit, done. If not, there is $e \in N$ such that $N - e$ is a dependent. Repeat this work until get a circuit. $\square$

Theorem 1.6. *A circuit is a minimal submatroid contained in no bases, i.e., containing at least one element from each base complement. A base is a maximal submatroid containing no circuit. A base complement is a minimal submatroid containing at least one element from each circuit.*

Proof. By the previous theorem, N is ‘contained in no bases’ iff it is dependent, and N is ‘containing no circuit’ iff it is independent. Then the first two statements are obvious. The last one is obtained by taking the complement operator for each set in the second statement. $\square$

We can observe the reciprocal relation ship between circuits and base complements. Do we derive a dual concept of a matroid from this?

Definition 1.5.

$$\Delta(N, N') = r(N + N') - r(N).$$

Proposition 1.7.

$$\Delta(A, B + C) = \Delta(A + C, B) - \Delta(A + C, C).$$

Proof.

$$\begin{aligned}
\delta(A, B + C) &= r(A + B + C) - r(A) \\
&= r(A + B + C) - r(A + C) - r(A) + r(A + C) \\
&= \Delta(A + C, B) - \Delta(A + C, C).
\end{aligned}$$

□

Lemma 1.8.

$$\Delta(N + e_2, e_1) \leq \Delta(N, e_1).$$

Proof. $\Delta(N + e_2, e_1) = r(N + e_1 + e_2) - r(N + e_2)$ and $\Delta(N, e_1) = r(N + e_1) - r(N)$. So the given statement is equivalent with

$$\begin{aligned}
\Delta(N, e_1 + e_2) &= r(N + e_1 + e_2) - r(N) \\
&\leq (r(N + e_1) - r(N)) + (r(N + e_2) - r(N)) \\
&= \Delta(N, e_1) + \Delta(N, e_2).
\end{aligned}$$

$\Delta(N, e_1 + e_2)$ is possible 0, 1, or 2. If it is 0, nothing to do. If it is 2, we can easily deduce that $\Delta(N, e_1) = \Delta(N, e_2) = 1$ since

$$\begin{array}{ccccc}
& & r(N + e_1) & & \\
& \nearrow +1 & & \searrow +1 & \\
r(N) & & & & r(N + e_1 + e_2) \\
& \searrow +1 & & \nearrow +1 & \\
& & r(N + e_2) & &
\end{array}$$

by the axiom (R2).

For $\Delta(N, e_1 + e_2) = 1$, suppose that $\Delta(N, e_1) = \Delta(N, e_2) = 0$. Then $\Delta(N, e_1 + e_2) = 0$ by (R3), which is a contradiction. Therefore, the given inequality holds. □

Before this lemma, we did not use the axiom (R3).

Lemma 1.9.

$$\Delta(N + N', e) \leq \Delta(N, e).$$

Proof. Denote N' as $e_1 + e_2 + \dots + e_k$ where $k = \rho(N')$. Then inductively

$$\Delta(N + N', e) \leq \Delta(N + e_1 + \dots + e_{k-1}, e) \leq \dots \leq \Delta(N, e)$$

by the previous lemma. □

Theorem 1.10.

$$\Delta(N + N_2, N_1) \leq \Delta(N, N_1),$$

or equivalently,

$$r(N + N_1 + N_2) \leq r(N + N_1) + r(N + N_2) - r(N).$$

Proof. Consider an induction on $\rho(N_1)$. For $\rho(N_1) = 1$, it is a case of the previous lemma.

$$\begin{aligned}\Delta(N + N_2, N_1 + e) &= \Delta(N + N_2 + e, N_1) - \Delta(N + N_2, e) \\ &\leq \Delta(N + e, N_1) - \Delta(N, e) \\ &= \Delta(N, N_1 + e).\end{aligned}$$

The equalities are guaranteed by the earlier proposition. The middle inequality holds by the induction hypothesis. $\square$

The above theorem is equivalent with

$$r(N_1 + N_2) \leq r(N_1) + r(N_2) - r(N_1 \cap N_2).$$

The interpretation of these lemmas and theorem is when $N \subseteq N' (\subseteq M)$, adding elements does not increase a rank of N' more than a rank of N , i.e., $r(N' + e_1 + \dots + e_k) - r(N') \leq r(N + e_1 + \dots + e_k) - r(N)$. This agree with setting M as a set of column vectors for a given matrix A , and r as a dimension of a spanning space. (Increase of a dimension after adding some vectors in a bigger space is smaller than increase of a dimension after adding the vectors in a smaller space.)

2 Week01-2, 2019.09.04. Wed

Example 2.1 (Basic classes of matroids). Continue to exhibit classic examples.

3. *Graphic matroid* (or *cycle matroid*). Let $G = (V, E)$ be a graph. Let $M := M(G)$ be the cycle matroid of G if it is a matroid on E (i.e., $E(M) = E$) where X is indep (i.e., $X \in \mathcal{I}$) iff X contains no edge set of a cycle of G .

We say that a matroid M is *graphic* if $M = M(G)$ for some graph G .

4. *Transversal matroid* (or *matching matroid*). Let G be a simple bipartite graph with a bipartition (A, B) . Letting $M(E) = A$, and $X \in \mathcal{I}$ iff $X \subseteq A$ is matchable to B .

5. *Partition matroid*. Let $E = \coprod_{i=1}^r E_i$, where each E_i is finite. X is independent iff $|X \cap E_i| \leq 1$ for each i . It is obviously a matroid.

One remark is that if $|E_1| = \dots = |E_r| = 1$, then it is $U_{r,r}$. Especially, we call it a *free matroid*.

Remark. Not every matroid is graphic. Consider $U_{2,4}$. $E(U_{2,4}) = \{a, b, c, d\}$. Let us try to represent it as a graph. Since $\{a, b, c\}$ is dependent, it is a cycle. Also, $\{a, b, d\}$ is a cycle. It implies that c and d are parallel edges. So $\{c, d\}$ is a cycle of length two, i.e., $\{c, d\} \notin \mathcal{I}$. It is a contradiction.

Not every matroid is a binary matroid. Also, $U_{2,4}$ is a counter-example. Suppose it is $\mathbb{F}_2$ -representable. Let e_a, e_b, e_c, e_d be column vectors corresponding to a, b, c, d , respectively. Then $e_a + e_b + e_c = e_a + e_b + e_d = 0$. It deduces $e_c + e_d = 0$, which is a contradiction.

Remark. Any $U_{r,n}$ is representable over a big field.

Proof. ??? □

Remark. There is a matroid which is not representable over any fields. V_8 matroid(?).

Proposition 2.1. *A graphic matroid $M(G)$ is a matroid.*

Proof. (I1) and (I2) are trivial. Now consider (I3). Let $X, Y \in \mathcal{I}$ with $|X| < |Y|$. Let $C_1, C_2, \dots, C_l$ be the components of a subgraph (V, X) of G . If Y contains an edge e joining C_i and C_j with $i \neq j$. Then $X + e \in \mathcal{I}$. So WMA there are no such edges. It implies that $Y \subseteq \coprod_i E(G[V(C_i)])$. Since Y contains no cycles,

$$|Y| \leq \sum_i (|V(E_i)| - 1) = |X|.$$

It is a contradiction. □

Proposition 2.2. *A transversal matroid is a matroid*

Proof. (I1) and (I3) are obvious. Showing (I2) is similar with the proof after Example 1.3. So I will roughly describe a proof. Pick matchings M and N covering X and Y , respectively. WMA $|M| = |X| < |Y| = |N|$. Consider $G[M \Delta N]$. Each component is a path or a cycle of which edges are alternatively in M or N . Since $|M| < |N|$, there is a path P such that $|P \cap M| < |P \cap N|$. Consider $M \Delta E(P)$. It is a matching covering $X' := X + a$ where $a \in Y - X$. □

Proposition 2.3. *Every transversal matroid is representable over a big field.*

Proof. Let G be a simple bipartite graph with a bipartition (A, B) . Let $(A, \mathcal{I})$ be a transversal matroid satisfying $X \in \mathcal{I}$ iff $X \subseteq A$ is matchable to B . Consider an incidence matrix of G such that rows are labeled by elements of B , and columns are labeled by elements of A . Put 0 for each entry (b, a) where a, b are not adjacent. Put a non-zero undefined value $x_{b,a}$ for an entry (b, a) where a, b are adjacent. We want that X is matchable to B iff the columns of X are linearly independent. The last statement is equivalent that the columns of X are rank $|X|$ iff there is a set of $|X|$ rows inducing a $|X| \times |X|$ non-singular submatrix. Note that a square matrix is non-singular iff its determinant is non-zero.

A matching $M = \{a_1 b_1, \dots, a_k b_k\}$ in a complete bipartite graph with a bipartition (A, B) corresponds to a set of entries $\{(b_1, a_1), \dots, (b_k, a_k)\}$ of a $B \times A$ incidence matrix of the complete bipartite graph. When consider an original bipartite graph G , a matching M covering X with size $|X|$ exactly corresponds to a set of entries $\{(b_1, a_1), \dots, (b_k, a_k)\}$ where each value of entry is non-zero.

Here, a non-trivial fact (we skip a proof) is that we can give non-zero $x_{b,a}$'s satisfying any two product $\prod_i x_{b_i, a_i}$ and $\prod_j x_{b'_j, a'_j}$ are different over a big field $\mathbb{F}$. Moreover, we can find $x_{b,a}$'s such that summations of $\pm \prod_i x_{b_i, a_i}$'s are non-zero.

This fact implies that $\det S = \sum_{\sigma} (-1)^{\text{sgn} \sigma} \prod_i \pi_{i\sigma(i)} \neq 0$ iff there exists a matching covering X , where S is a square submatrix chosen in the first paragraph and π_{ij} is a value of (i, j) -entry of S . Therefore, the transversal matroid $(A, \mathcal{I})$ is representable over a big field $\mathbb{F}$. $\square$

What are the interesting problems in this course? Here are some motivating questions in the matroid theory.

- How to find a maximum weight independent set?
- How to find a largest common independent set in two matroids (on a same ground set)?
- How to find a largest set that is a union of k sets $X_1, \dots, X_k$ where X_i is independent in the i -th matroid M_i ?
- When is a matroid representable over the binary field?
- When is a matroid graphic?
- What is the relation between $M(G)$ and $M(G^*)$, where G is a plane graph?

The first question is related to the problem finding a maximum(or minimum) weight spanning tree in a connected graph G with a weighted edge-set. It is solved by a greedy algorithm. We will show that a matroid can be defined by a structure holding a greedy algorithm.

As mentioned in the previous lecture, we can imagine a matroid as a graph or a vector space. When we consider a matroid as a vector space, its dimension goes up very quickly.

Let us think about representing low-rank matroids.

Definition 2.1 (Affine geometry). Let $v_1, v_2, \dots, v_n \in \mathbb{F}^m$ be affinely independent whenever $\sum_i c_i v_i = 0$ with $\sum_i c_i = 0$ implies $c_i = 0$ for all i .

Note that $v_1, \dots, v_n \in \mathbb{F}^m$ are affinely independent iff $(1, v_1), \dots, (1, v_n) \in \mathbb{F}^{m+1}$ are linearly independent.

Definition 2.2. Let M be an *affine matroid* on E if $X \subseteq E$ is independent iff X is affinely independent.

Proposition 2.4. *An affine matroid is a matroid.*

Proof.

$\square$

Proposition 2.5. *An affine matroid over $\mathbb{F}$ is representable over $\mathbb{F}$.*

Proof. By the earlier note about relation between affine independency and linearly independency, an affine matroid on $E = \{v_1, \dots, v_n\} \subseteq \mathbb{F}^m$ is $M(A)$, where

$$A = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ v_1 & v_2 & \cdots & v_n \end{pmatrix},$$

an $(m+1) \times n$ matrix over $\mathbb{F}$.

$\square$

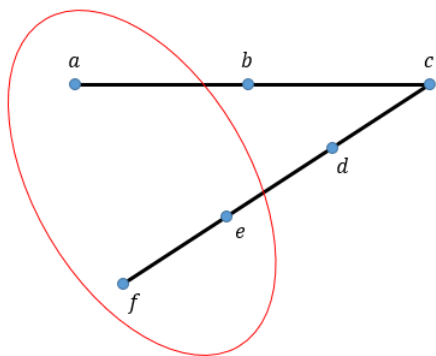
The converse holds, i.e., every linear matroid is an affine matroid. (?)
 How to draw matroids of rank ≤ 3 on the plane?

- A dependent set of rank 2 corresponds to a line segment,
- a dependent set of rank 1 corresponds to a set of multiple points at the same location, and
- a loop is outside of the picture.

Here, $e \in E$ is a loop if $\{e\}$ is dependent, i.e., it is an element of rank 0. The name of a loop is originated from the graph theory.

Example 2.2. $U_{2,n}$ with $n \geq 3$ is drawn on the plane by n colinear points. $U_{3,n}$ is drawn on the plane by n points without line segments.

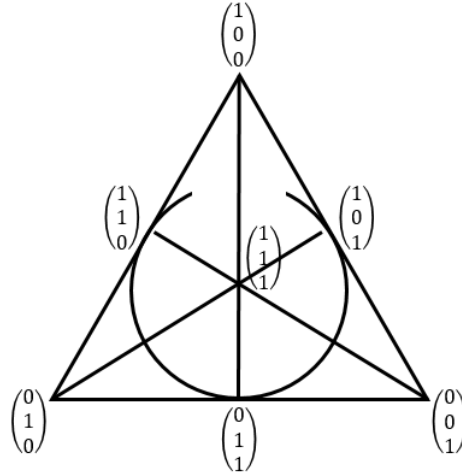
Consider a drawing of points a, b, c, d, e, f such that a, b, c are colinear and c, d, e, f are colinear. Then $\{a, b, f\}$ is independent.



Example 2.3 (Fano matroid). Let

$$A = \begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix}$$

over the binary field. Then we call $M = M(A)$ a *Fano matroid*. Its drawing is not embeddable on $\mathbb{R}^2$. (Moreover, not embeddable on any $\mathbb{R}^n$.)



Can we recover a matroid from a drawing on the plane? What rules are needed?

3 Week02-1, 2019.09.09. Mon

Chapter 2. Cryptomorphisms.

Remind that the three axioms of a matroid $M = (E, \mathcal{I})$:

- (I1) $\emptyset \in \mathcal{I}$,
- (I2) $X \in \mathcal{I}, Y \subseteq X \Rightarrow Y \in \mathcal{I}$, and
- (I3) $X, Y \in \mathcal{I}, |X| < |Y| \Rightarrow \exists e \in Y - X$ s.t. $X \cup \{e\} \in \mathcal{I}$.

We will explore several equivalent definitions of the matroid.

Chapter 2.1. Circuit (= minimally dependent set)

Proposition 3.1. *If $\mathcal{C}$ is a collection of all circuits in a matroid, then*

- (C1) $\emptyset \notin \mathcal{C}$,
- (C2) $X, Y \in \mathcal{C}, X \subseteq Y \Rightarrow X = Y$, and
- (C3) $X, Y \in \mathcal{C}, e \in X \cap Y, X \neq Y \Rightarrow \exists D \in \mathcal{C}$ s.t. $D \subseteq X \cup Y - e$.

We call (C3) the (weak) circuit elimination axiom.

Proof. (C1) $\Leftarrow$ (I1). (C2) is trivial from the definition (minimality). Now we will show (C3). Suppose (C3) does not hold, i.e., there are X, Y, e such that (C3) fails. Then $X \cup Y - e$ is independent, and it is a maximal independent set in $X \cup Y$. Since $X, Y \in \mathcal{C}$ and $X \neq Y, X \not\subseteq Y$. So there is $f \in X - Y$. $X - f$ is independent. Then there is a maximal independent set Z in $X \cup Y$

containing $X - f$. Since X is dependent, $Z \not\supseteq X$, i.e., $Z \not\supseteq f$. Note that $|Z| = |X \cup Y - e| (= |X \cup Y| - 1)$ by the Theorem 1.1: For $X \subseteq E$, all maximal independent subset of X has the same size. It implies that $Z = X \cup Y - f$, so $Z \supseteq Y$. It is a contradiction. $\square$

Proposition 3.2. *If $\mathcal{C}$ is a collection of subsets of a finite set E satisfying (C1), (C2), (C3), and let $\mathcal{I} := \{X \subseteq E : Y \not\subseteq X, \forall Y \in \mathcal{C}\}$, then $(E, \mathcal{I})$ is a matroid with $\mathcal{C}$ as the collection of circuits.*

Proof. (I1) $\Leftarrow$ (C1). (I2) holds trivially from the construction of $\mathcal{I}$. Now we will show (I3). Suppose that (I3) does not hold, i.e., there are $X, Y \in \mathcal{I}$ with $|X| < |Y|$ such that $X + e \notin \mathcal{I}$ for all $e \in Y - X$. Choose a pair (X, Y) so that $|X \cap Y|$ is maximized. Since $X + e \notin \mathcal{I}$ for $e \in Y - X$, there is $C_e \in \mathcal{C}$ such that $e \in C_e \subseteq X + e$. Note that such C_e is unique since if there are two different $C_e, C'_e \in \mathcal{C}$ such that $e \in C_e, C'_e \subseteq X + e$, then we can find $D \in \mathcal{C}$ such that $D \subseteq C_e \cup C'_e - e \subseteq X$ by (C3), which is a contradiction. Since $Y \in \mathcal{I}$, $C_e \not\subseteq Y$. So there is $(e \neq) f \in C_e - Y$. Let $X' := X + e - f$. Then $X' \in \mathcal{I}$ since if not, i.e., there is $D \in \mathcal{C}$ such that $D \subseteq X' \subset X + e$ and $e \in D$, then by the earlier observation $D = C_e \ni f$, which is a contradiction. Since $|X' \cap Y| > |X \cap Y|$ ($X' \cap Y = X \cap Y + e$), (I3) holds for (X', Y) by our choice of (X, Y) . There is $(e, f \neq) g \in Y - X'$ such that $X' + g = X + e + g - f \in \mathcal{I}$. Since also $g \in Y - X$, there is a unique $C_g \in \mathcal{C}$ such that $g \in C_g \subseteq X + g$. $C_g \ni f$ since if not, $C_g \subseteq X' + g$, which is a contradiction. Remind that $C_e, C_g \in \mathcal{C}$, $f \in C_e \cap C_g$, and $C_e \neq C_g$. Then by (C3) there is $D \in \mathcal{C}$ such that $D \subseteq C_e \cup C_g - f \subseteq X' + g$, which is a contradiction.

By the construction of $\mathcal{I}$, any $C \in \mathcal{C}$ is not in $\mathcal{I}$, i.e., dependent in the matroid $M := (E, \mathcal{I})$. For any $e \in C$, $C - e$ is not in $\mathcal{C}$ by (C2), and moreover, $C - e \in \mathcal{I}$. It implies that C is a minimal dependent set, i.e., a circuit in M . Hence $\mathcal{C}$ is a collection of circuits (but we do not know it contains all circuits). Let D be a circuit in M . Since $D \notin \mathcal{I}$, there is $C \in \mathcal{C}$ such that $C \subseteq D$. By the minimality (both C, D are minimal dependent sets), $D = C \in \mathcal{C}$. Therefore, $\mathcal{C}$ is the collection of circuits. $\square$

Chapter 2.2. Base (= maximally independent set)

Proposition 3.3. *If $\mathcal{B}$ is a collection of all bases of a matroid, then*

(B1) $\mathcal{B} \neq \emptyset$, and

(B2) $B_1, B_2 \in \mathcal{B}$, $e \in B_1 - B_2 \Rightarrow \exists f \in B_2 - B_1$ s.t. $B_1 - e + f \in \mathcal{B}$.

We call (B2) the base exchange axiom.

Proof. (B1) $\Leftarrow$ (I1). Now we will show (B2). $B_1 - e$ is an independent set with cardinality $= |B_1| - 1 < |B_2|$ (remind all bases have the same size). By (I3) there is $f \in B_2 - (B_1 - e) = B_2 - B_1$ such that $B_1 - e + f$ is an independent set. Because of its cardinality, it is a base. $\square$

Proposition 3.4. *If $\mathcal{B}$ is a collection of subsets of a finite set E satisfying (B1) and (B2), and $\mathcal{I} := \{X \subseteq E : X \subseteq B \text{ for some } B \in \mathcal{B}\}$, then $(E, \mathcal{I})$ is a matroid with $\mathcal{B}$ as the collection of bases.*

Proof. (I1) $\Leftrightarrow$ (B1). By the construction of $\mathcal{I}$, (I2) is obvious. Now we will show (I3). Let $X, Y \in \mathcal{I}$ with $|X| < |Y|$. There are $X', Y' \in \mathcal{B}$ such that $X \subseteq X'$ and $Y \subseteq Y'$. By (B2), every set in $\mathcal{B}$ has the same size. (Let $B_1, B_2 \in \mathcal{B}$ with $|B_1| < |B_2|$. Choose a pair (B_1, B_2) so that $|B_1 \cap B_2|$ is maximized. By (B2), $B_1 \subsetneq B_2$ is impossible since for $e \in B_2 - B_1$ we cannot find $f \in B_1 - B_2 (= \emptyset)$ such that $B_2 - e + f \in \mathcal{B}$. So there is $e \in B_1 - B_2$. By (B2), there is $f \in B_2 - B_1$ such that $B_1' = B_1 - e + f \in \mathcal{B}$. $|B_1'| = |B_1| < |B_2|$ and $|B_1' \cap B_2| > |B_1 \cap B_2|$, which contradicts to our choice of (B_1, B_2) .) So $|X'| = |Y'|$. Choose a pair (X', Y') so that $|X' \cap Y'|$ is maximized. Suppose $(Y - X) \cap X' = \emptyset$. Note that X' is a disjoint union of X , $(Y - X) \cap X'$, $X' \cap Y' - X \cup Y$, and $X' - Y' - X$. Similarly, Y' is a disjoint union of Y , $(X - Y) \cap Y'$, $X' \cap Y' - X \cup Y$, and $Y' - X' - Y$. Note that $X' \cap Y' - X \cup Y = X' \cap Y' - X$ by the assumption, $(Y - X) \cap X' = \emptyset$.

$$\begin{aligned} |X'| &= |X| + |(Y - X) \cap X'| + |X' \cap Y' - X \cup Y| + |X' - Y' - X| \\ &= |X| + |X' \cap Y' - X| + |X' - Y' - X|, \\ |Y'| &= |Y| + |(X - Y) \cap Y'| + |X' \cap Y' - X \cup Y| + |Y' - X' - Y| \\ &\geq |Y| + |X' \cap Y' - X| + |Y' - X' - Y|. \end{aligned}$$

Since $|X'| = |Y'|$ and $|X| < |Y|$, we can derive that $|X' - Y' - X| \geq |Y| - |X| + |Y' - X' - Y| > 0$. There is $e \in Y' - X' - X$. By (B2), there is $f \in Y' - X'$ such that $X'' := X' - e + f \in \mathcal{B}$. Then $|X'' \cap Y'| > |X' \cap Y'|$, which is a contradiction. So the choice of (X', Y') so that $|X' \cap Y'|$ is maximized implies $(Y - X) \cap X' \neq \emptyset$. There is $e \in (Y - X) \cap X'$ such that $X + e \subseteq X'$ (so $X + e \in \mathcal{I}$). Therefore, (I3) holds.

By the construction of $\mathcal{I}$, it is obvious that every $B \in \mathcal{B}$ is a base in the matroid $M := (E, \mathcal{I})$. (Remind that all sets in $\mathcal{B}$ has the same size, so no set in $\mathcal{B}$ is a proper subset of another set in $\mathcal{B}$.) Hence $\mathcal{B}$ is a collection of bases in M (but it is not sure that $\mathcal{B}$ contains all bases). Let A be a base in M . Since it is in $\mathcal{I}$, there is $B \in \mathcal{B}$ such that $A \subseteq B$. Since B is independent and A is maximally independent, $A = B \in \mathcal{B}$. Therefore, $\mathcal{B}$ is the collection of bases in M . $\square$

The *fundamental circuit* of $e \notin B$ with respect to $B \in \mathcal{B}$ is a unique circuit C_e contained in $B + e$. Note that $e \in C_e$. The uniqueness is from (C3) (see the proof of the Proposition 3.1). This terminology is from the *fundamental cycle* of $e \notin E(T)$ with respect to a tree T in a graph G . Remind that T and $e(\notin E(T))$ determines a unique cycle containing e .

4 Week02-2, 2019.09.11. Wed

Chapter 2.3. Rank

Remind that $r_M(X) := \max\{|I| : I \subseteq X, I \in \mathcal{I}\}$ for a matroid $M = (E, \mathcal{I})$ is the rank of X . By (I3), all maximal independent subsets of X have size $r_M(X)$. We call r_M the rank function of M . (We omit a subscription so write simply r when a given matroid is clear.)

Proposition 4.1. *The rank function r of a matroid satisfies*

- (R1) $0 \leq r(x) \leq |X|$,
- (R2) $X \subseteq Y \Rightarrow r(X) \leq r(Y)$, and
- (R3) $r(X) + r(Y) \geq r(X \cap Y) + r(X \cup Y)$.

We call (R3) the submodular inequality.

Proof. (R1) and (R2) are trivial from the definition of r . Let I be an independent set contained in $X \cap Y$ with $|I| = r(X \cap Y)$. Let J be a maximal independent set containing I contained in $X \cup Y$. Then $|J| = r(X \cup Y)$.

$$r(X) + r(Y) \geq |X \cap J| + |Y \cap J| = |I| + |J| = r(X \cap Y) + r(X \cup Y),$$

i.e., (R3) holds. $\square$

Proposition 4.2. *If $r : 2^E \rightarrow \mathbb{Z}$ satisfies (R1), (R2), (R3), and $\mathcal{I} := \{X \subseteq E : r(X) = |X|\}$, then $(E, \mathcal{I})$ is a matroid with the rank function r .*

Proof. (I1) is directly from (R1) since $0 \leq r(\emptyset) \leq |\emptyset| = 0$. Let $X \subseteq Y$ and $Y \in \mathcal{I}$, i.e., $r(Y) = |Y|$. By (R3),

$$r(X) + r(Y - X) \geq r(\emptyset) + r(Y) = |Y|.$$

By (R1), $r(X) \leq |X|$ and $r(Y) \leq |Y - X|$, so $r(X) + r(Y - X) \leq |Y|$. It implies that $r(X) = |X|$, i.e., $X \in \mathcal{I}$. Hence (I2) holds.

Before showing (I3), we claim a lemma: If $r(X + e) = r(X)$ for all $e \in Z$, then $r(X \cup Z) = r(X)$. Its proof is following. Let W be a maximal subset of Z such that $r(X \cup W) = r(X)$. If $W = Z$, done. WMA $W \subsetneq Z$. Let $e \in Z - W$. Then by (R3),

$$r(X + e) + r(X \cup W) \geq r(X) + r(X \cup W + e).$$

By the assumptions, $r(X + e) = r(X \cup W) = r(X)$. It implies that $r(X) \geq r(X \cup (W + e))$, which contradicts to our choice of W .

Now we will show (I3). Let $X, Y \in \mathcal{I}$ and $|X| < |Y|$. Suppose that $X + e \notin \mathcal{I}$ for all $e \in Y - X$. Note that $|X| = r(X) \leq r(X + e) \leq |X + e|$ by (R1) and (R2). Since $X + e \notin \mathcal{I}$, i.e., $r(X + e) < |X + e|$, $r(X + e) = r(X)$. Then by the above lemma,

$$|Y| = r(Y) \leq r(X \cup (Y - X)) = r(X) = |X|.$$

It is a contradiction.

We claim that r is the rank function of $M := (E, \mathcal{I})$. Let I be a maximal independent subset of X in M , for a given $X \subseteq E$. WTS $r(X) = |I| (= r_M(X))$. By the maximality, $r(I + e) < |I + e|$ for any $e \in X - I$. Then by the above lemma, $r(X) = r(I \cup (X - I)) = r(I) = |I|$. $\square$

One question that can we deduce that

$$0 \geq r(X \cup Y \cup Z) - r(X) - r(Y) - r(Z) + r(X \cup Y) + r(X \cup Z) + r(Y \cup Z) - r(X \cap Y \cap Z)$$

from the submodular inequality and others??? Or, can we deduce more general one???

Example 4.1. Consider a graphic matroid $M = M(G)$ for a graph $G = (V, E)$. Then

$$\begin{aligned} r_M(X) &= |E(\text{a union of a spanning tree of each component of } (V, X))| \\ &= \sum_{C: \text{a component of } (V, X)} (|V(C)| - 1) \\ &= |V| - \#(\text{components of } (V, X)). \end{aligned}$$

Here, (V, X) is a subgraph of G .

Chapter 2.4. Closure

Definition 4.1. Let $M = (E, \mathcal{I})$ be a matroid. For $X \subseteq E$, the *closure* (or *span*) of X is

$$cl_M(X) := \{e \in E : r_M(X \cup \{e\}) = r_M(X)\}.$$

We can omit a subscription of cl_M (so simply write cl) if there is no confusion.

From the definition, it is obvious that $X \subseteq cl(X)$.

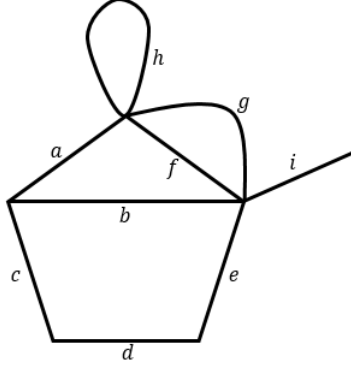
Example 4.2. Consider a vector matroid $M = M(A)$ where A is an $R \times E$ matrix over a field. Let $X \subseteq E$. We can think X as a set of column vectors of $A[X]$. Also, we can regard $e \in E$ as a column vector in A . If $e \in \text{span}X$, then $\dim(\text{span}(X + e)) = \dim(\text{span}X)$, and vice versa. Note that $r_M \equiv \dim \text{span}$, so for the vector matroid M ,

$$cl_M(X) = \{e \in E : e \in \text{span}X\}.$$

Example 4.3. Let $M = M(G)$ be a graphic matroid, where $G = (V, E)$ is a graph described in below. $E = \{a, b, c, d, e, f\}$. Then

$$cl_M(\{a, b, h\}) = \{a, b, h, g, j\}.$$

Adding g or j to $\{a, b, h\}$ does not affect on the number of components.



Lemma 4.3. $r_M(cl_M(X)) = r_M(X)$.

Proof. It is equivalent with the claim in the proof of the Proposition 4.2: If $r_M(X + e) = r_M(X)$ for all $e \in Z$, then $r_M(X \cup Z) = r_M(Z)$. $\square$

Proposition 4.4. *The closure cl of a matroid satisfies*

- (CL1) $X \subseteq cl(X)$ for all $X \subseteq E$,
- (CL2) $X \subseteq Y \Rightarrow cl(X) \subseteq cl(Y)$,
- (CL3) $cl(cl(X)) = cl(X)$, and
- (CL4) $\exists f \in cl(X + e) - cl(X) \Rightarrow e \in cl(X + f)$.

Proof. (CL1) is trivial from the definition. Let $X \subseteq Y$. Suppose that there is $e \in cl(X) - Y$ (otherwise, it is directly deduced that $cl(X) \subseteq Y \subseteq cl(Y)$). Then

$$r(X + e) + r(Y) \geq r(X) + r(Y + e)$$

by the submodular inequality, (R3). Since $r(X + e) = r(X)$, $r(Y) \geq r(Y + e)$, i.e., $e \in cl(Y)$. So (CL2) holds. Let $e \in cl(cl(X))$, i.e., $r(cl(X) + e) = r(cl(X))$. By the previous lemma, $r(cl(X)) = r(X)$. Then

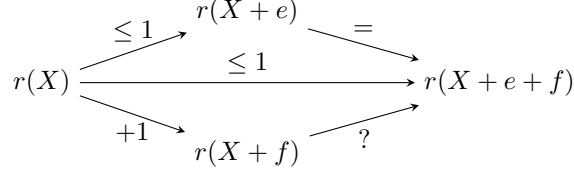
$$r(X) \leq r(X + e) \leq r(cl(X) + e) = r(X).$$

It implies $r(X + e) = r(X)$, i.e., $e \in cl(X)$. So $cl(cl(X)) \subseteq cl(X)$. The axiom (CL3) holds.

Now WTS (CL4). Assume its sufficient condition. Then we get

$$\begin{aligned} r(X + e + f) &= r(X + e), \\ r(X + f) &= r(X) + 1. \end{aligned}$$

Note that by the submodular inequality, $r(X) + r(e) \geq r(X \cap \{e\}) + r(X + e) \geq r(X + e)$, so $r(X + e) - r(X) \leq r(e) \leq 1$.



The above diagram means that

$$\begin{aligned}
r(X+e+f) - r(X+f) + 1 &= r(X+e+f) - r(X+f) + r(X+f) - r(X) \\
&= r(X+e+f) - r(X) \\
&= r(X+e+f) - r(X+e) + r(X+e) - r(X) \\
&\leq 1.
\end{aligned}$$

So $r(X+e+f) - r(X+f) = 0$, i.e., $e \in cl(X+f)$. $\square$

Proposition 4.5. *If $cl : 2^E \rightarrow 2^E$ satisfies (CL1), (CL2), (CL3), (CL4), and $\mathcal{I} := \{X \subseteq E : e \notin cl(X-e), \forall e \in X\}$, then $(E, \mathcal{I})$ is a matroid whose closure function is cl .*

Proof. (I1) is trivial. (Since $\nexists e \in \emptyset$, so the statement of $\mathcal{I}$ vaguely holds for $\emptyset$. It implies that $\emptyset \in \mathcal{I}$.) Let $X \subseteq Y$ and $Y \in \mathcal{I}$. For all $e \in Y$, $e \notin cl(Y-e)$ by the construction of $\mathcal{I}$. Let $f \in X$. By (CL2), $cl(X-f) \subseteq cl(Y-f)$. So $f \notin cl(X-f)$ (since $f \notin cl(Y-f)$), and it implies that $X \in \mathcal{I}$. (I2) holds.

Now WTS (I3). Let $X, Y \in \mathcal{I}$ and $|X| < |Y|$. We claim that if $f \in Y - cl(X)$, then $X+f \in \mathcal{I}$. The proof of the claim is following. Suppose $e \in X+f$. If $e = f$, then $e = f \notin cl(X) = cl((X+f)-e)$. If $e \neq f$, then we can suppose $e \in cl(X+f-e)$. Remind that $X \in \mathcal{I}$, so $e \notin cl(X-e)$. Then by (CL4), $f \in cl((X-e)+e) = cl(X)$, which is a contradiction. So $e \notin cl(X+f-e)$. Therefore, the claim is proved.

So WMA $cl(X) \supseteq Y$ (otherwise, (I3) holds by the above claim). Let choose a counter-example (X, Y) of (I3) with maximum $|X \cap Y|$. Let $e \in Y - X$. If $X \subseteq cl(Y-e)$, then $cl(X) \subseteq cl(cl(Y-e)) = cl(Y-e)$ by (CL3). Since $e \in Y - X \subseteq cl(X)$ and $e \notin cl(Y-e)$ (since $Y \in \mathcal{I}$), it is a contradiction. So $X \not\subseteq cl(Y-e)$. So WMA there is $f \in X - Y$ (since if not, then $X \subset Y$, so (I3) holds) with $f \notin cl(Y-e)$.

We claim that $Y+f-e \in \mathcal{I}$. If not, then there is $g \in Y+f-e$ so that $g \in cl(Y+f-e-g)$. If $g = f$, then $g = f \notin cl(Y-e)$, so $g \notin cl(Y-e-g)$. If $g \neq f$, then $g \notin cl(Y-e-g)$ since $Y-e \in \mathcal{I}$ by (I2). So $g \notin cl(Y-e-g)$. By (CL4), $f \in cl((Y-e-g)+g) = cl(Y-e)$, which is a contradiction. Therefore, the claim holds.

Let $Y' := Y+f-e \in \mathcal{I}$. Then a pair (X, Y') satisfies (I3) by the maximum condition for $|X \cap Y| (= |X \cap Y'| - 1)$. So there is $e' \in Y' - X (\subset Y - X)$ so that $X+e' \in \mathcal{I}$. Therefore, (I3) holds.

So we know that $M := (E, \mathcal{I})$ is a matroid. Now WTS

$$cl_M(X) = cl(X)$$

for each $X \subseteq E$. Suppose $e \in cl_M(X) - cl(X)$. Let r be the rank function of M . Then $r(X + e) = r(X)$ since $e \in cl_M(X)$. Let I be a maximal independent subset of X . $cl(I) \subseteq cl(X)$ by (CL2). $I + e$ is dependent (note that $x \notin cl(X)$ so $x \notin I$) since

$$r(X) = r(X + e) \geq r(I + e) \geq r(I) = |I| = r(X).$$

So there is $f \in I + e$ such that $f \in cl(I + e - f)$. It is impossible that $e = f$ since if not, $f = e \notin cl(X) = cl((I + e) - f)$. So $f \in I$, and $f \notin cl(I - f)$ since $I \in \mathcal{I}$. Then by (CL4), $e \in cl((I - f) + f) = cl(I) \subseteq cl(X)$, which is a contradiction. So $cl_M(X) \subseteq cl(X)$.

Before proving the other direction, we claim a lemma: If $I \in \mathcal{I}$ ($\mathcal{I}$ is define in the problem) and $e \in E - I$ such that $I + e \notin \mathcal{I}$, then $e \in cl(I)$. Its proof is following. There is $f \in I + e$ such that $f \in cl(I + e - f)$. If $f = e$, then $e = f \in cl((I + e) - f) = cl(I)$. If $f \neq e$, then $f \in I$ so $f \notin cl(I - f)$. By (CL4), $e \in cl((I - f) + f) = cl(I)$.

For the other direction, suppose $e \in cl(X)$. WMA $e \notin X$. Let I be a maximal independent subset of X . $cl(I) \subseteq cl(X)$. $I + e' \notin \mathcal{I}$ for all $e' \in X - I$ by the maximality. By the above lemma, $X - I \subseteq cl(I)$ so $X \subseteq cl(I)$. By (CL2) and (CL4), $cl(X) \subseteq cl(cl(I)) = cl(I)$ so $cl(X) = cl(I)$. It implies that $e \in cl(X) = cl(I) = cl((I + e) - e)$. So $I + e \notin \mathcal{I}$, i.e., dependent. Then

$$|I| = r(I) \leq r(I + e) < |I + e| = |I| + 1.$$

It implies that $r(I) = r(I + e)$, i.e., $e \in cl_M(I) \subseteq cl_M(X)$. We can conclude that $cl(X) \subseteq cl_M(X)$. $\square$

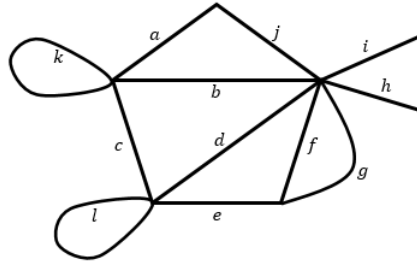
(The proof of the Proposition 4.5 is completed in the lecture Week03-1.)

5 Week03-1, 2019.09.16. Mon

Chapter 2.5. Flat (or Closed set)

Definition 5.1. $X \subseteq E$ is a *flat* (or a *closed set*) if $cl(X) = X$. X is *spanning* if $cl(X) = E(M)$. X is a *hyperplane* if it is a flat of rank $r(M) - 1$.

Example 5.1. Let G be a below graph with labels on edges. Let $M := M(G)$ be a graphic (or cycle) matroid.



(Flats of rank 0) = $cl(\emptyset)$ = (the set of all loops) = $\{k, l\}$,
(Flats of rank 1) = $cl(\text{non-loop edge})$
= (sets of all edges joining 2 adj. vertices union w/ all loops)
= $\{a, k, l\}, \{f, g, k, l\}, \dots$,
(Flats of rank 2) = $\{a, c, k, l\}, \{a, b, j, k, l\}, \{d, e, f, g, k, l\}, \dots$.

Let $V_1, V_2, \dots, V_{n-r}$ be a partition of $V = V(G)$, where $n = |V|$ and $0 \geq r \geq n - 1$. Then $F := \coprod_i E(G[V_i])$ is a flat of rank r in $M = M(G)$.

Example 5.2. Let $M := M(K_6)$ be a graphic matroid. How many hyperplanes? It is

$$\binom{6}{1} + \binom{6}{2} + \frac{1}{2} \binom{6}{3}.$$

First term means the number of (K_1, K_5) pairs, the second means the number of (K_4, K_2) , and the last means the number of (K_3, K_3) . Generally, $M(K_n)$ has the below number of hyperplanes,

$$\binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{k}$$

if $n = 2k + 1$, and

$$\binom{n}{1} + \binom{n}{2} + \dots + \binom{n}{k-1} + \frac{1}{2} \binom{n}{k}$$

if $n = 2k$.

Definition 5.2. We say X spans Y if $Y \subseteq cl(X)$, and X spans e if $e \in cl(X)$.

Lemma 5.1. X spans e if and only if $cl(X + e) = cl(X)$.

Proof. ‘If’ part is trivial. Now we will show ‘only if’ part. Let $e \in cl(X)$. Then $X + e \subseteq cl(X)$. By (CL2) and (CL3), $cl(X + e) \subseteq cl(cl(X)) = cl(X)$. So $cl(X + e) = cl(X)$. $\square$

The above lemma still holds if we replace an element e to a set Y . The proof is almost same.

Lemma 5.2. The followings hold:

1. X is spanning $\Leftrightarrow r(X) = r(M)$,
2. X is a hyperplane $\Leftrightarrow X$ is a maximal set not spanning, and
3. X is a flat of rank $k \Leftrightarrow X$ is a maximal set of rank k .

Proof. 1. X is spanning, i.e., $cl(X) = E(M)$. Then $r(X) = r(cl(X)) = r(E(M)) = r(M)$. Here the first equality holds by the Lemma 4.3. Conversely, suppose $r(M) = r(X)$. For any $e \in E(M) - X$,

$$r(E(M)) \geq r(X + e) \geq r(X) = r(M).$$

It implies that $r(X + e) = r(X)$, i.e., $e \in cl(X)$. So $E(M) = cl(X)$.

2. Let X be a hyperplane. Then $r(X) = r(M) - 1$, so $cl(X) = X \subsetneq E(M)$, i.e., X is not spanning. Let Y be a set containing X and not spanning. Then for any $e \in Y$,

$$r(X) \leq r(X + e) < r(M) = r(X) + 1.$$

It implies that $r(X) = r(X + e)$, i.e., $e \in cl(X)$. $Y \subseteq X$. So X is a maximal set not spanning. Conversely, let X be a maximal set not spanning. Then $r(cl(X)) = r(X) < r(M)$. By the maximality, $r(X + e) = r(M)$ for any $e \in E(M) - X$. It implies two facts: First, $(E(M) - X) \cap cl(X) = \emptyset$, i.e., $cl(X) \subseteq X$. Second, $r(X) = r(M) - 1$ since $r(X + e) \leq r(X) + 1$. Therefore, X is a flat of rank $r(M) - 1$, i.e., a hyperplane.

3. Let X be a flat of rank k , i.e., $X = cl(X)$ and $r(X) = k$. Let Y be a set or rank k containing X . Then for any $e \in Y$,

$$k = r(X) \leq r(X + e) \leq r(Y) = k,$$

so $e \in cl(X)$. It implies that $Y \subseteq X$. Therefore, X is a maximal set of rank k . Conversely, let X be a maximal set of rank k . Let $e \in cl(X)$, i.e., $r(X + e) = r(X)$. Then by the maximality, $X = X + e$, i.e., $e \in X$. Hence $X = cl(X)$. $\square$

Lemma 5.3. *Let $e \notin X$. $e \in cl(X)$ iff there is a circuit $C \subseteq X + e$ containing e .*

Proof. Let I be a maximal independent set of X . Since $e \in cl(X) - X$,

$$|I| = r(I) = r(X) = r(cl(X)) \geq r(I + e)$$

and $|I + e| = |I| + 1$. It implies that $I + e$ is dependent. So there is a circuit C in $I + e$ (so in $X + e$). It must contain e . Conversely, $e \in cl(C - e)$ since $r(C) = r(C - e)$ (note that $C - e$ is independent and C is dependent). So $e \in cl(C - e) \subseteq cl(X)$. $\square$

Proposition 5.4 (Strong circuit elimination axiom). *Let C_1 and C_2 be distinct circuits, and $e \in C_1 \cap C_2$, $f \in C_1 - C_2$. Then there is a circuit $C_3 \subseteq (C_1 + C_2) - e$ such that $f \in C_3$.*

Proof. Since $C_2 - e$ is independent and C_2 is dependent, $r(C_2 - e) = r(C_2)$, i.e., $e \in cl(C_2 - e)$. Similarly, $f \in cl(C_1 - f)$.

$$cl((C_1 + C_2) - e - f) \supseteq cl(C_2 - e) \ni e.$$

Then $cl((C_1 + C_2) - e - f) \supseteq (C_1 + C_2) - f$, so by (CL2) and (CL3)

$$cl((C_1 + C_2) - e - f) = cl((C_1 + C_2) - f) \supseteq cl(C_1 - f) \ni f.$$

By the previous lemma, there is a circuit C_3 in $((C_1 + C_2) - e - f) + f = (C_1 + C_2) - e$ containing f . $\square$

6 Week03-2, 2019.09.18. Wed

Proposition 6.1. *Let $M = (E, \mathcal{I})$ be a matroid and $\mathcal{F}$ be the collection of all flats. Then*

(F1) $E \in \mathcal{F}$,

(F2) $F_1, F_2 \in \mathcal{F} \Rightarrow F_1 \cap F_2 \in \mathcal{F}$, and

(F3) $F \in \mathcal{F}$ and $\{F_1, F_2, \dots, F_k\}$ is the collection of all minimal members of $\mathcal{F}$ containing F properly $\Rightarrow \{F_1 - F, F_2 - F, \dots, F_k - F\}$ is a partition of $E - F$.

Proof. (F1): By (CL1), $E \subseteq cl(E) (\subseteq E)$. Hence $cl(E) = E$, i.e., $E \in \mathcal{F}$.

(F2): Suppose not, i.e., there is $e \in cl(F_1 \cap F_2) - (F_1 \cap F_2)$. By symmetry, let us assume $e \notin F_1$. There is a circuit $C \subseteq (F_1 \cap F_2) + e$ containing e by Lemma 5.3. Then $e \in C \subseteq F_1 + e$, so $e \in cl(C - e) \subseteq cl(F_1) = F_1$. It is a contradiction.

(F3): Let $F_1, F_2, \dots, F_k$ be a collection of minimal flats containing F properly. Then $F_i \cap F_j = F$ for $i \neq j$ by the minimality. Note that for any $e \in E - F$, $cl(F + e)$ is a flat minimally containing F properly. It is obvious that $cl(F + e)$ is a flat containing F properly. The minimality is guaranteed by $r(cl(F + e)) = r(F + e) = r(F) + 1$. Hence $cl(F + e) = F_i$ for some i . Therefore, $\{F_1 - F, F_2 - F, \dots, F_k - F\}$ is a partition of $E - F$. $\square$

What is an intuition of (F3)? For a vector matroid M , let F be a flat of rank 1, i.e., it is a set of vectors contained in a given line. Then each F_i is a plane containing the line corresponds to F . Then we can easily observe that $\{F_1 - F, \dots, F_k - F\}$ is a partition of $E - F$. For a graph G and a graphic matroid $M = M(G)$, a flat F of rank k is a subgraph $G[F]$ with $r(M) - k + 1 = |V(G)| - k$ components which are induced subgraphs of G . Then $F_i - F$ corresponds to an edge-cut between two different components.

Proposition 6.2. *If $\mathcal{F}$ is a collection of subsets of a finite set E satisfying (F1), (F2), (F3), and let $cl(X) = \bigcap_{X \subseteq F \in \mathcal{F}} F$ for all $X \subseteq E$, then there is a matroid on E whose closure function is cl .*

Proof. By (F1), $cl(X)$ is well-defined. We will show that cl satisfies the closure axioms.

(CL1): By its construction, $X \subseteq cl(X)$.

(CL2): Also by its construction, if $X \subseteq Y$ then $cl(X) \subseteq cl(Y)$.

(CL3): $cl(X) \in \mathcal{F}$ by (F2). So $cl(cl(X)) \subseteq cl(X)$.

(CL4): Let $f \in cl(X + e) - cl(X)$. Let $F := cl(X) \in \mathcal{F}$. By (F3), there is $\{F_1, F_2, \dots, F_k\} \subseteq \mathcal{F}$ of which members minimally contain F properly, and $\{F_1 - F, F_2 - F, \dots, F_k - F\}$ is a partition of $E - F$. WLOG $e \in F_1 - F$. Then $F \subsetneq cl(X + e) \subseteq F_1$ and $cl(X + e) \in \mathcal{F}$, so $cl(X + e) = F_1$ by the minimality. It implies that $f \in F_1 - F$. Then $F \subsetneq cl(X + f) \subseteq F_1$ and $cl(X + f) \in \mathcal{F}$, so $cl(X + f) = F_1 \ni e$.

Therefore, cl defines a matroid $M = (E, \mathcal{I})$, where $\mathcal{I} = \{X \subseteq E : e \notin cl(X - e), \forall e \in X\}$, by Proposition 4.5. In addition, the closure function of M is cl . Let $\mathcal{F}_M$ be the collection of flats of M , i.e., $\mathcal{F}_M = \{X \subseteq E : X = cl(X)\}$. WTS $\mathcal{F} = \mathcal{F}_M$.

If $F \in \mathcal{F}$, then $cl(F) \subseteq F$ by the definition of cl . Hence $cl(F) = F$, i.e., $F \in \mathcal{F}_M$.

Let $F \in \mathcal{F}_M$. By (F2), $(F =)cl(F) \in \mathcal{F}$. □

Chapter 2.6. Greedy Algorithm and Matroids

Definition 6.1 (Greedy algorithm). An algorithm described in below is called the *greedy algorithm*.

Input: A finite set E , a collection $\mathcal{F} \subseteq 2^E$, and a weight function $\omega : E \rightarrow \mathbb{R}$.

Goal: Find $X \in \mathcal{F}$ s.t. $\omega(X) = \sum_{e \in X} \omega(e)$ is maximized.
(i.e., $\omega(X) \geq \omega(X')$ for all $X' \in \mathcal{F}$.)

Algorithm: 1. Start with $X_0 = \emptyset$.
2. Sort E so that $\omega(e_1) \geq \omega(e_2) \geq \dots \geq \omega(e_n)$, where $n = |E|$.
3. For each $i = 1, 2, \dots, n$,

$$X_i = \begin{cases} X_{i-1} + e_i, & \text{if there is a set } Y \text{ in } \mathcal{F} \text{ so that } X_{i-1} + e_i \subseteq Y, \\ X_{i-1}, & \text{otherwise.} \end{cases}$$

Output: X_n .

Note that we know that $X_n \subseteq Y$ for some $Y \in \mathcal{F}$. Moreover, we can always guarantee that $X_n \in \mathcal{F}$. Suppose not, i.e., X_n is properly contained in Y for all $Y \in \mathcal{F}$ containing X_n . Choose any such Y . Let i be the maximum number so that $e_i \in X_n$. Let $e_j \in Y - X_n$. Then $j \neq i$. It is impossible that $j < i$ by the third step of the algorithm. However, $j > i$ is also impossible. Therefore, we can conclude that $X_n \in \mathcal{F}$.

Remind that the above is just an algorithm. We do not know that the goal is really achieved or not, i.e., possibly, X_n is not optimal, i.e., possibly, there is $X' \in \mathcal{F}$ such that $\omega(X_n) < \omega(X')$.

Note that we can replace $X_{i-1} + e_i \subseteq Y$ with $X_{i-1} + e_i = Y \cap \{e_1, \dots, e_i\}$ in the third step in the algorithm. Obviously, the new condition, $X_{i-1} + e_i = Y \cap \{e_1, \dots, e_i\}$, deduces the old condition, $X_{i-1} + e_i \subseteq Y$. In addition, we can easily check that the old condition implies the new condition. If not, there is $Y \in \mathcal{F}$

such that $Y \cap \{e_1, \dots, e_i\} \neq X_{i-1} + e_i \subseteq Y$. So there is $e_j \in (Y \cap \{e_1, \dots, e_{i-1}\}) - X_{i-1}$. However, it implies that $X_{j-1} + e_j \subseteq Y$, so $X_{j-1} + e_j = X_j \subseteq X_{i-1}$. It is a contradiction.

One example of applying the greedy algorithm is a spanning tree of a connected graph G . See *Modern Graph Theory* by Béla Bollobás, page 11-14.

Proposition 6.3. *The greedy algorithm always finds an optimum solution if $\mathcal{F}$ is a set of all bases of some matroid M .*

Proof. Suppose not, i.e., there is a base Y such that $\omega(Y) > \omega(X_n)$. Let i be the minimum such that $Y \cap \{e_1, \dots, e_i\} \neq X_n \cap \{e_1, \dots, e_i\} = X_i$. Choose Y so that i is maximized. Note that by our choice of i , $Y \cap \{e_1, \dots, e_{i-1}\} = X_{i-1}$. (If $i = 1$, then $Y \cap \emptyset = X_0$.)

Case 1, $e_i \notin X_i$. Then $X_{i-1} + e_i$ is dependent. It implies $e_i \notin Y$ since if not, $X_{i-1} + e_i \subseteq Y$, a contradiction. Hence $Y \cap \{e_1, \dots, e_i\} = X_i$, which is a contradiction.

Case 2, $e_i \in X_i$. Then $e_i \notin Y$. There is the fundamental circuit C of e_i with respect to Y . Since X_i is independent, $C \not\subseteq X_i = (Y \cap \{e_1, \dots, e_{i-1}\}) + e_i$. There is $f \in C - X_i = C - \{e_1, \dots, e_i\}$. Then $f = e_j$ for some $j > i$, so $\omega(f) \leq \omega(e_i)$. In addition, $Y - f + e_i$ is independent since it does not contain a circuit. $Y - f + e_i$ is a base. Let $Y' := Y - f + e_i$. Then $\omega(Y') \geq \omega(Y)$, and $i' > i$ where i' be the smallest number such that $Y' \cap \{e_1, \dots, e_{i'}\} \neq X_{i'}$. It contradicts to our choice of Y . $\square$

Proposition 6.4. *Let E be a finite set and $\mathcal{I}$ be a set of subset of E satisfying (I1), (I2), and (G): The greedy algorithm finds an optimum solution for all $\omega : E \rightarrow \mathbb{R}$ when $\mathcal{F}$ is the set of all maximal members of $\mathcal{I}$. Then $(E, \mathcal{I})$ is a matroid.*

Proof. WTS (I3) holds. Let $X, Y \in \mathcal{I}$ with $|X| < |Y|$. Suppose $X + e \notin \mathcal{I}$ for all $e \in Y - X$. It implies $X \subsetneq Y$, i.e., $X - Y \neq \emptyset$. Let $\epsilon > 0$ be arbitrary. Let us define a weight function $\omega : E \rightarrow \mathbb{R}$ as

$$\omega(e) = \begin{cases} 1 + \epsilon, & \text{if } e \in X, \\ 1, & \text{if } e \in Y - X, \\ 0, & \text{otherwise.} \end{cases}$$

Let $X_n \in \mathcal{F}$ be an output of the greedy algorithm with the above inputs. X_i 's are defined also in a procedure of the algorithm. Here, it is okay that we choose an arbitrary sorting of E in the second step of the algorithm. By (G), X_n is optimal, i.e., $\omega(X_n) \geq \omega(X)$ for all $X' \in \mathcal{F}$. Then $X = X_{|X|} \subseteq X_n$ since $X \in \mathcal{I}$. In addition, $X_n \subseteq X \cup (E - Y)$ since $X + e \notin \mathcal{I}$ for all $e \in Y - X$. We can check that

$$\begin{aligned} \omega(X_n) &= (1 + \epsilon)|X|, \\ \omega(Y) &= |Y - X| + (1 + \epsilon)|X \cap Y| = |Y| + \epsilon|X \cap Y|. \end{aligned}$$

Since $\omega(X_n) \geq \omega(Y') \geq \omega(Y)$ where $Y \subseteq Y' \in \mathcal{F}$,

$$\epsilon|X - Y| \geq |Y| - |X| > 0.$$

Remind that $\epsilon > 0$ is arbitrary. Choose small $\epsilon < \frac{|Y| - |X|}{|X - Y|}$. Then it contradicts to the above inequality. Therefore, we can conclude that $\mathcal{I}$ satisfies (I3), so $(E, \mathcal{I})$ is a matroid. $\square$

6.1 Homework 2.3, Hyperplane Axioms

Proposition 6.5. *Let E be a finite set and let $\mathcal{H}$ be a collection of subsets of E . Prove that $\mathcal{H}$ is the set of all hyperplanes of a matroid if and only if the following three properties hold.*

(H1) $E \notin \mathcal{H}$.

(H2) If $H_1, H_2 \in \mathcal{H}$ and $H_1 \subseteq H_2$, then $H_1 = H_2$.

(H3) For all distinct H_1 and H_2 in $\mathcal{H}$ and for all $x \in E$, there exists $H \in \mathcal{H}$ with

$$(H_1 \cap H_2) + x \subseteq H.$$

Proof. ($\Rightarrow$) Let M be a matroid with a ground set E . Let $\mathcal{H}$ be the collection of all hyperplanes of M . Let r be the rank function of M . We will show $\mathcal{H}$ satisfies (H1), (H2), and (H3).

Since $r(E) = r(M)$, E is not a hyperplane, i.e., $E \notin \mathcal{H}$. So (H1) holds.

Let $H_1, H_2 \in \mathcal{H}$ and $H_1 \subseteq H_2$. Suppose there is $e \in H_2 - H_1$. Then

$$r(M) - 1 = r(H_1) \leq r(H_1 + e) \leq r(H_2) = r(M) - 1.$$

It implies that $e \in cl(H_1) = H_1$, which is a contradiction. Hence $H_1 = H_2$. (H2) holds.

Let H_1, H_2 be a distinct hyperplanes, and $x \in E$. Denote $A := (H_1 \cap H_2) + x$. By the submodular inequality,

$$r(H_1 \cap H_2) + r(H_1 \cup H_2) \leq r(H_1) + r(H_2) = 2r(M) - 2.$$

Since $H_1 \cup H_2 \supsetneq H_1$, $r(H_1 \cup H_2) = r(M)$. So $r(H_1 \cap H_2) \leq r(M) - 2$. Note that $r(X + e) \leq r(X) + r(\{e\}) \leq r(X) + 1$ for any $X \subseteq E$ and $e \in E$. So $r(A) \leq r(M) - 1$. It also implies that through we add elements of $E - A$ into A one-by-one, we can find a set $B \subseteq E - A$ such that $r(A + B) = r(M) - 1$. Choose $H := cl(A + B)$. Then it is a flat of rank $r(cl(A + B)) = r(A + B) = r(M) - 1$, which contains A . So (H3) holds.

($\Leftarrow$) We will show the opposite direction. Let $\mathcal{H}$ be a collection of subsets of E satisfying (H1), (H2), and (H3). Let $\delta : 2^E \rightarrow 2^E$ be a function defined by

$$\delta(X) = \bigcap_{\substack{H \in \mathcal{H} \\ X \subseteq H}} H.$$

If there is no such $H \in \mathcal{H}$ satisfying $X \subseteq H$, then we define $\delta(X) = E$. First, we claim that δ satisfies the closure axioms.

By its construction, δ satisfies (CL1) and (CL2), i.e., $X \subseteq \delta(X)$, and if $X \subseteq Y$ then $\delta(X) \subseteq \delta(Y)$.

Let $X \subseteq E$. If $H \in \mathcal{H}$ contains X , then it must contain $\delta(X)$ by its construction. It implies that $\delta(X) \supseteq \delta(\delta(X))$ for $\delta(X) \subsetneq E$. If $\delta(X) = E$, i.e., there is no $H \in \mathcal{H}$ containing X , then $\delta(\delta(X)) = \delta(E) = E = \delta(X)$. Hence (CL3) holds.

Now we will show that δ satisfies (CL4). Let $f \in \delta(X + e) - \delta(X)$. Suppose $e \notin \delta(X + f)$. Then

1. $[f \in \delta(X + e)]$ implies that [if there is $H_1 \in \mathcal{H}$ with $H_1 \supseteq X + e$, then $H_1 \ni f$].
2. $[f \notin \delta(X)]$ implies that [there is $H_3 \in \mathcal{H}$ s.t. $H_3 \supseteq X$ and $H_3 \not\ni f$].
3. $[e \notin \delta(X + f)]$ implies that [there is $H_2 \in \mathcal{H}$ s.t. $H_2 \supseteq X + f$ and $H_2 \not\ni e$].

If there is no $H_3 \in \mathcal{H}$ containing X , then $\delta(X) = E \ni f$. So there is $H_3 \in \mathcal{H}$ containing X . If all such H_3 contains f , then $f \in \delta(X)$. Therefore, (2) holds. Similarly, (3) holds. Note that $f \in H_2$ and $f \notin H_3$, so H_2 and H_3 are distinct.

By (H3), there is $H_1 \in \mathcal{H}$ such that $H_1 \supseteq (H_2 \cap H_3) + e$ (so $H_1 \supseteq X + e$). By the above argument $e, f \in H_1$. $f \in H_2$ but $e \notin H_2$. $f \notin H_3$ so $e \notin H_3$ by (1). Hence H_1, H_2, H_3 are distinct.

Let us say that a triple (G_1, G_2, G_3) is *good* if

- $G_1, G_2, G_3 \in \mathcal{H}$,
- $X + e + f \subseteq G_1$ (equivalently, $X + e \subseteq G_1$ by (1)),
- $X + f \subseteq G_2$ and $e \notin G_2$, and
- $X \subseteq G_3$ and $e, f \notin G_3$.

The existence of a good triple is already shown. Now, choose a good triple (G_1, G_2, G_3) so that $|G_1 \cap G_2 \cap G_3|$ is maximized. Among such candidates, choose (G_1, G_2, G_3) so that $|G_1 \cap G_2|$ is maximized. Also, among such candidates, choose (G_1, G_2, G_3) so that $|G_1 \cap G_3|$ is maximized. Let us denote a such good triple as (I_1, I_2, I_3) . In other words, (I_1, I_2, I_3) is a good tripe satisfying

- $|I_1 \cap I_2 \cap I_3| \geq |G_1 \cap G_2 \cap G_3|$ for any good triple (G_1, G_2, G_3) ,
- $|I_1 \cap I_2| \geq |G_1 \cap G_2|$ for any good tripe (G_1, G_2, G_3) with $|G_1 \cap G_2 \cap G_3| = |I_1 \cap I_2 \cap I_3|$, and
- $|I_1 \cap I_3| \geq |G_1 \cap G_3|$ for any good tripe (G_1, G_2, G_3) with $|G_1 \cap G_2 \cap G_3| = |I_1 \cap I_2 \cap I_3|$ and $|G_1 \cap G_2| = |I_1 \cap I_2|$.

WTS $(I_2 \cap I_3) - I_1 = \emptyset$. By (H3), there is $I'_1 \in \mathcal{H}$ containing $(I_2 \cap I_3) + e$. (Possibly, $I'_1 = I_1$.) Then $X + e \subseteq I'_1$, so (I'_1, I_2, I_3) is a good triple. Suppose $(I_2 \cap I_3) - I_1 \neq \emptyset$. Then $|I'_1 \cap I_2 \cap I_3| > |I_1 \cap I_2 \cap I_3|$. It contradicts to our choice of (I_1, I_2, I_3) .

By (H2), there is $g \in I_2 - I_1 (= I_2 - (I_1 \cup I_3))$. (Otherwise, $I_2 \subsetneq I_1$.) Also, by (H2), there is $h \in I_3 - I_1 (= I_3 - (I_1 \cup I_2))$. (Otherwise, $I_3 \subsetneq I_1$.) By (H3), there is $H \in \mathcal{H}$ containing $(I_1 \cap I_2) + h$. Note that $X + f \subseteq H$.

Case 1: $e \notin H$. Let $I'_2 := H$. By (H3), there is $I'_1 \in \mathcal{H}$ containing $(I'_2 \cap I_3) + e$. Then $X + e \subseteq I'_1$, $X + f \subseteq I'_2$, and $e \notin I'_2$. So (I'_1, I'_2, I_3) is a good triple. And $I'_1 \cap I'_2 \cap I_3 \supseteq (I_1 \cap I_2 \cap I_3) + h$. It contradicts to our choice of (I_1, I_2, I_3) .

Case 2: $e \in H$. Let $I'_1 := H$. Note that (I'_1, I_2, I_3) is a good triple satisfying $I'_1 \cap I_2 \cap I_3 \supseteq I_1 \cap I_2 \cap I_3$, and $I'_1 \cap I_2 \supseteq I_1 \cap I_2$.

Sub-case 2.1: $I'_1 \supseteq I_1 \cap I_3$. Then $I'_1 \cap I_3 \supseteq (I_1 \cap I_3) + h$. It contradicts to our choice of (I_1, I_2, I_3) .

Sub-case 2.2: $\exists i \in (I_1 \cap I_3) - I'_1$. Then I_1 and I'_1 are distinct. By (H3), there is $I''_1 \in \mathcal{H}$ containing $(I_1 \cap I'_1) + g$. Remind that $(I_1 \cap I_2) + e \subseteq I_1 \cap I'_1$, so $X + e + f \subseteq I_1$. A tripe (I''_1, I_2, I_3) is good. However, $I''_1 \cap I_2 \cap I_3 \supseteq I_1 \cap I_2 \cap I_3$, and $I''_1 \cap I_2 \supseteq (I_1 \cap I_2) + g$. It contradicts to our choice of (I_1, I_2, I_3) .

Hence we can conclude that the basic assumption, $e \notin \delta(X + f)$, fails via above case analysis. So δ satisfies (CL4): $f \in \delta(X + e) - \delta(X)$ implies that $e \in \delta(X + f)$.

Let M be a matroid on a ground set M defined by setting its closure function as δ . Let $\mathcal{H}_M$ be the collection of hyperplanes on M . WTS $\mathcal{H}_M = \mathcal{H}$.

For any $H \in \mathcal{H}$, $\delta(H) \subseteq H$ by the definition of δ . By (CL1), $\delta(H) = H$. So H is a flat on M . By (H1), $H \neq E$. Let $e \in E - H$. By (H2), there is no $H' \in \mathcal{H}$ containing $H + e$. It implies that $\delta(H + e) = E$. It holds for any $e \in E - H$. So H is a maximal set not spanning, i.e., H is a hyperplanes on M , i.e., $H \in \mathcal{H}_M$.

Let $G \in \mathcal{H}_M$, i.e., G is a maximal set not spanning, i.e., $\delta(G) = G \subsetneq E$ and $\delta(G + e) = E$ for any $e \in E - G$. Since $\delta(G) \subsetneq E$, there is $H \in \mathcal{H}$ such that $G \subseteq H$. Suppose there is $f \in H - G$. Then $G + f \subseteq H$, so $E = \delta(G + f) \subseteq H$. By (H1), $H \neq E$. This is a contradiction. Hence $G = H \in \mathcal{H}$.

Therefore, we can conclude that $\mathcal{H} = \mathcal{H}_M$. $\square$

7 Week05-1, 2019.09.30. Mon

Chapter. Symmetric Greedy Algorithm and Delta-matroids

Definition 7.1 (Symmetric greedy algorithm). An algorithm described in below is called the *symmetric greedy algorithm*.

Input: A finite set E , a collection $\mathcal{F} \subseteq 2^E$, and a weight function $\omega : E \rightarrow \mathbb{R}$.

Goal: Find $X \in \mathcal{F}$ s.t. $\omega(X) = \sum_{e \in X} \omega(e)$ is maximized.
(i.e., $\omega(X) \geq \omega(X')$ for all $X' \in \mathcal{F}$.)

Algorithm: 1. Sort E so that $|\omega(e_1)| \geq |\omega(e_2)| \geq \dots \geq |\omega(e_n)|$, where $n = |E|$.

2. Start with $J_0 = \emptyset$. For each $i = 1, 2, \dots, n$,

$$J_i = \begin{cases} J_{i-1} + e_i, & \text{if } (*), \\ J_{i-1}, & \text{otherwise.} \end{cases}$$

(*): For $\omega(e_i) \geq 0$, $J_{i-1} + e_i$ can be extended to a member of $\mathcal{F}$, i.e., $\exists B \in \mathcal{F}$ so that $B \cap \{e_1, e_2, \dots, e_i\} = J_{i-1} + e_i$. For $\omega(e_i) < 0$, J_{i-1} cannot be extended to a member of $\mathcal{F}$ not containing e_i , i.e., ???

Output: J_n .

Definition 7.2. We say $(E, \mathcal{F})$ is a delta-matroid if E is a finite set, and $\mathcal{F}$ is a collection of subsets of E satisfying

(D1) $\mathcal{F} \neq \emptyset$, and

(D2) $X, Y \in \mathcal{F}$ and $e \in X \triangle Y \Rightarrow \exists f \in X \triangle Y$ s.t. $X \triangle \{e, f\} \in \mathcal{F}$.

We call (D2) the symmetric greedy axiom.

Note that $X \triangle Y := (X - Y) \cup (Y - X)$.

The symmetric greedy axiom, (D2), is similar with the base exchange axiom, (B2), but not equivalent.

Remark.

Example 7.1. One trivial example of delta-matroids, which is not a matroid, is $(E, 2^E)$ for a finite set E .

8 Week05-2, 2019.10.02. Wed

Chapter. Dual Matroids

Let $\mathcal{B}$ be the collection of bases a given matroid M . Remind that

(B2) $X, Y \in \mathcal{B}$, $e \in X - Y \Rightarrow \exists f \in Y - X$ so that $X - e + f \in \mathcal{B}$, and

(B2') $X, Y \in \mathcal{B}$, $e \in Y - X \Rightarrow \exists f \in X - Y$ so that $X - f + e \in \mathcal{B}$.

Let $\mathcal{B}^* := \{E - B : B \in \mathcal{B}\}$ ($= c(\mathcal{B})$), the collection of complements of members of $\mathcal{B}$. Then it is trivial that $\mathcal{B}^*$ satisfies (B1). Consider (B2) once again. $X, Y \in \mathcal{B}$ and $e \in X - Y$. There is $f \in Y - X$ such that

$$X - e + f \in \mathcal{B}.$$

We observe that

$$(E - X) - f + e \in \mathcal{B}^*.$$

In addition, $E - X, E - Y \in \mathcal{B}^*$, $e \in (E - Y) - (E - X)$, and $f \in (E - X) - (E - Y)$. Hence (B2) for $\mathcal{B}$ corresponds to (B2') for $\mathcal{B}^*$. It implies that $\mathcal{B}^*$ defines a matroid.

Definition 8.1. The dual matroid M^* of a matroid M is a matroid on $E(M)$ having $\mathcal{B}^*$ as the collection of bases of M^* when $\mathcal{B}$ is the collection of bases of M .

By the definition, it is directly deduced that $(M^*)^* = M$.

Example 8.1. There are easy examples of dual matroids.

1. $(U_{r,n})^* = U_{n-r,n}$.
2. $M(K_3)^* = M(K_2^3)$. Here K_2^3 is a graph on two vertices, which has three parallel edges. Note that K_2^3 is a dual graph of $K_3 = \Delta$. It is a reason why Whitney developed the concept of matroids. He wanted to generalize the concept of dual on planar graphs.

Definition 8.2. We say that

- X is coindependent in M if X is independent in M^* ,
- X is a cobase in M if X is a base in M^* ,
- X is a cocircuit in M if X is a circuit in M^* ,
- X is a cohyperplane in M if X is a hyperplane in M^* , and
- e is a coloop in M if e is a loop in M^* .

Proposition 8.1. X is a cocircuit of M iff $E - X$ is a hyperplane of M .

Proof. The followings are equivalent.

- X is a cocircuit of M
- $\Leftrightarrow X$ is a circuit of M^*
- $\Leftrightarrow X$ is a minimal set which is not a subset of a base of M^*
- $\Leftrightarrow X$ is a minimal set intersecting with every base of M
- $\Leftrightarrow E - X$ is a maximal set containing no base of M
- $\Leftrightarrow E - X$ is a maximal non-spanning set
- $\Leftrightarrow E - X$ is a hyperplane.

□

Example 8.2. Let G be a connected plane graph. Let $M := M(G)$. Let $E - X$ be a hyperplane of M . It is an edge-set of a subgraph $G[E - X]$ of G , which consists of two component, and each component is an induced subgraph of G . By the previous proposition, X is a cocircuit of M . In the graph sense, X is a cycle of G^* , or equivalently X is an edge-cut of G . More precisely, it is an edge-cut between two components of $G[E - X]$.

Proposition 8.2 (Rank of M^*).

$$r_{M^*}(X) = |X| - r(M) + r_M(E - X).$$

Proof.

$$\begin{aligned} r_{M^*}(X) &= \max_{B^* \in \mathcal{B}^*} |X \cap B^*| \\ &= \max_{B \in \mathcal{B}} |X - B| \\ &= \max_{B \in \mathcal{B}} (|X| - |X \cap B|) \\ &= |X| - \min_{B \in \mathcal{B}} |X \cap B| \\ &= |X| - \min_{B \in \mathcal{B}} (|B| - |B \cap (E - X)|) \\ &= |X| - r(M) + \max_{B \in \mathcal{B}} |B \cap (E - X)| \\ &= |X| - r(M) + r_M(E - X). \end{aligned}$$

□

In particular, if $X \in \mathcal{B}$, i.e., $E - X \in \mathcal{B}^*$, then

$$r_{M^*}(X) = r_M(E - X).$$

Note that $r(M) + r(M^*) = |E|$.

We can also define dual matroids in the sense of the rank function.

Proposition 8.3. *Let M be a matroid on a finite set E , and r be the rank function of M . Let define $r^* : 2^E \rightarrow \mathbb{Z}$ as*

$$r^*(X) = |X| - r(M) + r(E - X).$$

Then r^ satisfies the rank axioms, and a matroid defined by r^* is same with the dual matroid M^* of M .*

Proof. ???

□

One important question in dual matroids is following: Suppose that M is representable over $\mathbb{F}$. Can we argue that M^* is representable over $\mathbb{F}$? The answer is positive.

Theorem 8.4. *If M is representable over $\mathbb{F}$, then M^* is representable over $\mathbb{F}$. Of course, the converse holds.*

Proof. Let A be a matrix over $\mathbb{F}$ satisfying $M = M(A)$. Note that the row operation does not affect on the independence/dependence of column vectors. Also changing of ordering of column vectors does not affect. WMA

$$A = \begin{pmatrix} I_r & A_0 \end{pmatrix},$$

where $r = r(M)$, $n = |E(M)|$, and A_0 is an $r \times (n - r)$ -matrix. In this case, we say that A is of the *standard form* (or the *standard representation*) of the vector matroid M . Let define

$$A^* := \begin{pmatrix} A_0^T & I_{n-r} \end{pmatrix},$$

an $(n - r) \times n$ -matrix. We claim that $M^* = M(A^*)$.

Denote $E := E(M) = E(M^*)$. Let $F \subseteq E$ corresponding to I_r of A , and $G \subseteq E$ corresponding to A_0 of A . Then F corresponds to A_0^T of A^* , and $G \subseteq E$ corresponds to I_{n-r} of A^* .

Let $X \subseteq E$. Then

X is a base of $M = M(A)$

$\Leftrightarrow A[X]$, a submatrix of A consisting of column vectors corresponding to X , is an invertible $r \times r$ -matrix.

Let $s = |F \cap X|$. Note that

$$A[X] = \begin{pmatrix} 0_{r-s,s} & B \\ I_s & C \end{pmatrix}.$$

where B is an $(r - s) \times (r - s)$ -matrix, and C is a $s \times (r - s)$ -matrix. In addition, the above is equivalent that

B is an invertible matrix.

Similarly, we observe that

$E - X$ is a base of $M(A^*)$

$\Leftrightarrow A^*[E - X]$, a submatrix of A^* consisting of column vectors corresponding to $E - X$, is an invertible $(n - r) \times (n - r)$ -matrix

$\Leftrightarrow B'$ is an invertible $(r - s) \times (r - s)$ -matrix,

where B' is an $(r - s) \times (r - s)$ -matrix, and D' is a $(n - 2r + s) \times (r - s)$ -matrix with

$$A^*[E - X] = \begin{pmatrix} B' & 0_{r-s,n-2r+s} \\ D' & I_{n-2r+s} \end{pmatrix}.$$

For example, by taking X as last s elements from F and first $r - s$ elements from G , we can consider A and A^* as like below

$$A = \begin{pmatrix} I_{n-r} & 0 & B & D \\ 0 & I_r & C & * \end{pmatrix}$$

and

$$A^* = \begin{pmatrix} B' & C' & I_{r-s} & 0 \\ D' & * & 0 & I_{n-2r+s} \end{pmatrix}.$$

We can confirm that $B' = B^T$. In addition, it holds for any choice of X .

Therefore, we can conclude that

$E - X$ is a base of M^*

$\Leftrightarrow X$ is a base of $M = M(A)$

$\Leftrightarrow B$ is invertible

$\Leftrightarrow E - X$ is a base of $M(A^*)$,

i.e., the collection of bases of $M(A^*)$ is same with $\mathcal{B}^*$, the collection of bases of M^* . It implies that $M(A^*) = M^*$, so M^* is representable over $\mathcal{F}$. $\square$

Note that for a plane graph G , if C is a cycle of G and D is an edge-cut of D , i.e., D is a cycle of G^* , then $|C \cap D|$ is always even. We can prove a weaker version of this fact for matroids.

Proposition 8.5. *If C is a circuit and D is a cocircuit, then $|C \cap D| \neq 1$.*

Proof. Suppose $|C \cap D| = 1$, and denote $\{e\} = C \cap D$. By the earlier proposition, $E - D$ is a hyperplane. So $E - D$ does not span $e \in D$.

By the assumption, $|(E - D) \cap C| = |C| - 1$. Since C is a circuit, $C - e = C - D$ spans $e \in C$, i.e., $C \subseteq cl(C - D)$. It implies that $E - D (\supseteq C - D)$ spans e by (CL2). It is a contradiction. $\square$

What is an example that $|C \cap D| = 2k + 1$ for some $k \geq 1$???

By the above proposition, for any element e of M , it is impossible that e is a loop and a coloop simultaneously.

Definition 8.3. Let H be a *hypergraph* if it is a pair (V, E) with $E \subseteq 2^V$ and $\emptyset \notin E$. We call $V(H) := V$ the vertex set, and $E(H) := E$ the (hyper)edge set of H .

Obviously, we can understand $(E, \mathcal{I} - \{\emptyset\})$ as a hypergraph when $M = (E, \mathcal{I})$ is a matroid.

Definition 8.4. Let $H = (V, E)$ be a hypergraph. We say that H is a *clutter* if no edge is a proper subset of another edge. In other words, E is an antichain under the inclusion, $\subseteq$. Usually, we say that E is a clutter on V .

Example 8.3. Let M be a matroid. The followings are clutters on $E(M)$.

- $\mathcal{B}(M)$, the collection of bases of M ,
- $\mathcal{C}(M)$, the collection of circuits of M ,
- $\mathcal{H}(M)$, the collection of hyperplanes of M ,
- $\mathcal{B}^*(M)$, the collection of cobases of M ,
- $\mathcal{C}^*(M)$, the collection of cocircuits of M , and
- $\mathcal{H}^*(M)$, the collection of cohyperplanes of M .

When the given matroid M is clear, we can omit M , and simply write them as $\mathcal{B}$, $\mathcal{C}$, $\mathcal{H}$, $\mathcal{B}^*$, $\mathcal{C}^*$, and $\mathcal{H}^*$.

Definition 8.5. Let $H = (V, E)$ be a hypergraph. Let $b(H)$ be a hypergraph on V , whose edges are minimal sets intersecting every edge of H . We call it the *blocker* of H .

If V is clear, then we can write it as $b(E)$, and also we can understand it as a set-system on V , i.e., understand the hypergraph $b(E)$ as its edge set.

Since we choose minimal sets, $b(H)$ is a clutter on $V(H)$.

For a convenience, we do not consider a unfruitful case that $\mathcal{I} = \mathcal{B} = \{\emptyset\}$ for a matroid M .

Proposition 8.6. $b(\mathcal{B}(M)) = \mathcal{C}^*(M)$.

Proof. See the first four equivalent statements of the proof of Proposition 8.1. $\square$

Theorem 8.7 (Edmonds-Fulkerson, '70). *Let H be a clutter. Then*

$$b(b(H)) = H.$$

Proof. In this proof, we understand each hypergraphs as its edge set. Since the vertex set of hypergraphs H , $b(H)$, $b(b(H))$ is $V(H)$, it does not make a confusion.

I referred the below proof in chapter 9, *Extremal Combinatorics with Applications in Computer Science* by Stasys Jukna.

Let us say that $X \subseteq V(H)$ is a blocking set of H if X intersects with every $Y \in H$. Then $b(H)$ is the collection of minimal blocking sets of H . Note that each member of H is a blocking set of $b(H)$.

Let $X \in b(b(H))$. Since it is a minimal blocking set of $b(H)$, there is no member of H properly contained in X . Suppose $X \not\subseteq H$. Then by the previous observation, for any $Y \in H$, $Y - X \neq \emptyset$. So we can choose $x_Y \in Y - X$ for each $Y \in H$. Let $Z := \{x_Y : Y \in H\}$. Obviously, Z is a blocking set of H , and there is $Z' \subseteq Z$ such that $Z' \in b(H)$. Then $Z' \cap X = \emptyset$, it contradicts to $X \in b(b(H))$. Therefore, $X \in H$, so $b(b(H)) \subseteq H$.

Let $X \in H$. Remind that it is a blocking set of $b(H)$. Then there is $X' \subseteq X$ such that $X' \in b(b(H))$. Since H is a clutter, $X = X' \in H$. Hence $H \subseteq b(b(H))$. $\square$

It implies that $\mathcal{B}(M) = b(\mathcal{C}^*(M))$.

Note that for a general hypergraph H , the above theorem does not holds. However, it is always true that $b(b(H)) \subseteq H$.

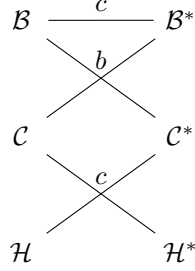
Definition 8.6. Let $H = (V, E)$ be a hypergraph. Let $c(H)$ be a hypergraph on V , of which edge set is

$$\{V - X : X \in E\}.$$

We call it the *complement* of H .

If V is clear, then we can write it as $c(E)$, and also we can understand it as a set-system on V , i.e., understand $c(E)$ as its edge set.

By the definition of the dual matroid, Proposition 8.1, 8.6, and Theorem 8.7, we can summarize relations between clutters induced by a matroid M .



9 Week06-1, 2019.10.07. Mon

Chapter. Duals of a cycle matroid

Definition 9.1. A matroid M is *cographic* if M^* is graphic.

Proposition 9.1. $M(K_5)$ is not cographic.

Proof. Suppose $M(K_5)$ is cographic, i.e., there is a graph G so that $(M(K_5))^* = M(G)$. The collection of circuits of $M(G)$ (cycles of G) does not change even though we identify vertices for each component of G . So WMA G is connected.

$$r(M(G)) = |E(M(K_5))| - r(M(K_5)) = \binom{5}{2} - 4 = 6.$$

It implies that $|E(G)| = |E(M(G))| = 10$ and $|V(G)| = |r(M(G))| + 1 = 7$. Then the average degree $d(G)$ of G is $\frac{2 \times 10}{7} < 3$. There is a vertex v of degree 1 or 2. Note that deleting incident edges of a vertex induces a hyperplane of $M(G)$. Since a cocircuit is a complement of a hyperplane, the set of incident edges of v is a cocircuit of $M(G)$. So it is a circuit of $M(K_5)$, i.e., there is a cycle of length 1 or 2 in G . It is a contradiction. $\square$

Proposition 9.2. $M(K_{3,3})$ is not cographic.

Proof. Suppose that $M(K_{3,3})$ is cographic, i.e., there is a graph G so that $(M(K_{3,3}))^* = M(G)$. WMA G is connected.

$$r(M(G)) = |E(M(K_{3,3}))| - r(M(K_{3,3})) = 9 - 5 = 4.$$

It implies that $|E(G)| = |E(M(G))| = 9$ and $|V(G)| = |r(M(G))| + 1 = 5$. Then the average degree $d(G)$ of G is $\frac{2 \times 9}{5} < 4$. There is a vertex v of degree ≤ 3 . The set of incident edges of v is a cocircuit of $M(G)$. So there is a cycle of length ≤ 3 in G , which is a contradiction. $\square$

What is a cocircuit of $M(G)$? Remind that

$$r_M(X) = |V(G)| - \#(\text{components of the subgraph } (V(G), X) \text{ of } G)$$

for $X \subseteq E(G) = E(M(G))$. (It holds whether G is connected or not.) Suppose G is connected. Let F be a flat of rank $r(M(G)) - k = |V(G)| - 1 - k$, and H be a hyperplane of $M(G)$. Then F induces $k + 1$ components, and H induces 2 components, i.e., subgraphs $(V(G), F)$ and $(V(G), H)$ of G have $k + 1$ and 2 components, respectively. From the fact that a cocircuit is a complement of a hyperplane, i.e., $\mathcal{C}^* = c(\mathcal{H})$, $E(G) - H$ is a cocircuit, and $E(G) - F$ is a union of cocircuits of $M(G)$. Moreover,

X is a cocircuit of $M(G)$

$\Leftrightarrow E(G) - X$ is a hyperplane of $M(G)$

$\Leftrightarrow E(G) - X$ is a maximal edge-set inducing two components of G

$\Leftrightarrow X$ is a minimal edge-cut of G .

Remind that $\mathcal{C}^* = b(\mathcal{B})$.

X is a cocircuit of $M(G)$, i.e., X is a minimal edge-cut of G

$\Leftrightarrow X$ is a minimal set intersecting with every base of $M(G)$

$\Leftrightarrow X$ is a minimal set intersecting with every spanning tree of G .

Definition 9.2. A *bond* of a graph G is a minimal nonempty edge-cut of G .

Definition 9.3. The *bond matroid* $M^*(G)$ of a graph G is a matroid on $E(G)$ whose circuits are bonds of G .

Note that the collection of circuits of $M^*(G)$ (bonds of G) does not change even though we identify vertices for each component of G . So WMA G is connected when we define a bond matroid.

By the above observation, $\mathcal{C}(M^*(G)) = \mathcal{C}^*(M(G)) (= \mathcal{C}(M(G)^*))$. In addition, $E(M^*(G)) = E(M(G)) = E(G)$. It implies that

$$(M(G))^* = M^*(G),$$

i.e., the dual of the cycle matroid is a bond matroid, and vice versa. So every bond matroid is cographic.

If G is a planar graph then $M(G)$ is a both graphic and cographic. It is from some facts of graph theory. Let G be a plane graph, and G^* be its geometric dual. (There is the natural one-to-one corresponding between $E(G)$ and $E(G^*)$.) Then

X is an edge-set of a cycle of G

$\Leftrightarrow X^*$ is a minimal nonempty edge-cut of G^* .

It implies that $\mathcal{C}(M(G)) = \mathcal{C}(M^*(G^*))$. In addition, if G is connected, then $(G^*)^* = G$. Therefore,

$$\begin{aligned} M(G) &= M^*(G^*), \\ M^*(G) &= M(G^*). \end{aligned}$$

(The second holds by $M^*(G) = M(G)^* = (M^*(G^*))^* = (M(G^*))^{**} = M(G^*)$.) We can interpret them as, first, cycles of G correspond to bonds of G^* , and second, bonds of G correspond to cycles of G^* . Moreover, the converse holds.

Proposition 9.3. *If $M(G)$ is both graphic and cographic then G is planar.*

It will be proved in later. See Proposition 10.3.

Chapter. Matroid Minors

Remind *graph minors*. There are three operations: vertex deletion $G \setminus v$, edge deletion $G \setminus e$ and edge contraction G/e . We say that H is a minor of G if H is obtained from G by contracting edges and deleting edges/vertices, i.e., a finite sequence of the three operations.

Definition 9.4. For a set $T \subseteq E = E(M)$, we define $M \setminus T$ as a matroid on $E - T$ such that

$I(\subseteq E - T)$ is independent in $M \setminus T$ if and only if it is independent in M .

Simply, when $M = (E, \mathcal{I})$,

$$M \setminus T = (E - T, \{I \subseteq E - T : I \in \mathcal{I}\}).$$

We call it a *deletion*. When $T = \{e\}$, we can write $M \setminus e = M \setminus \{e\}$.

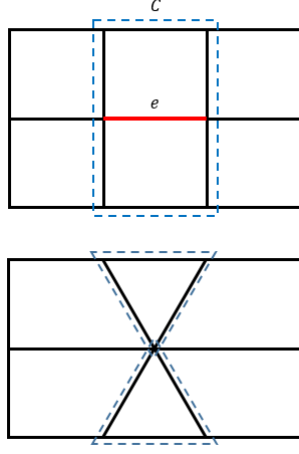
We define

$$M/T = (M^* \setminus T)^*.$$

It is a *contraction*. Here, $E(M/T) = E(M^* \setminus T) = E(M^*) - T = E - T$.

In graph sense, for a connected plane graph G , a contraction in G corresponds to a deletion in G^* . Hence the concept of minors in graph theory agrees with the concept of minors in matroid theory.

Example 9.1. Why we do not define a matroid minor by using circuits? Let G be a below first graph. C is a cycle emphasized by blue dotted line, and e is an edge emphasized by red bold line. Then G/e is a second figure. We can check that C is no longer a cycle in G/e , but it is a union of cycles. This example shows why defining a matroid contraction as the way of describing circuits is hard.



We will discuss more about circuits of a matroid contraction in later.

Proposition 9.4 (Ranks of $M \setminus T$ and M/T). *For $X \subseteq E - T$,*

$$\begin{aligned} r_{M \setminus T}(X) &= r_M(X), \\ r_{M/T}(X) &= r_M(X \cup T) - r_M(T). \end{aligned}$$

Proof. Denote $E = E(M)$. The first equation is obvious by the definition of $M \setminus T$. In addition, it implies that

$$r(M \setminus T) = r_M(E - T).$$

Remind Proposition 8.2: $r_{M^*}(X) = |X| - r(M) + r_M(E(M) - X)$.

$$\begin{aligned} r_{M/T}(X) &= |X| - r(M^* \setminus T) + r_{M^* \setminus T}((E - T) - X) \\ &= |X| - r_{M^*}(E - T) + r_{M^*}(E - T - X) \\ &= |X| - (|E - T| - r(M) + r_M(T)) \\ &\quad + (|E - T - X| - r(M) + r_M(X \cup T)) \\ &= r_M(X \cup T) - r_M(T). \end{aligned}$$

□

Proposition 9.5. *Let B_T be a maximal independent subset of T in M . Then*

$$X \text{ is independent in } M/T \Leftrightarrow X \cup B_T \text{ is independent in } M.$$

Proof. ($\Leftarrow$) If $X \cup B_T$ is independent in M , then

$$r_M(X \cup B_T) = |X \cup B_T| = |X| + |B_T| = |X| + r_M(T).$$

Hence

$$r_{M/T}(X) = r_M(X \cup T) - r_M(T) \geq r_M(X \cup B_T) - r_M(T) = |X|.$$

It implies that X is independent in M/T .

($\Rightarrow$) If X is independent in M/T , then

$$r_M(X \cup T) = r_{M/T}(X) + r_M(T) = |X| + |B_T|.$$

Let Y be a maximal independent subset of $X \cup T$ containing B_T . By the maximality of B_T , $Y \cap T = B_T$. In addition,

$$|Y - B_T| + |B_T| = |Y| = r_M(Y) = r_M(X \cup T) = |X| + |B_T|.$$

It implies that $|Y - B_T| = |X|$, so $Y - B_T = X$, i.e., $Y = X \cup B_T$. Hence $X \cup B_T$ is independent in M . $\square$

In a graph sense, we can interpret the previous proposition as follow: Let G be a graph, and $E = E(G)$. Let $T \subseteq E$. Then B_T is a maximal tree in T . X is independent in G/T if and only if X is a tree in G/T . Possibly, a vertex of the tree X can be identified by T/T . Then we can observe that $X \cup B_T$ is also a tree in M . The converse is similarly done. In addition, we can check that the maximality of B_T needs to guarantee the converse.

Sub-chapter. Circuits of M/T

Proposition 9.6. *The collection of circuits of M/T is equal to the collection of minimal nonempty sets in $\{C - T : C \text{ is a circuit of } M\}$.*

We checked in Example 9.1 that a member of $\{C - T : C \text{ is a circuit of } M\}$ is possibly not a circuit of M/T .

Lemma 9.7. *Let $\mathcal{A}$ and $\mathcal{B}$ be hypergraphs on a same vertex set. Suppose that the followings hold:*

1. *For each $A \in E(\mathcal{A})$, there is $B \in E(\mathcal{B})$ contained in A .*
2. *For each $B \in E(\mathcal{B})$, there is $A \in E(\mathcal{A})$ contained in B .*

Then $m(\mathcal{A}) = m(\mathcal{B})$.

Here, $m(\mathcal{H}) = (V, m(E))$ for a hypergraph $\mathcal{H} = (V, E)$, where $m(E)$ be the collection of minimal sets of E . Obviously, $m(\mathcal{H})$ is a clutter.

Proof. Let $A \in E(m(\mathcal{A})) = m(E(\mathcal{A}))$. By the first condition, there is $B \in E(\mathcal{B})$ such that $B \subseteq A$. WMA $B \in m(E(\mathcal{B}))$ by taking a minimal set contained in the old $B \in \mathcal{B}$. By the second condition, there is $A' \in E(\mathcal{A})$ such that $A' \subseteq B$. Then $A' \subseteq B \subseteq A$. By the minimality of A , $A' = A$, so $A = B \in m(E(\mathcal{B}))$. It implies that $m(\mathcal{A}) \subseteq m(\mathcal{B})$, i.e., $E(m(\mathcal{A})) \subseteq E(m(\mathcal{B}))$. Similarly, $m(\mathcal{B}) \subseteq m(\mathcal{A})$. $\square$

Similarly, we can prove that if $\mathcal{A}$ and $\mathcal{B}$ are two hypergraphs on a same vertex set satisfying

- 1'. *for each $A \in E(\mathcal{A})$, there is $B \in E(\mathcal{B})$ containing A , and*

2'. for each $B \in E(\mathcal{B})$, there is $A \in E(\mathcal{A})$ containing B ,

then $M(\mathcal{A}) = M(\mathcal{B})$. Here, $M(\mathcal{H}) = (V, M(E))$ for a hypergraph $\mathcal{H} = (V, E)$, where $M(E)$ be the collection of maximal sets of E . Obviously, $M(\mathcal{H})$ is a clutter.

Proof of Proposition 9.6. Let $\mathcal{C}(M/T)$ be the collection circuits of M/T . Let

$$\mathcal{A} = \{C - T : C \text{ is a circuit of } M, C - T \neq \emptyset\}.$$

Note that $m(\mathcal{C}(M/T)) = \mathcal{C}(M/T)$ since it is a clutter. Note that $m(\mathcal{A})$ is equal to the collection of minimal nonempty sets in $\{C - T : C \text{ is a circuit of } M\}$. In addition, both $\mathcal{C}(M/T)$ and $\mathcal{A}$ are edge sets of hypergraphs on $E(M) - T = E(M/T)$.

By the lemma, ETS

1. if $C - T \in \mathcal{A}$ is nonempty then $C - T$ contains a circuit of M/T , and
2. if C' is a circuit of M/T then it contains $D - T \in \mathcal{A}$.

To prove the first statement, ETS $C - T \in \mathcal{A}$ is dependent in M/T .

$$\begin{aligned} r_{M/T}(C - T) &= r_M((C - T) \cup T) - r_M(T) \\ &= r_M(C \cup T) - r_M(T) \\ &\leq r_M(C) - r_M(C \cap T) \\ &= |C| - 1 - |C \cap T| \\ &= |C - T| - 1. \end{aligned}$$

The inequality holds by (R3), the submodular inequality. Note that since $C - T \neq \emptyset$, $C \cap T$ is independent in M . From the above inequality, we can deduce that $C - T$ is dependent in M/T .

Now we will prove the second statement. Let C' be a circuit of M/T . Let B_T be a maximal independent subset of T in M . By the earlier proposition, X is independent in M/T if and only if $X \cup B_T$ is independent in M . Hence $C' \cup B_T$ is dependent in M . There is a circuit D of M such that $C' \cup B_T$. Obviously, $D - T = D - B_T \subseteq C'$. Since B_T is independent in M , $D \not\subseteq B_T$, i.e., $D - T$ is nonempty, i.e., $D - T \in \mathcal{A}$. $\square$

Some people may define a contraction of a matroid as this way. In other words, it might be possible defining M/T by giving its circuit as the collection of minimal nonempty sets in $\{C - T : C \text{ is a circuit of } M\}$.

What is the collection $\mathcal{C}(M \setminus T)$ of circuits of $M \setminus T$?

$$\mathcal{C}(M \setminus T) = \{C \in \mathcal{C}(M) : C \cap T = \emptyset\}.$$

Let $C \subseteq E(M) - T$. Then

$$C \in \mathcal{C}(M \setminus T)$$

$$\Leftrightarrow C \cap (E(M \setminus T) - I) \text{ for any } I \in \mathcal{I}(M \setminus T)$$

$$\Leftrightarrow C \cap (E(M) - I) \text{ for any } I \in \mathcal{I}(M)$$

$$\Leftrightarrow C \in \mathcal{C}(M).$$

(The remaining proof of Proposition 9.6 was done in October 14th.)

10 Week07-1, 2019.10.14. Mon

Sub-chapter. Closure in M/T

Proposition 10.1. $cl_{M/T}(X) = cl_M(X \cup T) - T$.

Proof. Let $e \in E(M) - T$. Then

$$\begin{aligned} r_{M/T}(X + e) - r_{M/T}(X) &= (r_M((X + e) \cup T) - r_M(T)) - (r_M(X \cup T) - r_M(T)) \\ &= r_M((X \cup T) + e) - r_M(X \cup T). \end{aligned}$$

It implies that $e \in cl_{M/T}(X)$ if and only if $e \in cl_M(X \cup T) - T$. $\square$

Without doubt, the closure function of $M \setminus T$ satisfies that

$$cl_{M \setminus T}(X) = cl_M(X) - T.$$

(For $e \in E(M) - T$, $r_{M \setminus T}(X + e) - r_{M \setminus T}(X) = r_M(X + e) - r_M(X)$.)

Sub-chapter: Minor

Definition 10.1. Let M, N be matroids. N is a minor of M if

$$N = M \setminus T_1 / T_2$$

for some $T_1, T_2 \subseteq E(M)$ with $T_1 \cap T_2 = \emptyset$.

The above definition of a matroid-minor is enough because of a below proposition.

Proposition 10.2. Let M be a matroid. Let T_1, T_2 be disjoint subsets of $E(M)$. Then

$$\begin{aligned} M \setminus T_1 \setminus T_2 &= M \setminus (T_1 \cup T_2), \\ M / T_1 / T_2 &= M / (T_1 \cup T_2), \\ M / T_1 \setminus T_2 &= M \setminus T_2 / T_1. \end{aligned}$$

Proof. The first one is obvious from the definition of the deletion. The second one is a dualized problem of the first. More precisely,

$$\begin{aligned} M/T_1/T_2 &= (M^* \setminus T_1)^*/T_2 \\ &= (M^* \setminus T_1 \setminus T_2)^* \\ &= (M^* \setminus (T_1 \cup T_2))^* \\ &= M/(T_1 \cup T_2). \end{aligned}$$

The last equality is proved by comparing rank functions. For any $X \subseteq E(M) - (T_1 \cup T_2)$,

$$\begin{aligned} r_{M/T_1 \setminus T_2}(X) &= r_{M/T_1}(X) \\ &= r_M(X \cup T_1) - r_M(T_1) \\ &= r_{M \setminus T_2}(X \cup T_1) - r_{M \setminus T_2}(T_1) \\ &= r_{M \setminus T_2/T_1}(X). \end{aligned}$$

□

Sub-chpater. Minors of graphic matroids

Let G be a graph, and e be an edge of G . Then we can easily check that

$$\begin{aligned} M(G \setminus e) &= M(G) \setminus e, \\ M(G/e) &= M(G)/e \end{aligned}$$

by comparing circuits. First,

$$\begin{aligned} C &\in \mathcal{C}(M(G \setminus e)) \\ \Leftrightarrow C &\text{ is a cycle of } G \setminus e \\ \Leftrightarrow C &\text{ is a cycle of } G \text{ not containing } e \\ \Leftrightarrow C &\in \mathcal{C}(M(G)) \text{ and } e \notin C \\ \Leftrightarrow C &\in \mathcal{C}(M(G)) \setminus e. \end{aligned}$$

Second, ... ???

Hence simply we can write $M(G \setminus T_1/T_2) = M(G) \setminus T_1/T_2$ for disjoint $T_1, T_2 \subseteq E(G)$.

This is the motivation why we define minors of matroids in this way. Moreover, it implies that

every minor of a graphic matroid is graphic.

Note that if N is a minor of M , then N^* is a minor of M^* since

$$\begin{aligned} N &= M \setminus T_1/T_2 \\ \Rightarrow N^* &= (M \setminus T_1/T_2)^* = (M \setminus T_1)^* \setminus T_2 = M^*/T_1 \setminus T_2. \end{aligned}$$

Hence we can deduce that

every minor of a cographic matroid is cographic.

In fancy words, we say that the matroid properties, graphic and cographic, are closed under the matroid-minor.

Proposition 10.3. *$M(G)$ is cographic if and only if G is planar.*

Proof. ($\Leftarrow$) Remind that $M(G)^* = M(G^*)$ whenever G is planar.

($\Rightarrow$) Remind that $M(K_5)$ and $M(K_{3,3})$ are not cographic. Since the cographic property is hereditary, a cographic matroid $M(G)$ has no minor isomorphic to $M(K_5)$ nor $M(K_{3,3})$. By the previous observation, $M(G \setminus T_1 / T_2) = M(G) \setminus T_1 / T_2$ for disjoint $T_1, T_2 \subseteq E(G)$. Note that deleting or adding isolated vertices in G does not affect on $M(G)$. Hence G has no minor isomorphic to K_5 nor $K_{3,3}$. By the Kuratowski's theorem (or Kuratowski-Wagner theorem), G is planar. $\square$

Theorem 10.4 (Tutte). *M is graphic iff M has no minors isomorphic to $U_{2,4}$, $M^*(K_5)$, $M^*(K_{3,3})$, F_7 nor F_7^* .*

Here, F_7 is the Fano matroid. Remind that it is a vector matroid induced by

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix}$$

over the binary field.

We will not cover the proof of the Tutte's theorem in this lecture.

Sub-chapter. Minors of representable matroids

Proposition 10.5. *Every minor of $\mathbb{F}$ -representable matroids is $\mathbb{F}$ -representable.*

Proof. Let $M = M(A)$ be a vector matroid, where A is a matrix over the field $\mathbb{F}$. ETS $M \setminus e$ and M / e are $\mathbb{F}$ -representable for each $e \in E := E(M)$. WMA A is standard form, see Theorem 8.4.

$$A = \begin{pmatrix} I_r & A_0 \end{pmatrix},$$

where $r = r(M)$, $n = |E|$, and A_0 is an $r \times (n - r)$ -matrix. Moreover we will show that we can conserve the structure of standard form A in $M \setminus e$ and M / e .

For a convenience, let k be an integer such that k -th column of A corresponds to e . Denote i -th row of A_0 as β_i for $1 \leq i \leq r$. Let $A_0(i)$ be a submatrix of A_0 deleting the i -th row β_i .

First, consider $M \setminus e$. Without doubt, $M \setminus e = M(A \setminus e)$ where $A \setminus e$ is a submatrix of A deleting a column corresponding to e . Remaining works are about conserving a standard form. If $k > r$, $A \setminus e$ is still a standard form. When $k \leq r$, we can check that $A \setminus e$ is not a standard form. However, we can obtain a new

standard form A' of M by the row operations and swapping columns (which do not affect on the column space) such that

$$A' = (I_r \quad A'_0),$$

and e corresponds to a column in A'_0 , unless $\beta_k = (0 \ 0 \ \cdots \ 0)$. In a case of $k \leq r$ and $\beta_k = 0_{1 \times (n-r)}$ (actually, it is equivalent that e is a coloop), $M(A) \setminus e = M(A \setminus e) = M(A/e)$, where A/e is a submatrix of A deleting both k -th row and column. In addition, A/e is a standard form.

Second, consider M/e . Remind that $M/e = (M^* \setminus e)^*$, and $M^* = M(A^*)$ where

$$A^* = (A_0^T \quad I_{n-r}) = (\beta_1^T \quad \beta_2^T \quad \cdots \quad \beta_r^T \quad I_{n-r}).$$

Moreover, column indexing of both A and A^* is E with same ordering. When $k \leq r$, by the earlier observation, $M^* \setminus e = M(A^* \setminus e)$ where $A^* \setminus e$ is a submatrix of A^* deleting β_k^T , i.e., $A^* \setminus e = (A_0(k)^T \quad I_{n-r})$. In addition, we can check that $M/e = (M^* \setminus e)^* = M((A^* \setminus e)^*)$ where

$$(A^* \setminus e)^* := (I_{r-1} \quad A_0(k))$$

is a standard form. When $k > r$, as like above, we can obtain a new standard form $(A^*)'$ of M^* by the row operations and swapping columns (which do not affect on the column space) such that

$$(A^*)' = ((A_0^T)' \quad I_{n-r}),$$

and e corresponds to a column in $(A_0^T)'$, unless $(k-r)$ -row of A_0^T is zero (i.e., $(k-r)$ -column of A_0 is zero, i.e., e is a loop in M). In a case of $k > r$ and $(k-r)$ -row of A_0^T is zero, we can check that $M(A^*) \setminus e = M(A^* \setminus e) = M(A^*/e)$, where A^*/e is a submatrix of A deleting k -row and $(k-r)$ -column. Hence $M/e = M(A^*/e)^* = M((A^*/e)^*)$ where

$$(A^*/e)^* = (I_r \quad A_0[k-r]),$$

and $A_0[k-r]$ is a submatrix of A_0 deleting $(k-r)$ -th column. □

Remark. If e is a loop or a coloop then

$$M \setminus e = M/e.$$

First, when e is a loop, it is simply proved by comparing circuits,

$$\mathcal{C}(M \setminus e) = \mathcal{C}(M) - \{e\} = \mathcal{C}(M/e).$$

Second, when e is a coloop in M , i.e., e is a loop in M^* , $M^* \setminus e = M^*/e$. Hence

$$M/e = (M^* \setminus e)^* = (M^*/e)^* = (M^{**} \setminus e)^{**} = M \setminus e.$$

Remark. Minors of transversal matroids are not necessarily transversal. For example, let $G = (V = [4], E = \{a, a', b, b', c, c', d\})$ where edges a, a' are incident with vertices 1,2, b, b' are incident 2,3, c, c' are incident with 3,4, and d is incident with 4,1. Then we can easily check that $M(G)$ is transversal, but $M(G/d)$ is not transversal.

Chapter. Matroid Intersection

One basic question on matroid theory is following: Let M_1 and M_2 be two matroids on E . Find a set X independent in both M_1 and M_2 with maximizing $|X|$. Why we consider this? By the greedy algorithm, we can maximize a given weight for one matroid. When we think two matroid (or more), can we do like that?

Example 10.1 (Maximum matching in a bipartite graph). Let G be a bipartite graph with a bipartition (A, B) . Let M_1 be a partition matroid on E whose partition is given by the vertices of A , i.e., $e, f \in E$ are in a same partition set if e, f are incident with a same vertex in A . Let M_2 be a partition matroid on E whose partition is given by the vertices of B . Then $X \subseteq E$ is independent in both M_1 and M_2 if and only if X is a matching in G . In addition, it is a well-known problem finding a maximum matching in a bipartite graph. (I remember that we can do it in poly-time. (?))

Example 10.2 (Intersection of three matroids is hard). Let G be a directed graph. Let M_1 be a matroid on $E(G)$ such that X is independent if every vertex has ≤ 1 incoming edge in X . Let M_2 be a matroid on $E(G)$ such that X is independent if every vertex has ≤ 1 outgoing edge in X . Let M_3 be a cycle matroid of G as a undirected graph. Then X is independent in M_1, M_2, M_3 and $|X| = |V(G)| - 1$ if and only if it is a directed path passing through all vertices if and only if it is a (directed) Hamiltonian path. (Note that X is independent in M_1, M_2 if and only if it is directed path or directed cycle. Note that X is independent with $|X| = |V(G)| - 1$ in M_3 if and only if it is a spanning tree in the underlying graph of G .) Note that an algorithm finding a Hamiltonian path is NP-complete.

Theorem 10.6 (Matroid intersection theorem). *Let $M_1 = (E, \mathcal{I}_1)$ and $M_2 = (E, \mathcal{I}_2)$ be matroids with rank functions r_1 and r_2 , respectively. Then*

$$\max_{I \in \mathcal{I}_1 \cap \mathcal{I}_2} |I| = \min_{X \subseteq E} (r_1(X) + r_2(E - X)).$$

(Moreover, there is an efficient algorithm finding $I \in \mathcal{I}_1 \cap \mathcal{I}_2$ achieving the above equation. Here, the oracle is about rank function.)

Proof. (Unfortunately, this proof gives an exponential-time algorithm.)

($\leq$) If I is a common independent set of M_1 and M_2 , then

$$\begin{aligned} r_1(X) &\geq |X \cap I|, \\ r_2(E - X) &\geq |(E - X) \cap I|. \end{aligned}$$

Hence $r_1(X) + r_2(E - X) \geq |X \cap I| + |(E - X) \cap I| = |I|$.

($\geq$) Let $k = \min_{X \subseteq E} (r_1(X) + r_2(E - X))$. We will prove this problem by an induction on $|E|$. Suppose that there is no e so that $\{e\}$ is independent in both M_1 and M_2 . Then $\emptyset$ is the largest independent set in both M_1 and M_2 , i.e., $\max_{I \in \mathcal{I}_1 \cap \mathcal{I}_2} |I| = 0$. In addition, we can easily check that $k = 0$. Let X be the set of loops in M_1 . Then $E - X$ is the set of loops in M_2 by the given assumption (there is no e so that $\{e\} \in \mathcal{I}_1 \cap \mathcal{I}_2$). Hence $r_1(X) = r_2(E - X) = 0$.

Now WMA there is e so that $\{e\}$ is independent in both M_1 and M_2 . If there is a common independent set of size k in $M_1 \setminus e$ and $M_2 \setminus e$, then it is also a common independent set in M_1 and M_2 , so the proof is done. So WMA every common independent set in $M_1 \setminus e$ and $M_2 \setminus e$ is of size $< k$, i.e., by the induction hypothesis, there is $S \subseteq E - e$ so that

$$r_1(S) + r_2((E - e) - S) = r_{M_1}(S) + r_{M_2}((E - e) - S) \leq k - 1.$$

Let M be a matroid, and $e \in E(M)$ not be a loop. If J is an independent set in M/e , then $J + e$ is an independent set in M . Suppose not, i.e., $J + e$ is dependent in M , i.e., there is a circuit C in M contained in $J + e$. Then $\emptyset \neq C - e \subseteq J$ since e is not a loop (note that possibly $e \notin C$). It implies that $C - e$ contains a circuit of M/e , so J is a dependent set in M/e . It is a contradiction, so we can conclude that $J + e$ is an independent set in M .

If there is a common independent J set of size $k - 1$ in M_1/e and M_2/e , then $J + e$ is a common independent set of size k in M_1 and M_2 by the above observation, so the proof is done. So WMA every common independent set in M_1/e and $M_2/2$ is of size $< k - 1$, i.e., by the induction hypothesis, there is $T \subseteq E - e$ so that

$$\begin{aligned} & r_1(T + e) + r_2(E - T) - 2 \\ &= r_1(T + e) - r_1(\{e\}) + r_2(((E - e) - T) + e) - r_2(\{e\}) \\ &= r_{M_1/e}(T) + r_{M_2/e}((E - e) - T) \\ &\leq k - 2. \end{aligned}$$

Then

$$\begin{aligned} & 2k - 1 \\ &\geq r_1(S) + r_2(E - e - S) + r_1(T + e) + r_2(E - T) \\ &= (r_1(S) + r_1(T + e)) + (r_2(E - e - S) + r_2(E - T)) \\ &\geq r_1(S \cap T) + r_1(S + T + e) + r_2(E - (S + T + e)) + r_2(E - (S \cap T)) \\ &= (r_1(S \cap T) + r_2(E - (S \cap T))) + (r_1(S + T + e) + r_2(E - (S + T + e))) \\ &\geq k + k = 2k. \end{aligned}$$

Here, the second inequality holds by applying the submodular inequality twice. It is a contradiction. Therefore, we can conclude that there is a common independent set of size k in both $M_1 \setminus e$ and $M_2 \setminus e$, or a common independent set of size $k - 1$ in both M_1/e and M_2/e . It makes the proof complete. $\square$

Remark (Kőnig's theorem). Remind that Kőnig's theorem in graph theory: For a bipartite graph G , $\nu(G) = \tau(G)$ where $\nu(G)$ is the maximum size of a matching in G , and $\tau(G)$ is the minimum size of a set of vertices hitting every edge (we usually call this set a *vertex cover* in G). We can prove this theorem using Theorem 10.6, the matroid intersection theorem.

Note that $\nu(G) \leq \tau(G)$ is obvious. (Give a minimum vertex cover C in G , and a maximum matching M . Then a vertex in C is incident with at most one edge in M since M is the matching. It implies that $\nu(G) = |M| \leq |C| = \tau(G)$.)

Let G be a bipartite graph with a bipartition (A, B) . Let $M_1 = (E(G), \mathcal{I}_1)$ and $M_2 = (E(G), \mathcal{I}_2)$ be defined as like in Example 10.1. Let r_i be the rank function of M_i for $i = 1, 2$. Then by the matroid intersection theorem,

$$\begin{aligned}\nu(G) &= \max_{I \in \mathcal{I}_1 \cap \mathcal{I}_2} |I| \\ &= \min_{X \subseteq E} (r_1(X) + r_2(E - X)) \\ &= \min_{X \subseteq E} (\#(\text{vertices in } A \text{ hitting } X) + \#(\text{vertices in } B \text{ hitting } E - X)) \\ &\geq \tau(G).\end{aligned}$$

Check the last line.....

11 Week07-2, 2019.10.16. Wed

Lemma 11.1. *Let M be a matroid on $E = \{e_1, e_2, \dots, e_n\}$. Let $(m_1, m_2, \dots, m_n)$ be a finite sequence of positive integers, and*

$$E' = \{e_1^1, e_1^2, \dots, e_1^{m_1}, e_2^1, \dots, e_2^{m_2}, \dots, e_n^1, \dots, e_n^{m_n}\}.$$

Let M' be a pair $(E', \mathcal{I}')$ such that $I' \in \mathcal{I}'$ if and only if (i) I' contains at most one of e_i^k 's for varying $1 \leq k \leq m_i$, and (ii) the underlying set I of I' is an independent set in M . Here, the underlying set I of I' is a set of e_i 's satisfying $e_i^k \in I'$ for some k . Then M' is a matroid.

Proof. Obviously, $\mathcal{I}'$ satisfies (I1) and (I2). Let $X', Y' \in \mathcal{I}'$ with $|X'| < |Y'|$. Let X and Y be the underlying sets of X' and Y' , respectively. By (i), $|X| < |Y|$, so there is $e_i \in Y - X$ such that $X + e_i$ is independent in M . There is $e_i^k \in Y' - X'$. In addition, the underlying set of $X_i + e_i^k$ is $X + e_i$, so $X_i + e_i^k \in \mathcal{I}$. Therefore, $\mathcal{I}'$ satisfies (I3). $\square$

By considering a matroid-deletion, we can consider that $(m_1, m_2, \dots, m_n)$ be a finite sequence of non-negative integers in the above lemma.

Theorem 11.2 (Rado's theorem, 1962). *Let G be a bipartite graph with a bipartition (A, B) . Let N be a matroid on B with the rank function r . Then G has a matching M covering A such that $V(M) \cap B$ is independent in N if and only if $r(n_G(X)) \geq |X|$ for all $X \subseteq A$.*

Here, $n_G(X) = (\cup_{x \in X} n_G(x)) - X$ is the open neighbor set of X .

The Rado's theorem is a generalization of the Hall's theorem: G has a matching M covering A if and only if $|n_G(X)| \geq |X|$ for all $X \subseteq A$. Set a matroid N in the Rado's theorem as $(B, 2^B)$. It implies the Hall's theorem.

Proof. ($\Rightarrow$) It is trivial. (Let M_X be a matching whose edges are in M and incident with at least one vertex of X . Remind that M covers A , and $V(M) \cap B$ is independent in N . Then $r(n_G(X)) \geq r(V(M_X) \cap B) = |X|$.)

($\Leftarrow$) Denote $E := E(G)$. Let M_1 be a matroid on E such that X is independent in M_1 if and only if no 2 edges share a vertex in A . In other words, M_1 is a partition matroid on E whose partition is given by the vertices of A . Let M_2 be a matroid on E such that X is independent in M_2 if and only if no 2 edges share a vertex in B , and the vertices of B incident with edges in X for a independent set of N . We can consider M_2 as a matroid defined as like the previous lemma with $(m_1, \dots, m_{|B|})$ is a sequence of degrees of corresponding vertices in B . Hence we can observe that M_2 is actually a matroid by the previous lemma.

Let r_1 and r_2 be the rank functions of M_1 and M_2 , respectively. Then for $X \subseteq E$,

$$\begin{aligned} r_1(X) &= |Z|, \\ r_2(E - X) &\geq r(n_G(A - Z)) \geq |A - Z|, \end{aligned}$$

where $Z (= V(X) \cap A)$ is the set of vertices in A hitting edges in X . The first inequality in the second line since $E - X$ contains all edges incident with vertices of $A - Z$ (so $V(E - X) \cap B \supseteq n_G(A - Z)$). The second inequality in the second line holds by the given condition. Then

$$r_1(X) + r_2(E - X) \geq |Z| + |A - Z| = |A|.$$

By the matroid intersection theorem, M_1 and M_2 have a common independent set M of size $|A|$. We can check that M is a matching covering A such that $V(M) \cap B$ is independent in N , from the construction of M_1 and M_2 . $\square$

Theorem 11.3 (Generalized version of Rado's theorem). *Let G be a bipartite graph with a bipartition (A, B) . Let N be a matroid on B with the rank function r . Let $d \geq 0$ be fixed. Then G has a matching M covering $\geq |A| - d$ vertices in A (or B) such that $V(M) \cap B$ is independent in N if and only if $r(n_G(X)) \geq |X| - d$ for all $|X| \subseteq A$.*

Proof. The proof is almost same. Here, we notice some different parts.

($\Rightarrow$) Let M_X be a matching whose edges are in M and incident with at least one vertex of X . Then $r(n_G(X)) \geq r(V(M_X) \cap B) \geq |X| - d$.

($\Leftarrow$)

$$\begin{aligned} r_2(E - X) &\geq r(n_G(A - Z)) \geq |A - Z| - d, \\ r_1(X) + r_2(E - X) &\geq |A| - d. \end{aligned}$$

By the matroid intersection theorem, M_1 and M_2 have a common independent set M of size $|A| - d$. $\square$

By setting $N = (B, 2^B)$ in the above theorem, we can deduce a variation of Hall's theorem: Let G be a bipartite graph with bipartition (A, B) . Then G has a matching covering $|A| - d$ vertices of A if and only if $|n_G(X)| \geq |X| - d$ for all $X \subseteq A$. From this,

$$\begin{aligned}\nu(G) &= \max\{|X| : X \subseteq A \text{ matchable to } B\} \\ &= |A| - \min\{d \in \mathbb{Z}_{\geq 0} : |n_G(X)| \geq |X| - d \text{ for all } X\} \\ &= |A| - \max_{X \subseteq A} (|X| - |n_G(X)|) \\ &= \min_{X \subseteq A} (|A| - |X| + |n_G(X)|).\end{aligned}$$

(Here, note that $\max_{X \subseteq A} (|X| - |n_G(X)|) \geq 0$ by setting $X = \emptyset$.)

Remark (The rank function of transversal matroid). Let G be a bipartite graph with a bipartition (A, B) . Let M be a transversal matroid on A with respect to G . Remind that $X \subseteq A$ is an independent set in M if and only if it is matchable to B . From the previous observation, we can deduce that

$$r_M(Y) = \min_{X \subseteq Y} (|Y| - |X| + |n_G(X)|).$$

for any $Y \subseteq A$. (Consider the above variation of the Hall's theorem for an induced subgraph $G[Y \cup B]$).

Theorem 11.4 (Matroid union theorem). *Let $M_1, M_2, \dots, M_n$ be matroids on E . Let $\mathcal{I}_i$ be the collection of independent sets of M_i , and r_i be the rank function of M_i . Let $\mathcal{I} = \{I_1 \cup \dots \cup I_n : I_i \in \mathcal{I}_i\}$. Then $M = (E, \mathcal{I})$ is a matroid, and its rank function satisfies that*

$$r_M(X) = \min_{Y \subseteq X} (r_1(Y) + r_2(Y) + \dots + r_n(Y) + |X - Y|).$$

Here, remind that $r_M(X) = \max_{Z \subseteq X, Z \in \mathcal{I}} |Z|$. So the matrix union theorem is also regarded as a min-max type theorem.

Remark (LP duality). Why we consider these kinds of min-max problems? Notice the duality of the linear programming (LP). LP is a problem having a form of the following:

Objective: $\max c^T x$,

Restrictions: $Ax \leq b$, $x \geq 0$ and $x \in \mathbb{R}^n$.

Its dual is

Objective: $\max b^T y$,

Restrictions: $A^T y \geq c$, $y \geq 0$ and $y \in \mathbb{R}^m$.

In addition, the integer programming (IP) is same with LP except $x \in \mathbb{Z}^n$. Similarly, IP dual is same with LP dual except $y \in \mathbb{Z}^m$.

One surprising observation is the optimal solutions of LP and LP dual are exactly same. However, it does not generally hold for IP and IP dual.

... img ...

Researchers in combinatorial optimization work to find a condition that optimal solutions of IP and IP dual are same. Here, the matroid intersection theorem and the matroid union theorem (and their corollaries) are good examples of these kinds. König's theorem ($\nu(G) = \tau(G)$ for a bipartite graph G) is one of the most popular example.

Proof of Theorem 11.4. First, WTS $\mathcal{I}$ satisfies the independent set axioms. Obviously, $\mathcal{I}$ satisfies that (I1) and (I2). Let $X, Y \in \mathcal{I}$ with $|X| < |Y|$. Then $X = \cup_{i=1}^n I_i$ and $Y = \cup_{j=1}^n J_j$ where $I_i, J_j \in \mathcal{I}$ for each i, j . WMA $I_i \cap I_j = \emptyset$ and $J_i \cap J_j = \emptyset$ for all distinct i, j . (Set new I' as $I'_i = I_i - \cup_{t=1}^{i-1} I_t$, and do similarly for J_i .) Choose I_i 's and J_i 's so that $\sum_i |I_i \cap J_i|$ is maximized. There is k such that $|I_k| < |J_k|$ since $\sum |I_i| = |X| < |Y| = \sum |J_i|$, so there is $e \in J_k - I_k$ such that $I_k + e$ is independent in M_k . If $e \notin X$, then $X + e \in \mathcal{I}$ makes (I3) hold. So WMA $e \in X$. It implies that $e \in I_t$ for some $t \neq k$. Let us define

$$I'_i = \begin{cases} I_t - e, & \text{if } i = t, \\ I_k + e, & \text{if } i = k, \\ I_i, & \text{o.w..} \end{cases}$$

Then $I'_i \in M_i$ for each i , I'_i 's are disjoint, and $\cup I'_i = X$. In addition, $\sum |I'_i \cap J_i| = \sum |I_i \cap J_i| + 1$. It contradicts to our choice of I_i 's and J_i 's. Hence we can conclude that $\mathcal{I}$ satisfies (I3), so M is a matroid on E .

Second, WTS

$$r_M(X) = \min_{Y \subseteq X} (r_1(Y) + r_2(Y) + \dots + r_n(Y) + |X - Y|).$$

WMA $X = E$. (..... why?? ETS for $X = E$?) Let $E_1, E_2, \dots, E_n$ be n disjoint copies of E . Let $\pi_i : E_i \rightarrow E$ be the natural bijection. Let N_1 be a matroid on $\cup_{i=1}^n E_i$ such that X is independent in N_1 if and only if $\pi_i(X \cap E_i)$ is independent in M_i for all i . (We can easily show that N_1 is actually a matroid by checking the three independent axioms. Combining matroids on disjoint grounds sets always makes a matroid.) Let N_2 be a matroid on $\cup E_i$ such that X is independent in N_2 if and only if no two copies of some element of E are in X , i.e., $\pi_i(X \cap E_i) \cap \pi_j(X \cap E_j) = \emptyset$ for any distinct i, j . N_1 is a nothing but a partition matroid (of which partition is the collection of the set of copies in $\cup E_i$ of each element of E).

Let I is a common independent set of N_1 and N_2 . The the corresponding set $I' = \cup \pi_i(I \cap E_i)$ of elements of E is independent in M , and $|I'| = |I|$. It implies that

$$r_M(E) \geq \max_{I \in \mathcal{I}(N_1) \cap \mathcal{I}(N_2)} |I|.$$

Let $J \subseteq \mathcal{I}$ such that $r_M(E) = r_M(J)$. $J = \cup J_i$ for $J_i \in \mathcal{I}_i$. WMA J_i 's are disjoint. Let $I = \cup \pi_i^{-1}(J_i)$. Then I is a common independent set of N_1 and N_2 , so

$$r_M(E) \leq \max_{I \in \mathcal{I}(N_1) \cap \mathcal{I}(N_2)} |I|.$$

By Theorem 10.6, the matroid intersection theorem,

$$\begin{aligned} r_M(E) &= \min_{Y \subseteq \cup E_i} \left(r_{N_1}(Y) + r_{N_2}((\cup E_i) - Y) \right) \\ &= \min_{\forall i, Y_i \subseteq E_i} \left(r_{N_1}(\cup Y_i) + r_{N_2}((\cup E_i) - (\cup Y_i)) \right) \\ &= \min_{\forall i, Y_i \subseteq E_i} \left(\left(\sum r_i(\pi_i(Y_i)) \right) + |\cup (E - \pi_i(Y_i))| \right) \\ &= \min_{\forall i, Y_i \subseteq E_i} \left(\left(\sum r_i(\pi_i(Y_i)) \right) + |E - \cap \pi_i(Y_i)| \right). \end{aligned}$$

Take Y_i 's so that the minimum achieved. Set $Z = \cap \pi_i(Y_i)$. It implies that

$$\begin{aligned} &\min_{\forall i, Y_i \subseteq E_i} \left(\left(\sum r_i(\pi_i(Y_i)) \right) + |E - \cap \pi_i(Y_i)| \right) \\ &\geq \min_{Z \subseteq E} \left(\left(\sum r_i(Z) \right) + |E - Z| \right). \end{aligned}$$

In addition, by taking $Y_i = \pi_i^{-1}(Z)$ for each i , we can deduce that the above inequality is actually the equality. In conclusion,

$$r_M(E) = \min_{Z \subseteq E} \left(\left(\sum r_i(Z) \right) + |E - Z| \right).$$

□

Let M be a matroid on E . Its rank function is r . Let us write $M_1 \vee M_2$ to denote the union of two matroids M_1 and M_2 on E .

Corollary 11.4.1. *M has k bases whose union is E if and only if $k \cdot r(X) \geq |X|$ for all $X \subseteq E$.*

Proof. Let M_i 's be copies of M (on the same ground set). Then by Theorem 11.4, the matroid union theorem, (check first line....)

$$\begin{aligned} r_{M_1 \vee M_2 \vee \dots \vee M_k}(E) &= |E| \\ \Leftrightarrow \min_{X \subseteq E} (k \cdot r(X) + |E - X|) &\geq |E| \\ \Leftrightarrow \text{For all } X \subseteq E, k \cdot r(X) &\geq |X|. \end{aligned}$$

□

Corollary 11.4.2. *M has k disjoint bases if and only if $k \cdot r(X) + |E - X| \geq k \cdot r(M)$ for all $X \subseteq E$.*

Proof. □

We can interpret them in the graph sense. Let G be a graph.

Corollary 11.4.3. *G has k forests covering all edges if and only if $k(|T| - 1) \geq |E(G[T])|$ for all $T \subseteq V(G)$.*

Proof. □

12 Week09-1, 2019.10.28. Mon

Corollary 12.0.1 (Nash-Williams 1961; Tutte 1961). *Let G be connected. Then G has k edge-disjoint spanning trees if and only if for any partition of $V(G)$ into $P_1, P_2, \dots, P_s$ for some s ($P_i \neq \emptyset$ and $\cup P_i = V(G)$)*

$$\#(\text{edges having ends in distinct } P_i \text{'s}) \geq (s - 1)k.$$

Proof. ($\Rightarrow$) Trivial.

($\Leftarrow$) Our goal is showing that $k \cdot r(Y) + |E - Y| \geq k \cdot r(E)$ for any $Y \subseteq E$, where $E = E(G)$ and r is the rank function of $M(G)$. (Then by a corollary of the matroid union theorem, the proof is done.) WMA F is a flat (otherwise we adding edges not to increase its rank, and to decrease $|E - Y|$.) It implies that each component of $G[Y] = (V(G), Y)$ is an induced subgraph of G . Let us denote that $P_1, P_2, \dots, P_s$ are the vertex-sets of the components of $G[Y]$. Then

$$\begin{aligned} |E - Y| &= \#(\text{edges having ends in distinct } P_i \text{'s}) \geq (s - 1)k, \\ k \cdot r(Y) &= k(|V(G)| - s). \end{aligned}$$

Adding above two, we obtain

$$k \cdot r(Y) + |E - Y| \geq k(|V(G)| - 1) = k \cdot r(M(G)).$$

□

Corollary 12.0.2. *If G is $2k$ -edge-connected, then G has k edge-disjoint spanning trees.*

Proof. Let $P_1, \dots, P_s$ be a partition of $V(G)$. By $2k$ -edge-connectedness, there are at least $2k$ edges having ends in P_i and $V(G) \setminus P_i$ for each i . Then by double-counting,

$$\#(\text{edges having ends in distinct } P_i \text{'s}) \geq 2k \cdot s \cdot \frac{1}{2} = ks.$$

By the previous corollary, the proof is done. □

Does the above corollary give us a tight bound? How to prove it without a matroid argument (i.e., only graph sense proof)?

Note that if G has k edge-disjoint spanning trees, then G is k -edge-connected. For two distinct vertices of G , we get k paths in k edge-disjoint spanning trees. Then paths are edge-disjoint. By the Menger's theorem, the argument is completed.

Application on $\text{mad}(G)$, the maximum average degree of G .

$$\text{mad}(G) := \max_{H \subset G, |V(H)| \geq 1} \frac{\sum_{v \in V(H)} \deg_H(v)}{|V(H)|} = \max \frac{2|E(H)|}{|V(H)|}.$$

How can we decide $\text{mad}(G) \leq \alpha$ for some $\alpha \in \mathbb{Q}$. Is this decision problem solved in poly-time? Yes, it is by assuming that the algorithm of the matroid union theorem is poly-time.

Definition 12.1 (Bicircular matroid $B(G)$). Let define a matroid $B(G)$ on $E(G)$ satisfying one of below two: $X \subseteq E(G)$.

- X is a circuit iff X is a theta graph, two edge-disjoint cycles with exactly one common vertex, or two (vertex-)disjoint cycles and a path from a vertex in a cycle to a vertex in the other cycle.
- X is independent iff each component of the subgraph $(V(G), X)$ has at most 1 cycle.

We call $B(G)$ a *Bicircular matroid*.

We can easily check that the above two conditions are equivalent, and actually a bicircular matroid is a matroid.

Proposition 12.1. $B(G)$ is a matroid.

Proof. Our goal is showing that $B(G)$ satisfies the independent axioms. (I1) and (I2) holds obviously. Now WTS (I3). Suppose that X, Y is independent, $|X| < |Y|$, and $X \cup \{e\}$ is not independent for all $e \in Y - X$. WMA $E(G) = X \cup Y$. WMA G is connected (choose a component C of G so that $|X \cap E(C)| < |Y \cap E(C)|$). WMA $V(Y) \subseteq V(X)$ (otherwise there is an edge e of Y so that one of its ends is in $V(Y)$, and we can add it into X to make a larger independent set). Here $V(Y)$ is the set of vertices incident with Y .

Let X_1 be the union of the edge-set of the components of (V, X) having a cycle. (We call a graph having exactly one cycle a *unicyclic* graph.) Let $X_2 = X - X_1$, i.e., the union of edge-set of acyclic components of (V, X) . WMA there are no edges of $Y - X$, of which ends vertices are in X_1 and X_2 , or both are in X_2 (of course there are two sub-cases: two ends are in a same component, or not). (Otherwise a edge of a previous case can be added into X to make a larger independent set.) It implies that every edge of Y , of which ends are in $V(X_2)$, is in X_2 . In addition, the number of edges of Y , of which ends are in $V(X_1)$, is less or equal than $|V(X_1)|$ ($= |X_1|$) since each components of $(V(G), Y[V(X_1)])$ is acyclic (so $|E| = |V| - 1$) or unicyclic (so $|E| = |V|$). Hence $|Y| \leq |X_1| + |X_2| = |X|$, which is a contradiction. $\square$

Note that the rank of $B(G)$ is equal to $|V(G)| - \#(\text{acyclic components})$.

Proposition 12.2. *Let $k > 0$ and $l > 0$ be natural numbers. Let G_l be a graph obtained from G by replacing each edge with l parallel edges. Then the following are equivalent:*

- (1) $\text{mad}(G) \leq \frac{2k}{l}$
(i.e., $|E(H)|/|V(H)| \leq k/l$ for every subgraph H of G with $V(H) \neq \emptyset$),
- (2) $|F| \leq k \cdot r_{B(G_l)}(F)$ for all $F \subseteq E(G_l)$, and
- (3) $B(G_l)$ has k bases whose union is $E(G_l)$.

Proof. (2) $\Leftrightarrow$ (3) by a corollary of matroid union theorem.

(1) $\Rightarrow$ (2). Let $F \subseteq E(G_l)$. WMA F is a flat of $B(G_l)$. WMA F induces a connected graph. If F has at least 1 cycle, then F contains all l parallel edges whenever it contains one of them. Let H be a subgraph of G consisting of edges in F . Then

$$\begin{aligned} |F| &= l|E(H)|, \\ r_{B(G_l)}(F) &= |V(F)| = |V(H)|. \end{aligned}$$

Since $|E(H)|/|V(H)| \leq k/l$, we can conclude that $|F| \leq k \cdot r_{B(G_l)}(F)$.

If F has no cycles, then $r_{B(G_l)}(F) = |F|$. So $k \cdot r_{B(G_l)}(F) \geq |F|$.

(2) $\Leftarrow$ (1). Let H be an induced subgraph of G . Let $F = E(H_l)$ be the set of edges of G_l corresponding to edges in H . Then

$$\begin{aligned} r_{B(G_l)}(F) &= |V(H)| - \#(\text{acyclic components of } E(H_l)) \leq |V(H)|, \\ |F| &= l|E(H)|. \end{aligned}$$

Then $|F| \leq k \cdot r_{B(G_l)}(F) \leq k|V(H)|$, so $|E(H)|/|V(H)| \leq k/l$. $\square$

By the above proposition, a poly-time algorithm for the matroid union theorem implies that there is a poly-time algorithm for the decision problem: deciding whether $\text{mad}(G) \leq \alpha$ or not for a fixed $\alpha \in \mathbb{Q}$.

Note that a cycle matroid $M(G)$ is represented by a signed vertex-edge incident matrix of G over $\mathbb{Q}$ or $\mathbb{R}$, of which (v, e) -entry is 1 if v is an end of e , -1 if v is the other ends of e , and 0 otherwise. A bicircular matroid can be represented by a matrix replacing -1 to some other non-negative value in some field. There is a survey on this topic...

Remark (Equivalence of the matroid intersection/union theorems). Let M_1 and M_2 be matroids. How do we get a common independent set from the matroid union theorem?

A common independent set of size k exists

$$\Leftrightarrow |B_1 \cap B_2| \geq k \text{ for some } B_i \in \mathcal{B}(M_i), i = 1, 2,$$

$$\Leftrightarrow |B_1 \cup (E - B_2)| = |E| - |B_2 - B_1| = |E| - |B_2| + |B_1 \cap B_2| \geq |E| - |B_2| + k$$

$$\Leftrightarrow M_1 \vee M_2^* \text{ has rank } \geq |E| - r(M_2) + k.$$

From this, the matroid union theorem implies that the matroid intersection theorem. (???) We proved that the matroid union theorem from the matroid intersection theorem. Hence both theorems are equivalent.

13 Week09-1, 2019.10.30. Wed

Sub-chapter. Algorithm for the matroid union

Let $M_1 = (E, \mathcal{I}_1), M_2 = (E, \mathcal{I}_2), \dots, M_n = (E, \mathcal{I}_n)$ be matroids on E . Our goal is finding a maximal independent set in $M_1 \vee \dots \vee M_n$. It is enough to find disjoint n sets $X_1, X_2, \dots, X_n$ where $X_i \in \mathcal{I}_i$. First, find $s \notin \cup X_i$ such that $\cup X_i + s$ is independent in $\vee M_i$ if it exists.

For each i , let us define a directed graph $D_{M_i}(X_i)$ on E , i.e., its vertex set is E , so that

$$x \rightarrow y \text{ if } x \in X_i, y \notin X_i, \text{ and } X_i - x + y \in \mathcal{I}_i.$$

Let $D = D_{M_1}(X_1) \cup \dots \cup D_{M_n}(X_n)$, i.e., the graph induced by the union of (directed) edge-sets. Let $F_i = \{x \notin X_i : X_i + x \in \mathcal{I}_i\}$. Denote $X := \cup X_i$ and $F := \cup F_i$.

Lemma 13.1. *Let $s \in E - X$. Then $X + s$ is independent in $\vee M_i$ if and only if D has a directed path from F to s .*

Proof.

□

The previous lemma gives us an algorithm to find a maximal independent set in $\vee M_i$. Start from $X_0 = \emptyset$, and get independent $X_{k+1} = X_k + s_{k+1}$ for some $s_{k+1} \in E - X_k$ satisfying the necessity condition of the lemma (can find a directed path from F to s_{k+1}). Unfortunately, this algorithm has a bad time-efficiency since we check all elements of $E - X_k$ for each $k \leq r(\vee M_i)$ in a worst case. (Is it good? Time $r(\vee M_i) \times |E|$?) (Note that there is a good algorithm to find a shortest path in a graph. Poly-time?)

Remark. In 1986, Cunningham found an algorithm of time $O((r^{3/2} + n)mQ + r^{1/2}nm)$, where n is the number of matroids, $m = |E|$, $r = r(\vee M_i) \leq m$, and Q is a time to test whether a set is independent (i.e., a running time of an oracle).

Chapter. Connectedness

Note that the disjoint union of 2 matroids is a matroid. More precisely, if $M_1 = (E_1, \mathcal{I}_1)$ and $M_2 = (E_2, \mathcal{I}_2)$ are matroids with disjoint ground sets, then a new structure $M = (E, \mathcal{I})$ with $E = E_1 \cup E_2$ and $\mathcal{I} = \{X \cup Y : X \in \mathcal{I}_1, Y \in \mathcal{I}_2\}$ is a matroid. We can check easily that M satisfies the independent axioms. Denote it as $M_1 \oplus M_2$.

In a graph sense, a disconnected graph can be represent to the vertex-disjoint union of connected component. Moreover, the connectedness is defined as like this in a usual sense. We also define the connectedness of matroids following the

usual sense. However, it is little bit different with the connectedness in graph theory.

Definition 13.1. A matroid is *connected* if it cannot be written as the disjoint union of two non-trivial matroids.

Here a matroid is non-trivial means that its ground set is non-empty.

Proposition 13.2. A matroid M on E is connected if and only if $r(X) + r(E - X) > r(E)$ for all $\emptyset \neq X \subsetneq E$, where r is the rank function of M .

Proof. ($\Leftarrow$) Suppose M is disconnected, i.e., $M = M_1 \oplus M_2$ for some matroids M_1, M_2 on non-empty disjoint ground sets. Denote the rank function of M_i as r_i , and $E_i = E(M_i)$ for $i = 1, 2$. Then we can easily check that

$$r(Z) = r_1(Z \cap E_1) + r_2(Z \cap E_2).$$

Especially, $r(E) = r_1(E_1) + r_2(E_2)$. By taking $X = E_1 \notin \{\emptyset, E\}$, we can conclude that $r(E) = r_1(X) + r_2(E - X) = r(X) + r(E - X)$.

($\Rightarrow$) Suppose $r(X) + r(E - X) = r(E)$ for some $X \neq \emptyset, E$. We claim that

$$M = (M \setminus (E - X)) \oplus (M \setminus X).$$

ETS if $P \subseteq X$ and $Q \subseteq E - X$ are independent in M (i.e., independent in $M \setminus (E - X)$ and $M \setminus X$, respectively), then $P \cup Q$ is independent. By the submodular inequality,

$$\begin{aligned} r(P \cup Q) + r(X) &\geq r(P) + r(X \cup Q), \\ r(X \cup Q) + r(E - X) &\geq r(Q) + r(E). \end{aligned}$$

Therefore, we can conclude that $r(P \cup Q) \geq r(P) + r(Q) = |P| + |Q| = |P \cup Q|$. $\square$

Proposition 13.3. A matroid M is connected if and only if its dual M^* is connected.

Proof. Let r and r^* be the rank functions of M and M^* , respectively. Remind that $r^*(X) = |X| + r(E - X) - r(M)$.

$$\begin{aligned} &r^*(X) + r^*(E - X) - r^*(M) \\ &= (|X| + r(E - X) - r(M)) + (|E - X| + r(X) - r(M)) - (|E| - r(M)) \\ &= r(E - X) + r(X) - r(M). \end{aligned}$$

Therefore, the given statements holds by the previous proposition. $\square$

Theorem 13.4 (Tutte). If M is connected, $|E(M)| \geq 2$, and $e \in E(M)$, then $M \setminus e$ or M/e is connected.

Proof. Suppose not, i.e., both $M \setminus e$ and M/e are not connected. Then by an earlier proposition, there is a partition (non-empty disjoint subsets covering whole set) (X_1, Y_1) and (X_2, Y_2) of $E(M) - e$ such that $r_{M \setminus e}(X_1) + r_{M \setminus e}(Y_1) = r(M \setminus e)$ and $r_{M/e}(X_2) + r_{M/e}(Y_2) = r(M/e)$. Equivalently,

$$\begin{aligned} r(X_1) + r(Y_1) &= r(E - e), \\ r(X_2 + e) + r(Y_2 + e) &= r(E) + r(\{e\}), \end{aligned}$$

where $E = E(M)$ and r is the rank function of M .

Note that a connected matroid has no loops ($\nexists f \in E$ s.t. $r(f) = 1$) and no coloops ($\nexists f \in E$ s.t. $r(E - f) = r(E)$). If M has a loop f , then $M = M \setminus f \oplus M \setminus (E - f)$, so M is not connected. If M has a coloop f iff M^* has a loop f , then M^* is not connected iff M is not connected. Hence $r(E - e) = r(M)$ and $r(\{e\}) = 1$.

For a case that one of $X_1 \cap X_2, X_1 \cap Y_2, Y_1 \cap X_2, Y_1 \cap Y_2$ is empty, WMA $X_1 \cap Y_2$ is empty by swapping labels. It implies that $X_1 \cap X_2 = X_1 \neq \emptyset$ and $Y_1 \cap Y_2 = Y_2 \neq \emptyset$. Hence WMA $X_1 \cap X_2 \neq \emptyset$ and $Y_1 \cap Y_2 \neq \emptyset$.

From these observations,

$$\begin{aligned} & 2 \cdot r(M) - 1 \\ &= (r(X_1) + r(Y_1)) + (r(X_2 + e) + r(Y_2 + e)) \\ &= (r(X_1) + r(X_2 + e)) + (r(Y_1) + r(Y_2 + e)) \\ &\geq (r(X_1 \cap X_2) + r((X_1 \cup X_2) + e)) + (r(Y_1 \cap Y_2) + r((Y_1 \cup Y_2) + e)) \\ &= (r(X_1 \cap X_2) + r((Y_1 \cup Y_2) + e)) + (r(X_1 \cap X_2) + r((Y_1 \cup Y_2) + e)) \\ &\geq 2 \cdot (r(M) + 1). \end{aligned}$$

Here the first inequality is from the submodular inequality, and the second inequality holds since $(X_1 \cap X_2, (Y_1 \cup Y_2) + e)$ and $(Y_1 \cap Y_2, (X_1 \cup X_2) + e)$ are partitions of E , and M is connected. The above inequality derives a contradiction, so the proof is done. $\square$

(The remaining proof of Theorem 13.4 was done in November 06th.)

14 Week10-2, 2019.11.06. Wed

We say $x \sim y$ if $x = y$, or M has a circuit containing x and y . Note that $\sim$ is transitive, i.e., if $x \sim y$ and $y \sim z$ then $x \sim z$. (Check the Homework 2.1.) The reflexivity and symmetry are obvious. Hence $(E(M), \sim)$ is an equivalence relation.

Definition 14.1. A *component* of a matroid M is an equivalence class of $(E(M), \sim)$.

Proposition 14.1. A matroid M is connected if and only if for all distinct $x, y \in E(M)$, M has a circuit containing x and y , i.e., M has only one component.

Proof. ($\Rightarrow$) Suppose there are $x \neq y$ in $E(M)$ such that M has no circuits containing both x and y . Let $Z := [x]$, an equivalence class containing x . There is no circuit C of M such that $C \cap Z \neq \emptyset$ and $C - Z \neq \emptyset$. It implies that X is independent iff $X \cap Z$ and $X - Z$ are independent. (Forward direction is obvious. Reverse direction: Suppose X is dependent, then there is a circuit C contained in X . By the given condition, either $X \cap Z$ or $X - Z$ contains C .) Hence we can conclude that M is a disjoint union of two matroids $M \setminus Z$ and $M \setminus (E(M) - Z)$.

($\Leftarrow$) Suppose $r(X) + r(Y) = r(M)$ for some partition (X, Y) of $E(M)$, i.e., M is a disjoint union of two matroids M_1 and M_2 such that $\emptyset \neq X = E(M_1)$ and $\emptyset \neq Y = E(M_2)$. Then there is no circuit C such that $C \cap X \neq \emptyset$ and $C \cap Y \neq \emptyset$. (If there is such circuit C , then $C \cap X$ and $C \cap Y$ are independent in M since they are proper subset of C .)

$$\begin{aligned} r(C) + r(X) + r(Y) &\geq r(C \cap X) + r(C \cup X) + r(Y) \\ &\geq r(C \cap X) + r((C \cup X) \cap Y) + r(C \cup X \cup Y) \\ &= r(C \cap X) + r(C \cap Y) + r(M). \end{aligned}$$

It implies that $r(C) \geq r(C \cap X) + r(C \cap Y)$, so $r(C) = r(C \cap X) + r(C \cap Y) = |C \cap X| + |C \cap Y| = |C|$, i.e., C is independent. It is a contradiction.) Hence there is no circuit containing $x \in X$ and $y \in Y$. $\square$

Corollary 14.1.1. *Let G be a graph on > 2 vertices without isolated vertices. Then TFAE:*

- $M(G)$ is connected.
- For any two distinct edges, G has a cycle containing both edges.
- G is 2-connected, and has no loops.

Proof. The second equivalence is easily checked. (It is totally graph theory, so we skip. Remind that G is 2-connected if $|V(G)| > 2$, G is connected, and has no cut-vertex.)

The first equivalence is nothing, but a special case of the previous proposition. $\square$

Corollary 14.1.2. *M is connected if and only if for all distinct $x, y \in E(M)$, M has a cocircuit containing x and y .*

Proof. Remind that M is connected iff M^* is connected, and apply the previous proposition. $\square$

Proposition 14.2. *M is connected if and only if for all distinct $x, y \in E(M)$, M has a circuit or a cocircuit containing x and y .*

Proof. Actually, the earlier proposition and corollary imply that if M is connected, then for all distinct $x, y \in E(M)$, M has a circuit and a cocircuit containing x and y .

Now WTS the converse. Suppose $M = M_1 \oplus M_2$ with $E(M_1), E(M_2) \neq \emptyset$. Then we can easily check that $M^* = M_1^* \oplus M_2^*$. ($E(M) - X \subseteq E(M)$ is a base of M^* iff $X \subseteq E(M)$ is a base of M iff $X \cap E(M_1)$ and $X \cap E(M_2)$ are bases of M_1 and M_2 respectively iff $(E(M) - X) \cap E(M_1)$ and $(E(M) - X) \cap E(M_2)$ are bases of M_1^* and M_2^* respectively. Furthermore, check ranks also proves the claim. $r(M_1) + r(M_2) = r(M)$ iff $r(M_1^*) + r(M_2^*) = r(M^*)$.) Choose $x \in E(M_1) = E(M_1^*)$ and $y \in E(M_2) = E(M_2^*)$. Then there are no circuits nor cocircuits containing both x and y . $\square$

Proposition 14.3. *Let M be a connected matroid with $|E(M)| \geq 3$. Then for all $|X| = 3$, M has a circuit or a cocircuit containing X .*

It is not true that M has a circuit and a cocircuit containing X . For example, consider a theta graph (consider three paths, and identifying each starting and ending vertex) with joining three P_3 (a path of length 3, three edges and four vertices). Take X as three edges joining with a starting vertex. Then there is no circuit containing X , but there is a cocircuit containing X (X itself).

Proof. Induction on $|E(M)|$.

It is trivial for $|E(M)| = 3$. If $r(M) = 0$, then all elements are loops, so M is not connected. If $r(M) = 3$, then all elements are coloops, so M is not connected. For $r(M) = 1$ or 2 , WMA $r(M) = 1$ by the duality. Note that $r(M) = 1$, no loops, and no coloops implies that $M = U_{1,3}$. Then $E(M)$ is a cocircuit. (When $r(M) = 2$, a dual case of rank = 1, $M = U_{2,3}$. It implies that $E(M)$ is circuit.)

Now let us assume $|E(M)| > 3$. Choose $e \notin X$. By Theorem 13.4, $M \setminus e$ or M/e is connected. By symmetry (duality), WMA M/e is connected. If M/e has cocircuit $D \supseteq X$ then D is a cocircuit of M (D is a circuit of $M^* \setminus e$, so in M^*). If M/e has a circuit $C \supseteq X$ then C or $C + e$ is a circuit of M . (By Proposition 9.5, C is dependent in M/e iff $C + e$ is dependent in M . In addition, $C - f$ is independent in M/e iff $C - f + e$ is independent in M , so $C - f$ is independent in M for all $f \in C$. If C is dependent, then C is a circuit. If $C = (C + e) - e$ is independent, then $C + e$ is a circuit.) $\square$

Proposition 14.4. *Let M, N be connected matroids. Let $e \in E(M) - E(N)$. If N is a minor of M , then N is a minor of $M \setminus e$ or M/e which is connected.*

This proposition implies that we can choose a sequence $e_1, e_2, \dots, e_l$ of $E(M) - E(N)$ such that contracting or deleting e_i 's sequentially preserves connectedness.

It is possible that M/e (also $M \setminus e$) contains N , but not connected. For example, ... graph...

Proof. WMA $|E(M)| \geq 2$. Suppose one of $M \setminus e$ and M/e is connected (if not, the proof is done). By the duality, WMA $M \setminus e$ is not connected. By Theorem 13.4, M/e is connected.

Note that $(M_1 \oplus M_2) \setminus f = (M_1 \setminus f) \oplus M_2$ if $f \in E(M_1)$. Since $M_1^* \oplus M_2^* = (M_1 \oplus M_2)^*$ is easily concluded by Proposition 13.2, $(M_1 \oplus M_2) / f = (M_1 / f) \oplus M_2$ if $f \in E(M_2)$.

Let us denote $M \setminus e$ as $M_1 \oplus M_2$ with $|E(M_i)| > 0$ for $i = 1, 2$. Since N is connected, WMA Y is a minor of M_1 . Let $X = E(M_2)$. Then $M_1 = M \setminus e \setminus X$. Let r be the rank function of M . Since M is connected, $r(E(M_1) + e) + r(X) > r(M)$. Since $M \setminus e$ is not connected, $r(E(M_1)) + r(X) = r(M \setminus e)$. Since M is connected, it has no coloops (and no loops), so $r(M) = r(M \setminus e)$. Combining these, we deduce that $r(E(M_1) + e) > r(E(M_1))$. It implies that e is a coloop in $M \setminus X$. So $M_1 = (M \setminus X) \setminus e = (M \setminus X) / e = (M / e) \setminus X$. Hence the minor N of M_1 (and $M \setminus e$) is also the minor of M / e which is connected. $\square$

Definition 14.2. Let M be a matroid, and B be a base of M . Let us define a bipartite graph G_B on $E(M)$ such that x and y is adjacent if $x \in B$, $y \notin B$, and $B - x + y$ is a base. It is a bipartite graph with a bipartition $(B, E(M) - B)$. We call it the *fundamental graph* with respect to a base B .

Theorem 14.5. M is connected if and only if G_B is connected.

Proof. ($\Leftarrow$) By the definition of G_B , $x \in B$ and $y \notin B$ are adjacent in G_B iff $B + y$ has a circuit containing x , so $x \sim y$. If there is a path $(a_0 = a, a_1, a_2, \dots, a_l = b)$ in G_B , then $a = a_0 \sim a_1 \sim a_2 \sim \dots \sim a_l = b$, so $a \sim b$. Hence if G_B is connected, then M has only one component, i.e., M is connected.

($\Rightarrow$) Suppose G_B is not connected. Let H be a component of G_B . Let $B_1 = V(H) \cap B$, $Z_1 = V(H) - B$, $B_2 = (E(M) - V(H)) \cap B$ and $Z_2 = E(M) - V(H) - B$. They by Lemma 5.3, $cl(B_1) \supseteq Z_1$ and $cl(B_2) \supseteq Z_2$. It implies that $r(M) = r(B_1) + r(B_2) = r(B_1 \cup Z_1) + r(B_2 \cup Z_2)$. Since $(V(H) = B_1 \cup Z_1, E(M) - V(H) = B_2 \cup Z_2)$ is a partition of $E(M)$, we can conclude that M is not connected. $\square$

By this theorem, we can reduce a problem checking connectedness of a matroid to a problem checking connectedness of graph. Note that BFS or DFS for neighborhoods of vertex gives a poly-time(?) algorithm for graph connectedness. Hence there is a good algorithm for checking that matroid is connected.

14.1 Homework 2.1

Proposition 14.6. Let x, y, z be distinct elements in the ground set of a matroid. Prove that if there are a circuit containing x and y and a circuit containing y and z , then there is a circuit containing x and z .

Proof. Let C and D be a circuit containing x and z , respectively, such that $C \cap D \neq \emptyset$. The existence of a such pair (C, D) is guaranteed by the given condition (more precisely, the existence of y). Choose (C, D) so that $|C \cup D|$ is minimized. Suppose that C and D are distinct (so $z \notin C$ and $x \notin D$). Let $y' \in C \cap D$. By the strong circuit elimination axiom, there is a circuit $C' \subseteq (C \cup D) - y'$ containing x . Then it satisfies two properties: (i) $C' - C \neq \emptyset$ (so $C' \cap D \supseteq (C' - C) \cap D = C' - C \neq \emptyset$), and (ii) $C' - D = C - D$. If (i) does not hold,

then $C' \subseteq C - y'$ which contradicts to the minimality of circuits. If (ii) does not hold, then $|C' \cup D| = |C' - D| + |D| < |C - D| + |D| = |C \cup D|$, which contradicts to our choice of (C, D) . Also, we can obtain a circuit $D' \subseteq (C \cup D) - y'$ containing z by the strong circuit elimination axiom. By the symmetry, it satisfies (i') $D' - D \neq \emptyset$ (so $C \cap D' \neq \emptyset$), and (ii) $D' - C = D - C$. Then we can observe that C' and D' are circuits containing x and z , respectively, such that $C' \cap D' \supseteq C' \cap (D' - C) = C' \cap (D - C) = C' - C \neq \emptyset$ by (ii') and (i). However, $C' \cup D' \subseteq (C \cup D) - y'$ (so $|C' \cup D'| < |C \cup D|$), which contradicts to our choice of (C, D) . Therefore, we can conclude that $C = D$, so it is a circuit containing both x and z . $\square$

15 Week11-1, 2019.11.11. Mon

Sub-chapter. The number of elements in a connected matroid

Theorem 15.1. *If M is connected, $e \in E(M)$, every circuit of M containing e has $\leq c$ elements, and every cocircuit of M containing e has $\leq d$ elements, then $|E(M)| \leq (c - 1)(d - 1) + 1$.*

Lemma 15.2. *Let M be a connected matroid with $|E(M)| \geq 2$, and $e \in E(M)$. If every circuit containing e has $\leq c$ elements, then there exists a list of $l \leq c - 1$ cocircuits $D_1, \dots, D_l$ such that $e \in D_1 \cap \dots \cap D_l$ and $D_1 \cup \dots \cup D_l = E(M)$.*

Proof. Induction on c . Remind that a connected matroid has no loops and no coloops.

Let $c = 2$. Since M is connected, for every pair of two elements there is a circuit containing both. It implies that every element of M is parallel to e , i.e., for all $e \neq f \in E(M)$, $\{e, f\}$ is a circuit. So $cl(\{e\}) = E(M)$, and $r(M) = 1$. Therefore, $r(M^*) = |E(M)| - r(M) = |E(M)| - 1$, so $E(M)$ is a cocircuit of M .

Let $c > 2$. Let D be a cocircuit containing e . The existence of such D is followed: M^* is also connected, so there is a circuit of M^* containing e . Remind Theorem 13.4. There is a minor N on $E(M) - (D - e)$ of M , which is connected. Denote $N = M \setminus X / Y$ where $X \cup Y = D - \{e\}$ and $X \cap Y = \emptyset$. Note that $\mathcal{C}(N)$, the collection of circuits of N , is the collection of minimal members in $\{C - Y : C \text{ is a circuit of } M \text{ with } C \cap X = \emptyset\}$. By the previous note and Proposition 8.5, every circuit of N containing e has $\leq c - 1$ elements. By the induction hypothesis, N has $l \leq c - 2$ cocircuits $D_1, \dots, D_l$ containing e with $D_1 \cup \dots \cup D_l = E(N)$. It implies that M has a list of $\leq c - 2$ cocircuits $D'_1, \dots, D'_l$ containing e with $D'_1 \cup \dots \cup D'_l \supseteq E(M) - (D - e)$, where D'_i is recovered from D_i , so $D_i \subseteq D'_i \subseteq D'_i + X$ (since $N^* = M^* / X \setminus Y$). Add D to the list. It makes the proof complete. $\square$

This lemma proves the earlier theorem directly since $E(M) = (\cup_i (D_i - e)) + e$, where $D_1, \dots, D_l$ is as in the previous lemma.

Corollary 15.1.1. *Let G be 2-connected loopless graph, and $x, y \in V(G)$. Then $|E(G)|$, the number of edges in G , is equal or less than (the length of a largest path from x to y) $\times$ (the size of a largest edge-cut).*

Proof. Add a new edge e incident with x, y (possibly it can be a parallel edge), and denote $G' = G + e$. Let $M = M(G')$. Then size of a cycle of G' containing e is less than (the length of a largest path from x to y) + 1. Size of an edge-cut of G' containing e is less than (the length of a largest edge-cut) + 1. By applying the theorem with respect to M and e , the proof is finished. $\square$

Remark. Let M be a connected matroid. If all circuits have $\leq c$ elements, and all cocircuits have $\leq d$ elements, then $|E(M)| \leq \frac{1}{2}cd$. This fact is introduced in *Matroid Theory* by J. Oxley. This argument has no condition that circuits/cocircuits containing a fixed element e . Both the earlier theorem and this remarks are tight. (What are examples? Are they graphic?)

Chapter. Higher connectivity

Definition 15.1. A *separation* of a matroid $M = (E, \mathcal{I})$ is a partition (A, B) of $E(M)$ such that $A, B \neq \emptyset$. The *order* of a separation (A, B) is

$$\lambda_M(A) = r(A) + r(B) - r(M)$$

where r is the rank function of M . We call λ_M the *connectivity function* of M . If there is no confusion, we can omit the subscript M .

Note that $\lambda_M(A) = r(A) + r^*(A) - |A|$ where r^* is the rank function of M^* . In addition, note that $\lambda_M(A) = \lambda_M(A - E)$ and $\lambda_M(A) = \lambda_{M^*}(A)$.

Definition 15.2. A separation (A, B) of a matroid M is a *k-separation* if $\lambda_M(A) < k$ and $|A|, |B| \geq k$.

Note that if one of $|A|, |B|$ is $< k$, WLOG $|A| < k$, then $\lambda_M(A) = r(A) + r(B) - r(M) \leq r(A) < k$. This case is meaningless.

Note that the existence of k -separation does not implies the existence of l separation for both $k < l$ and $l < k$ (?). What are examples?

By the submodular inequality, $\lambda \geq 0$, so 0-separation does not exist. When (A, B) is a separation, it is an 1-separation of M iff $r(A) + r(B) = r(M)$ iff M is the disjoint union of $M \setminus A$ and $M \setminus B$. So M has an 1-separation iff it is not connected.

Definition 15.3. A matroid M is *k-connected* if it has no k' -separation for all $(1 \leq) k' < k$.

So M is 2-connected iff M is connected. In addition, any matroid M is 1-connected. If M is k -connected, then M is l -connected for any $l \leq k$.

Remark. What is a relation to the (vertex-) connectivity for graph? Let (A, B) be a separation of $M(G)$. Let S be the set of vertices incident with edges in A and edges in B at the same time. Let X be the set of vertices incident with only edges in A , and Y be a set of vertices with only edges in B .

$$\begin{aligned} r(A) &= |X \cup S| - \#(\text{components of } G[X \cup S]), \\ r(B) &= |Y \cup S| - \#(\text{components of } G[Y \cup S]), \\ r(M) &= |X \cup Y \cup S| - \#(\text{components of } G). \end{aligned}$$

Let us assume that G is connected (remind that $M(G)$ does not change since the list of circuits does not change), i.e., the number of components of G is 1. However, we cannot say that numbers of components of $G[X \cup S]$ and $G[Y \cup S]$ are 1. Then

$$\begin{aligned}\lambda(A) &= r(A) + r(B) - r(M) \\ &\leq (|X| + |S| - 1) + (|Y| + |S| - 1) - (|X| + |Y| + |S| - 1) \\ &= |S| - 1.\end{aligned}$$

If G is not k -connected and $|V(G)| > k$, then G has a vertex-set S of size $< k$ such that $G \setminus S$ is not connected. Let A be an edge-set of a component of $G \setminus S$ (or a union of edge-sets of some components at least one, but not all), and $B = E(G) - A$. By the previous observation, $\lambda(A) = \lambda(B) \leq |S| - 1$. However, we cannot say that (A, B) is a $|S|$ -separation of $M(G)$ since it is possible that $|A|$ or $|B|$ is $\leq |S| - 1$.

Choose a minimal $S \subseteq V(G)$ such that $G \setminus S$ is disconnected. Then for each vertex v in S there is an edge in A incident with v , and also there is an edge in B incident with v . (Otherwise, $S - v$ is still a vertex-cut of G .) Hence (A, B) are $|S|$ -separation in $M(G)$ in this case.

In short, if a graph G is not k -connected and $|V(G)| > k$, there is a l -separation (A, B) of $M(G)$ with $l < k$, so $M(G)$ is not k -connected (it is not $(l + 1)$ -connected, and possibly $l + 1 < k$). In other words, for a graph G on $> k$ vertices,

if $M(G)$ is k -connected, then G is k -connected.

Unfortunately, the converse does not holds. For example, consider any k -connected graph containing a loop with $k \geq 2$. Then $M(G)$ has a 1-separation because of the loop, so $M(G)$ is not 2-connected (and not l -connected for any $l \geq 2$).

Remind Corollary 14.1.1. It said that $M(G)$ is 2-connected iff G is a 2-connected graph without loops. We can argue a similar statement for a 3-connected graphic matroid. See Proposition 15.4. Before arguing the statement, we will prove a useful fact.

Proposition 15.3. *For $e \in E(M)$ and $X \subseteq E(M) - e$,*

$$\lambda_M(X) - 1 \leq \lambda_{M \setminus e}(X) \leq \lambda_M(X).$$

By the duality,

$$\lambda_M(X) - 1 \leq \lambda_{M/e}(X) \leq \lambda_M(X).$$

Proof. Denote $E = E(M)$.

$$\begin{aligned}\lambda_{M \setminus e}(X) &= r_{M \setminus e}(X) + r_{M \setminus e}(E - e - X) - r(M \setminus e) \\ &= r_M(X) + r_M(E - e - X) - r_M(E).\end{aligned}$$

Then

$$\lambda_M(X) - \lambda_{M \setminus e}(X) = r_M(E - X) - r_M(E - e - X) - r_M(E) + r_M(E - e).$$

By the submodular inequality, it is ≥ 0 . In addition, $0 \leq r_M(E - X) - r_M(E - e - X) \leq 1$ and $0 \leq r_M(E) - r_M(E - e) \leq 1$. It implies $\lambda_M(X) - 1 \leq \lambda_{M \setminus e}(X) \leq \lambda_M(X)$. $\square$

By taking complement, we can also deduce that

$$\lambda_M(X + e) - 1 \leq \lambda_{M \setminus e}(X) \leq \lambda_M(X + e),$$

and

$$\lambda_M(X + e) - 1 \leq \lambda_{M/e}(X) \leq \lambda_M(X + e).$$

Proposition 15.4. *Let G be a graph with $|V(G)| > 3$. TFAE:*

- $M(G)$ is 3-connected.
- G is 3-connected, loopless, and has no parallel edges.
- G is simple 3-connected.

Proof. The second and third statements are equivalent by the definition of simple graph.

Suppose $M(G)$ is 3-connected. Then by the earlier remark, G is 3-connected. G is loopless since if not, $M(G)$ is not 2-connected. If G has parallel edges e and f , let us take $A = \{e, f\}$ and $B = E(G) - A$. Then $\lambda(A) \leq r(A) + r(B) - r(A) = (2 - 1) + (|V(G)| - 1) - (|V(G)| - 1) = 1$, and $|B| \geq 2$ since $|V(G)| \geq 4$ and G is connected. So (A, B) is a 2-separation of $M(G)$, which is a contradiction.

Suppose $M(G)$ is not 3-connected. Then there is a separation (A, B) which is 1- or 2-separation. If it is a 1-separation, i.e., $M(G)$ is not 2-connected, then G is not 2-connected (so not 3-connected) or it has a loop by Corollary 14.1.1. If (A, B) is a 2-separation, then $|A|, |B| \geq 2$ and $\lambda(A) = 0, 1$.
.....? $\square$

In a graph, adding parallel edges does not affect on its connectivity. However, in a matroid, adding parallel elements ruins its connectivity.

Theorem 15.5. *Let M be a matroid, and $e \in E(M)$. If M is $(k+1)$ -connected, then $M \setminus e$ or M/e has no k -separation (A, B) with $|A|, |B| \geq 2k - 1$.*

Even though the $(k+1)$ -connectivity is not maintained under deletion and contraction, we can argue that for any k -separation of N , its one side is small (of size $< 2k - 1$), where N is one of $M \setminus e$ or M/e . In other words, there is no k -separation of which both sides are big.

For $k = 1$, it is nothing, but the same with Theorem 13.4. ($|A|, |B| \geq 2 \cdot 1 - 1 = 1$ says only that $|E(M)| \geq 2$.) Hence Theorem 15.5 is a direct generalization of Theorem 13.4.

Lemma 15.6. *Let M be a connected (2-connected) matroid. Let $e \in E(M)$ and $X, Y \subseteq E(M) - e$. Then*

$$\lambda_{M \setminus e}(X) + \lambda_{M/e}(Y) \geq \lambda_M(X \cap Y) + \lambda_M(X \cup Y \cup \{e\}) - 1.$$

Proof. Let r be the rank function of M , and $E = E(M)$. Note that

$$\lambda_{M \setminus e}(X) = r(X) + r(E - e - X) - r(E - e),$$

and

$$\lambda_{M/e}(Y) = r(Y + e) + r(E - Y) - r(E) - r(\{e\}).$$

Since M is connected, $r(\{e\}) = 1$ (no loops) and $r(E - e) = r(E)$ (no coloops). Then

$$\begin{aligned} & \lambda_{M \setminus e}(X) + \lambda_{M/e}(Y) \\ &= \left(r(X) + r(E - e - X) - r(E) \right) + \left(r(Y + e) + r(E - Y) - r(E) - 1 \right) \\ &= \left(r(X) + r(Y + e) \right) + \left(r(E - e - X) + r(E - Y) \right) - 2r(E) - 1 \\ &\geq \left(r(X \cap Y) - r(X + Y + e) \right) + \left(r(E - (X + Y + e)) + r(E - (X \cap Y)) \right) - 2r(E) - 1 \\ &= \left(r(X \cap Y) + r(E - (X \cap Y)) - r(E) \right) + \left(r(X + Y + e) + r(E - (X + Y + e)) - r(E) \right) - 1 \\ &= \lambda_M(X \cap Y) + \lambda_M(X + Y + e) - 1. \end{aligned}$$

□

Proof of Theorem 15.5. Suppose that $M \setminus e$ has a k -separation (X_1, X_2) with $|X_i| \geq 2k - 1$, $i = 1, 2$, and M/e has a k -separation (Y_1, Y_2) with $|Y_i| \geq 2k - 1$, $i = 1, 2$.

For a case that one of $X_i \cap Y_j$ for $1 \leq i, j, \leq 2$ has size $< k$, WMA $|X_1 \cap Y_2| < k$ by changing labels. So WMA $|X_1 \cap Y_1| \geq k$ and $|X_2 \cap Y_2| \geq k$. Note that $E(M) - (X_i \cap Y_i) \supseteq X_j \cap Y_j$ for $i \neq j$. By the lemma,

$$\begin{aligned} 2(k - 1) &\geq \lambda_{M \setminus e}(X_1) + \lambda_{M/e}(Y_1) \\ &\geq \lambda_M(X_1 \cap Y_1) + \lambda_M(X_1 \cup Y_1 \cup \{e\}) - 1 \\ &= \lambda_M(X_1 \cap Y_1) + \lambda_M(E(M) - (X_1 \cup Y_1 \cup \{e\})) - 1 \\ &= \lambda_M(X_1 \cap Y_1) + \lambda_M(X_2 \cap Y_2) - 1 \\ &\geq 2k - 1. \end{aligned}$$

The last inequality holds since M is $(k + 1)$ -connected (so has no k -separation). It gives a contradiction, so the proof is completed. □

16 Week11-2, 2019.11.13. Wed

Definition 16.1. Let M be a matroid. Let us define two new matroid $\text{si}(M)$ and $\text{co}(M)$. First, $\text{si}(M)$ is the *simplification* of M , which is the matroid obtained from M deleting all loops and parallel elements (circuit of size 2). Deleting parallel elements means that for each circuit of size 2, we delete one of elements of the circuit step-by-step. Second, $\text{co}(M)$ is the *cosimplification* of N , which is the matroid obtained from M contracting all coloops and coparallel elements (step-by-step deleting one elements of cocircuit of size 2). Equivalently, $\text{co}(M) = (\text{si}(M^*))^*$.

Corollary 16.0.1 (Bixby's lemma). *Let M be a 3-connected matroid, and $e \in E(M)$. Then $\text{si}(M/e)$ or $\text{co}(M \setminus e)$ is 3-connected.*

Proof. By Theorem 15.5, $M \setminus e$ or M/e has no 2-separation (A, B) with $|A|, |B| \geq 3$. By the duality (M^* is also 3-connected), WMA M/e has no such 2-separation.

Let $N = \text{si}(M/e)$. Suppose N is not 3-connected. It implies that N has a 1-separation or a 2-separation (A, B) . By adding loops and parallel elements to A, B , we obtain a separation (A', B') of M/e . Then $\lambda_{M/e}(A') = \lambda_N(A)$ since loops and parallel elements do not affect on the rank. Remind that $\lambda_M(A') \leq \lambda_{M/e}(A') + 1$.

First, consider a case that (A, B) is a 1-separation of N , so (A', B') is a 1-separation of M/e . Note that $\lambda_M(A)$ and $\lambda_M(A + e)$ are $\leq \lambda_{M/e}(A) + 1 < 2$. Since $(A', B' + e)$ is not a 2-separation of M , $|A'| \leq 1$ or $|B' + e| \leq 1$. Since $|A'|, |B'| \geq 1$, we can deduce that $|A'| = 1$. Similarly, $|B'| = 1$ since $(A' + e, B')$ is not a 2-separation of M . It implies that $A = A' = \{a\}$, $B = B' = \{b\}$ and $|E(M)| = 3$. Since $a, b \in E(N)$, i.e., a, b are neither loops nor parallel elements in N and M/e , $N = M/e$ and $r(N) = 2$. Moreover, $\{a, b\}$ is independent in M by Proposition 9.5. So $\{a, b\}$ is independent $M \setminus e$. It implies that $r_{M/e}(\{a\}) = r_{M/e}(\{b\}) = 1 = r(M \setminus e) - 1$, so a, b are coloops of $M \setminus e$. $\text{co}(M \setminus e)$ is the matroid on the empty ground set. It is obviously 3-connected.

Next, consider a case that (A, B) is a 2-separation of N , so (A', B') is a 2-separation of M/e . WMA $|A'| = 2$ by Theorem 13.4. Since $2 \leq |A| \leq |A'| = 2$, $A = A'$. So elements of A' are neither loops nor parallel elements in N and M/e . A' is an independent set in M/e . Note that $1 \geq \lambda_{M/e}(A') = r_{M/e}(A') + r_{M/e}(B') - r(M/e) = 2 + r_{M/e}(B') - r(M/e)$. So $r_{M/e}(B') \leq r(M/e) - 1$. $r_{M/e}^*(A') = |A'| + r_{M/e}(B') - r(M/e) = |A'| - 1$. It implies that M/e has a coloop or a cocircuit of size 2 (a cocircuit of size ≤ 2). So M has a cocircuit of size ≤ 2 . (If $M^* \setminus e$ has a circuit of size ≤ 2 , then M^* has a circuit of size ≤ 2 .) Note that 3-connectivity of M implies that M has neither coloops nor cocircuit of size 2. (Suppose not, i.e., there is a cocircuit of size ≤ 2 . Of course, since M is connected, every circuit and cocircuit has size ≥ 2 . Let C be a cocircuit of size 2. Let $D = E(M) - C$. Then D is a hyperplane, so $r(D) = r(M) - 1$. $\lambda(C) = r(C) + r(D) - r(M) = r(C) - 1 \leq 1$. So (C, D) is a 2-separation. It is a contradiction.) Hence we derive a contradiction. $\square$

In the graph theory course, we learned that if G is 3-connected, then a simplification of G/e or $G \setminus e$ is also 3-connected... (?) Tutte's theorem....

Is there a matroid analogue of Menger's theorem? Unfortunately, there is no matroid analogue of paths in graph. However, there is an analogue of Menger's theorem proved by Tutte.

Theorem 16.1 (Tutte's linking theorem). *Let M be a matroid, and $E = E(M)$. Let $A, B \subseteq E$ with $A \cap B = \emptyset$. Then*

$$\min_{A \subseteq X \subseteq E-B} \lambda_M(X) = \max_{\substack{N \text{ is a minor} \\ \text{of } M \text{ on } A \cup B}} \lambda_N(A).$$

LHS of the above equation corresponds to a min-cut part of Menger's theorem. Can we regard RHS as the concept of max-flow? (Whatever, this theorem has a min-max formula.)

Proof. ($\geq$) (Usually, the inequality of this direction is easy to derive for a min-max formula. Consider the max-flow min-cut theorem.) If N is a minor on $A \cup B$. Then $N = M \setminus S/T$ for some S, T where $S \cup T = E - (A \cup B)$ and $S \cap T = \emptyset$. We know that $\lambda_{M \setminus e} \leq \lambda_M$ and $\lambda_{M/e} \leq \lambda_M$ for any $e \in E$. Moreover, $\lambda_{M \setminus e}(Y) \leq \lambda_M(Y + e)$ and $\lambda_{M/e}(Y) \leq \lambda_M(Y + e)$ for any $Y \subseteq E - e$. Then we obtain for any X with $A \subseteq X \subseteq E - B$

$$\lambda_N(A) \leq \lambda_M(X).$$

($\leq$) ETS a claim: If $\lambda_M(X) \geq k$ for all $A \subseteq X \subseteq E - B$, then M has a minor N on $A \cup B$ such that $\lambda_N(A) \geq k$.

We will prove the claim by an induction on $|E(M)|$ for fixed A and B . The claim is trivial for $E = A \cup B$. Now consider for $|E| > |A \cup B|$. Let $e \in E(M) - (A \cup B)$. If $\lambda_{M \setminus e}(X) \geq k$ for all $A \subseteq X \subseteq E - e - B$, then we win by the induction. (There is a matroid N on $A \cup B$ which is a minor of $M \setminus e$ such that $\lambda_N(A) \geq k$. It is also a minor of M .) WMA $\lambda_{M \setminus e}(X_1) < k$ for some $A \subseteq X_1 \subseteq E - e - B$. Similarly, by the induction, WMA there is Y_1 such that $\lambda_{M/e}(Y_1) < k$ for some $A \subseteq Y_1 \subseteq E - e - B$. WMA M is connected. (Suppose $M = M_1 \oplus M_2 \oplus \dots \oplus M_n$ where M_i 's are connected matroids. Let E_i be the ground set of M_i . Then we can easily check that $\lambda_M(X) = \sum_i \lambda_{M_i}(X \cap E_i)$ since $r_M(X) = \sum_i r_{M_i}(X \cap E_i)$. Moreover, if N_i be a minor of M_i , then $\oplus_i N_i$ is a minor of M . Hence we can apply the theorem for each connected M_i to conclude the same result for a disconnected matroid.) Then by Lemma 15.6, we can deduce that

$$2(k-1) \geq \lambda_{M \setminus e}(X_1) + \lambda_{M/e}(Y_1) \geq \lambda_M(X_1 \cap Y_1) + \lambda_M(X_1 \cup Y_1 \cup \{e\}) - 1.$$

It implies that $\lambda_M(X_1 \cap Y_1)$ or $\lambda_M(X_1 \cup Y_1 \cup \{e\})$ is $\leq k-1$, which is a contradiction. (Hence $\lambda_{M \setminus e}$ or $\lambda_{M/e}$ is $\geq k$ on $\{X : A \subseteq X \subseteq E - e - B\}$. So by the induction, the proof is finished.) $\square$

Unfortunately, the proof written in the above gives us an exponential-time algorithm. (We need to contract or delete e in $E - (A \cup B)$, recursively.) (Algorithm for what? Find a minor N ?????)

We can also prove $(\leq)$ using Homework 4.2.(ii): When we denote $\kappa_M(A, B) = \min_{A \subseteq X \subseteq E-B} \lambda_M(X)$,

$$\kappa_M(A, B) = \max(\kappa_{M/e}(A, B), \kappa_{M \setminus e}(A, B)).$$

It has a similar concept with Proposition 14.4. If we apply this observation for an enumeration of $E - (A \cup B)$ step-by-step, we can find a minor N on $A \cup B$ such that $\kappa_M(A, B) = \kappa_N(A, B) = \lambda_N(A)$.

Chapter. Graphic Matroids

For a directed graph D , we define the incidence matrix $A_D = (a_{ij})_{i \in V(D), j \in E(D)}$ where $a_{ij} = 1$ if j is a non-loop incoming edge of i , $= -1$ if j is a non-loop outgoing edge of i , and $= 0$ otherwise (this case includes a case that j is a loop).

For example, let us consider a directed 4-cycle $D = (V, E)$ where $V = [4]$ and $E = \{a, b, c, d\}$ such that $a = (1, 2)$, $b = (2, 3)$, $c = (3, 4)$ and $d = (4, 1)$. Then

$$A_D = \begin{matrix} & \begin{matrix} a & b & c & d \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \end{matrix} & \begin{pmatrix} -1 & 0 & 0 & 1 \\ 1 & -1 & 0 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & -1 \end{pmatrix} \end{matrix}$$

Theorem 16.2. *If G is a graph, then $M(G)$ is representable over every field. In particular, $M(G) = M(A_D)$ for any orientation D of G .*

Proof. If C is a circuit of $M(G)$, then trivially C is dependent in $M(A_D)$ since sum of the columns corresponding to C is a zero vector. (So C contains a circuit of $M(A_D)$.)

If $C = \{e_1, e_2, \dots, e_m\}$ is a circuit of $M(A_D)$, then $\sum c_i e_i = 0$ for some non-zero $(c_1, \dots, c_m)$ where each e_i in the sum is regarded as column vectors of A_D corresponding to the element e_i in C . WMA all c_i 's are non-zero since if not, we can find a strictly smaller dependent set in $M(A_D)$. WMA no e_i is a loop in $M(A_D)$ (so in G). Every vertex is incident with ≥ 2 edges in C or no edges in C since if there is $e_i \in C$ of which v -row entry is non-zero, then there is at least one more distinct $e_j \in C$ of which v -row entry is non-zero. It implies that $G[C]$, the subgraph consisting of edges from C without isolated vertices, has the minimum degree ≥ 2 . So C contains a cycle of G .

By Lemma 9.7, we can conclude that $M(G) = M(A_D)$. $\square$

Definition 16.2. A matrix A is *totally unimodular* if every square submatrix has determinant 0 or ± 1 (over $\mathbb{R}$).

Of course, every entry of a totally unimodular matrix has value 0 or ± 1 .

Example 16.1. A_D is totally unimodular.

The proof is following: Let B be a square $m \times m$ submatrix of A_D . Consider rows of B . If there is a row with at most one non-zero entry, then computing determinant of a proper submatrix of B is enough to get determinant of B . Note

that for each column of A_D , there are exactly two non-zero entries. It implies that the number of non-zero entries of B is equal or less than $2m$. If there is a row of B with more than two non-zero entries, then we can find another row which has at most one non-zero entry by the above observation. So WMA every row of B has exactly two non-zero entries, i.e., B represents a cycle of D (do not consider a direction). By suitable interchanging of rows and columns, and multiplying -1 for certain columns, we get a new matrix

$$B' = \begin{pmatrix} -1 & 0 & 0 & \cdots & 0 & 1 \\ 1 & -1 & 0 & \cdots & 0 & 0 \\ 0 & 1 & -1 & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & -1 & 0 \\ 0 & 0 & 0 & \cdots & 1 & -1 \end{pmatrix}$$

such that $\det B' = \pm \det B$. Actually, we can compute $\det B' = B'_{1,1} \det B'[1,1] + (-1)^{n-1} B'_{1,n} \det B'[1,n] = (-1)^n + (-1)^{n-1} 1^n = 0$, where $B'_{r,c}$ is a value of (r, c) -entry of B' , and $B'[r, c]$ is a submatrix of B' deleting r -row and c -column. Therefore, we can conclude that A_D is totally unimodular.

Remark. If M is represented by A for some field, and A is a totally unimodular matrix, then it represents the same matroid over every field.

Note that determinants of square submatrices of A is computed over $\mathbb{R}$ when we determine whether a matrix is totally unimodular or not. Since determinant of every square submatrix A' of A is 0 or ± 1 , and $\text{char} \mathbb{R} = 0$, $\det_{\mathbb{F}} A' = \det_{\mathbb{R}} A'$ for any field $\mathbb{F}$. Here the subscript means that we compute a determinant over the field of subscript. In other words, the singularity of the submatrix does not change even though we change a base field.

Note that B be a submatrix consisted by some columns of A . A is an $m \times n$ matrix, and B is an $m \times l$ matrix. Let us find a condition that columns of B is independent. WMA $m \geq l$. If columns of B is dependent, then any $l \times l$ submatrix B' of B has determinant 0. Let us consider the converse. Any $l \times l$ submatrix B' of B has determinant 0. Then

It implies that the dependency of a set of columns of A is independent with a choice of base fields. ?????

Definition 16.3. A matroid is *regular* if it is representable by a totally unimodular matrix.

By the above remark, every regular matroid is representable over any fields. The converse holds. More surprisingly, if a matroid is binary and ternary, then it is representable over all fields, i.e., regular. (We will show it later.)

Example 16.2. Every graphic matroid is regular. See Theorem 16.2 and Example 16.1.

Example 16.3. The dual of a regular matroid is regular. We will show it later (see Proposition 22.3). It implies that every cographic matroid is regular.

16.1 Homework 4.2.(ii)

Proposition 16.3. For a matroid M on E with the rank function r , let $\lambda_M(X) = r(X) + r(E - X) - r(E)$. Given two disjoint sets S and T , the connectivity between S and T , denoted by $\kappa_M(S, T)$ is defined as

$$\min_{S \subseteq X \subseteq E - T} \lambda_M(X).$$

(ii) Show that

$$\kappa_M(S, T) = \max(\kappa_{M/e}(S, T), \kappa_{M \setminus e}(S, T))$$

for $e \in E(M) - (S \cup T)$.

Proof. We can compute $\lambda_{M/e}(X)$ and $\lambda_{M \setminus e}(X)$ for $X \subseteq E - e$ as

$$\begin{aligned} \lambda_{M/e}(X) &= r_{M/e}(X) + r_{M/e}(E - e - X) - r(M/e) \\ &= (r_M(X + e) - r_M(\{e\})) + (r_M(E - X) - r_M(\{e\})) - (r(M) - r_M(\{e\})) \\ &= r_M(X + e) + r_M(E - X) - r(M) - r_M(\{e\}), \end{aligned} \tag{1}$$

and

$$\begin{aligned} \lambda_{M \setminus e}(X) &= r_{M \setminus e}(X) + r_{M \setminus e}(E - e - X) - r(M \setminus e) \\ &= r_M(X) + r_M(E - e - X) - r_M(E - e). \end{aligned} \tag{2}$$

By the submodular inequality, $r_M(X + e) \leq r_M(X) + r_M(\{e\})$ and $r_M(E - e - X) + r(M) \leq r_M(E - X) + r_M(E - e)$. The first inequality and (1) implies that $\lambda_{M/e}(X) \leq \lambda_M(X)$, and the second inequality and (2) implies that $\lambda_{M \setminus e}(X) \leq \lambda_M(X)$ for $X \subseteq E - e$. Hence

$$\kappa_M(S, T) \geq \max\{\kappa_{M/e}(S, T), \kappa_{M \setminus e}(S, T)\}$$

for $e \in E - (S \cup T)$.

Suppose the above inequality is strict. Let $S \subseteq Y$, $Z \subseteq E - e - T$ such that $\lambda_{M/e}(Y) = \kappa_{M/e}(S, T) < \kappa_M(S, T)$ and $\lambda_{M \setminus e}(Z) = \kappa_{M \setminus e}(S, T) < \kappa_M(S, T)$. Then

$$\begin{aligned} r_M(Y + e) + r_M(E - Y) - r(M) - r_M(\{e\}) &= \lambda_{M/e}(Y) \\ &< \lambda_M(Y) \\ &= r_M(Y) + r_M(E - Y) - r(M). \end{aligned}$$

It implies that $r_M(Y) = r_M(Y + e)$ and $r_M(\{e\}) = 1$ (e is not a loop). In addition,

$$\begin{aligned} r_M(Z) + r_M(E - e - Z) - r_M(E - e) &= \lambda_{M \setminus e}(Z) \\ &< \lambda_M(Z) \\ &= r_M(Z) + r_M(E - Z) - r(M). \end{aligned}$$

It implies that $r_M(E - e) = r(M)$ (e is not a coloop) and $r_M(E - Z) = r_M(E - e - Z) + 1$. Moreover, it implies that $\kappa_M(S, T) = \lambda_M(Y) = \lambda_{M/e}(Y) + 1 = \kappa_{M/e}(S, T) + 1$ and $\kappa_M(S, T) = \lambda_M(Z) = \lambda_{M/e}(Z) + 1 = \kappa_{M/e}(S, T) + 1$.

Then we can deduce that

$$\begin{aligned}
& 2 \cdot \kappa_M(S, T) \\
&= \lambda_M(Y) + \lambda_M(Z) \\
&= r_M(Y) + r_M(E - Y) - r(M) + r_M(Z) + r_M(E - Z) - r(M) \\
&= (r_M(Y + e) + r_M(Z)) + (r_M(E - Y) + r_M(E - e - Z)) - 2 \cdot r(M) + 1 \\
&\geq r_M(Y \cap Z) + r_M((Y \cup Z) + e) + r_M(E - (Y \cup Z) - e) + r_M(E - (Y \cap Z)) - 2 \cdot r(M) + 1 \\
&= (r_M(Y \cap Z) + r_M(E - (Y \cap Z)) - r(M)) \\
&\quad + (r_M((Y \cup Z) + e) + r_M(E - (Y \cup Z) - e) - r(M)) + 1 \\
&\geq 2 \cdot \kappa_M(S, T) + 1.
\end{aligned}$$

It is a contradiction. Hence we can conclude that

$$\kappa_M(S, T) = \max\{\kappa_{M/e}(S, T), \kappa_{M \setminus e}(S, T)\}.$$

□

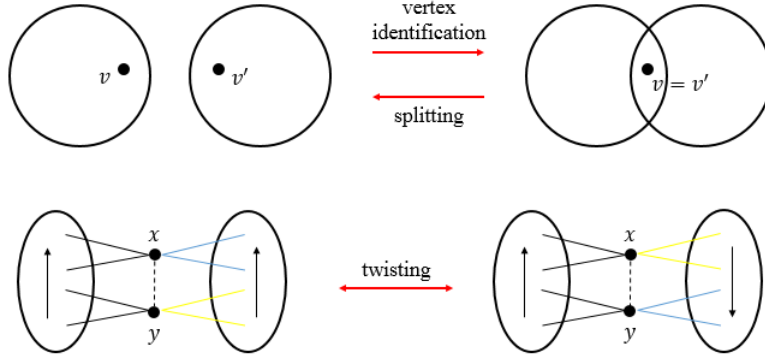
17 Week12-1, 2019.11.18. Mon

Sub-chapter. Whitney's 2-isomorphic theorem

When do we have $M(G) = M(H)$ for graphs G, H ? What are sufficient conditions?

- G and H are isomorphic.
- H is a vertex identification of G , i.e., G is a splitting of G .
- H is a twisting of G .

Here a vertex identification of G is that identification of two vertices in distinct components of G . Let $\{x, y\}$ be a minimal vertex-cut of size 2 in G . Let G_1 and G_2 be new components appearing by deleting x, y in G . Let H be a graph generating from G by reversing incidence of x, y and G_2 , i.e., $N_G(x) \cap V(G_2) = N_H(y) \cap V(G_2)$ and $N_G(y) \cap V(G_2) = N_H(x) \cap V(G_2)$. Then we call H a twisting of G .



Note that identification/splitting and twisting do not affect on lists of cycles, if we only focus on edge-sets of cycles.

We say G and H are *2-isomorphic* if an isomorphic copy of G can be obtained from H by a finite sequence of identification, splitting and twisting.

Theorem 17.1 (Whitney, 1933). *Let G and H be graphs (with no isolated vertices ?). Then $M(G)$ is isomorphic to $M(H)$ if and only if G is 2-isomorphic to H .*

Lemma 17.2. *Let G and H be graphs with no isolated vertices and no loops. If G is 3-connected and $M(G)$ is isomorphic to $M(H)$, then G is isomorphic to H .*

This lemma is a special case of Whitney's theorem. In addition, it implies that for any two distinct 3-connected graphs, their cycle matroids are distinct.

Proof. Claim: Let D be a cocircuit of $M(G)$. $M(G) \setminus D$ is connected if and only if for some vertex v of G , $D = \delta(v)$, the set of edges incident with v .

($\Rightarrow$) Note that $M(G) \setminus D = M(G \setminus D)$. D is an edge-cut of G , and it divides connected G into two connected graph. Since $M(G \setminus D)$ is connected, at least one component of $G \setminus D$ has no edges, i.e., D divides G into a vertex v and $G \setminus v$. So $D = \delta(v)$. (Here we do not use the 3-connectedness of G , but only use the connectedness of G .)

($\Leftarrow$) Let $D = \delta(v)$. Since G is 3-connected, $G \setminus v$ is 2-connected. So $M(G) \setminus D = M(G \setminus D) = M(G \setminus v)$ is connected by Corollary 14.1.1.

Now come back to the original problem. We can observe that $M(G)$ is connected, so $M(H)$ is connected since two matroids are isomorphic. It implies that H is (2-)connected by Corollary 14.1.1. Then

$$|V(G)| - 1 = r(M(G)) = r(M(H)) = |V(H)| - 1.$$

So $|V(G)| = |V(H)|$. In addition, $|E(G)| = |E(M(G))| = |E(M(H))| = |E(H)|$.

By the above claim, $M(G)$ has exactly $|V(G)|$ cocircuits D such that $M(G) \setminus D$ is connected, and it equals to $\delta_G(u)$ for some $u \in V(G)$. For a such D , $M(H) \setminus D$ is connected, so $D = \delta_H(v)$ for some vertex v of H . It holds since we do not

need 3-connectedness when we prove the forward direction of the above claim. Obviously, for distinct such cocircuits D and D' , deduced vertices v and v' of H are distinct. Hence the above observation gives a bijection between $V(G)$ and $V(H)$, and we can identify the vertex-sets of G and H . Moreover, we know that $\delta_G(v) = \delta_H(v)$. Note that an edge contains the information about its end-vertices. From this, we can conclude that G and H are isomorphic.

(In other words, the information $(\delta_G(v))_{v \in V(G)}$ is enough to reconstruct G uniquely. However, note that the information $(\deg_G v)_{v \in V(G)}$ is not enough to reconstruct G uniquely.) $\square$

Definition 17.1. Let M be a matroid, and C be a circuit of M . Let B be a *bridge* with respect to C (simply, a *C-bridge*) if it is a component of M/C .

For a graph G and a cycle C in G , an edge-set B is a *bridge* with respect to C (or *C-bridge*) if it is a C -bridge of $M(G)$. Equivalently, B is a block (a maximal 2-connected induced subgraph or an edge) of G/C (remind the definition of a component of a matroid, and a circuit of $M(G)$ is a cycle of G .)

Lemma 17.3. *If G is a simple 2-connected graph but not 3-connected, then G is a cycle or has a cycle C with a bridge B sharing exactly two vertices of C (when we consider them as edge-sets in G).*

Proof. We can prove it by an induction on $|E(G)|$. By the given condition, there exist distinct $x, y \in V(G)$ such that $G \setminus \{x, y\}$ is not connected. Let $H_1, H_2, \dots, H_r$ be the components of $G \setminus \{x, y\}$. Let H'_i be a subgraph of G obtained from H_i by recovering x, y and adding a new edge xy if it does not exist. In other words, $H'_i = G[V(H_i) \cup \{x, y\}]$ if there is an edge between x, y in G , or $= G[V(H_i) \cup \{x, y\}] + xy$ otherwise.

Note that each H'_i is 2-connected (why??? We can consider H'_i as $G/E(G \setminus (V(H_i) \cup \{x, y\})) / e$ where e is an edge between x and the contracted vertex ...), and $|E(H'_i)| < |E(G)|$.

If there is some H'_i which is 3-connected. Then $H'_i - xy$ is 2-connected. It has a cycle C containing x, y . Then H

If all H'_i are not 3-connected, then H'_i is a cycle or has a cycle C with a bridge B sharing exactly two vertices of C by the induction hypothesis. For the first case (all H'_i 's are cycles), $\square$

To assert Whitney's theorem, we need to divide cases. 3-connected case is derived by the earlier lemma. Not 2-connected case is derived by the identification/splitting and the induction. If a graph has a loop, then we can simply detach it. If a graph has a parallel edges, ...?

Proof of Theorem 17.1. ($\Leftarrow$) We already check that if G and H are 2-isomorphic, then $M(G)$ and $M(H)$ are isomorphic since $\mathcal{C}(M(G)) = \mathcal{C}(M(H))$.

($\Rightarrow$) Let us prove the theorem by an induction on $|E(G)|$.

Let $G_1, \dots, G_m$ be the blocks of G , and $H_1, \dots, H_n$ be the blocks of H . A component of $M(G)$ corresponds to a block of G since circuits of $M(G)$ corresponds to cycles of G . So $M(G) = \oplus M(G_i)$ and $M(H) = \oplus M(H_j)$. It

implies that $m = l$ and $G_i = H_i$ for each i by a rearrangement of subscripts. G and H are obtained by sequences of the vertex identification of G_i 's and H_i 's respectively. It implies that if G_i and H_i are 2-isomorphic for each i , then G and H are 2-isomorphic. In a case that a block G_i or H_i is an edge, then the other is also an edge. So WMA G and H are 2-connected. WMA G and H is not 3-connected by the earlier lemma.

..... WMA G is simple.

Let (A, B) be a separation of $M(G)$ (so $M(H)$). Suppose that in both G and H , exactly two vertices v, w meet A and B . Let G_1 be $G[A]$, a subgraph of G obtained from $(V(G), A)$ by deleting all isolated vertices. Let G_2 be $G[B]$. Let $H_1 = H[A]$ and $H_2 = H[B]$. We can observe that $M(G_1) = M(G) \setminus B$, $M(G_2) = M(G) \setminus A$, $M(H_1) = M(H) \setminus B$ and $M(H_2) = M(H) \setminus A$. So $M(G_i)$ is isomorphic to $M(H_i)$ for $i = 1, 2$. By the induction hypothesis, G_i and H_i are 2-isomorphic. By considering a twisting, we deduce that G and H are 2-isomorphic.

G is simple and 2-connected, but not 3-connected. By the previous lemma, G is a cycle, or has a cycle C with a bridge B sharing exactly two vertices of C . For a case that G is a cycle, then $M(G) = M(H)$ is a single circuit. So H is a cycle of same size with G , i.e., G and H are isomorphic (we do not care labelings of the edge-sets of G and H). Now assume G has a cycle C with a bridge B sharing exactly two vertices v, w of C (in G). Let B' be another bridge with respect to C . We claim that B' can share vertices of B only v, w . Since B and B' are blocks of G/C , it is impossible that they share more than one vertex. In addition, we can observe that every bridge with respect to C shares at least one vertex of C since G is 2-connected. (Skip details.) Suppose that in G , there is a vertex u distinct from v, w , which is shared by B and B' . Then B and B' shares u and the vertex corresponding to C in G/C . It is a contradiction.

Let $A = E(G) - B$. Then by the above claim and the our choice of B and C using the lemma, A and B shares exactly two vertices v and w in G . (How can we assure that it also happens in H ??? If then, we can conclude the proof by the earlier observation.)

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□

(The remaining proof of Theorem 17.1 was done in November 20th.)

18 Week12-2, 2019.11.20. Wed

Is there an algorithm deciding whether a matroid is graphic? In 1960, Tutte gave us a partial answer. He found an algorithm deciding whether a binary matroid given by a binary matrix is graphic.

Let $M = M(A)$ where A is a matrix over $GF(2)$. WMA M is connected. (How to find components of M , or to check whether M is connected? Greedily, check all $\lambda(X)$ for $X \subseteq E(M)$)

Observation 1: If D is a cocircuit of $M(G)$, G is connected, and $M \setminus D$ is connected, then $D = \delta_G(v)$ (the set of edges incident with v in G) for some vertex v . From this observation, we deduce a below fact. If M has three distinct cocircuits D_1, D_2, D_3 containing e such that $M \setminus D_i$ is connected for each i , then M is not graphic.

Definition 18.1. Let Y be a cocircuit of M , i.e., Y be a circuit of M^* . We say B is a Y -bridge (or a *bridge* with respect to Y) in M if B is a bridge with respect to Y in M^* . In other words, B is a component of $M \setminus Y$ (remind that the connectivity does not change taking dual).

Observation 2: Assume Y is a cocircuit of $M = M(G)$ such that $M \setminus Y$ is disconnected. Let B and B' be distinct Y -bridges.

19 Week13-1, 2019.11.25. Mon

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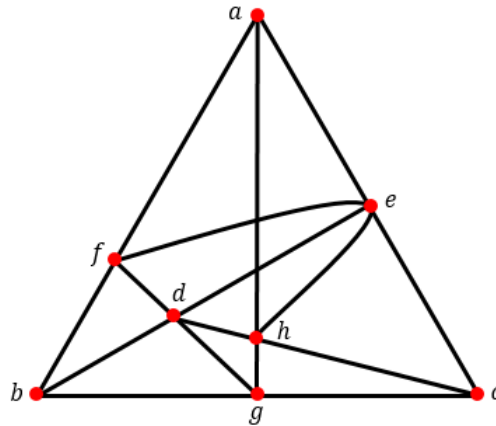
Chapter. Representable matroids

Proposition 19.1 (Proposition 1.5.6 of Matroid Theory by J. Oxley). ...

Proof.

□

Example 19.1 ($M_{\sqrt{-3}}$ matroid). $M_{\sqrt{-3}}$ is a matroid with a below geometric representation.



Its rank is 3 since its configuration is drawn in the plane, and has non-colinear three points. $\{a, b, c\}$ is a base of $M_{\sqrt{-3}}$. Let us find its matrix representation.

Start from a below matrix.

$$\begin{array}{cccccccc} & a & b & c & d & e & f & g & h \\ \left(\begin{array}{ccc|cccc} 1 & 0 & 0 & & & & & \\ 0 & 1 & 0 & & & * & & \\ 0 & 0 & 1 & & & & & \end{array} \right) \end{array}$$

From now on, we denote a non-zero entry as *. Possibly, *'s are distinct.

1. Since $\{a, b, d\}$, $\{b, c, d\}$, $\{c, a, d\}$ are independent sets, $d = \begin{pmatrix} * & * & * \end{pmatrix}^t$. By row operations and multiplying scalars to columns, WMA $d = \begin{pmatrix} 1 & 1 & 1 \end{pmatrix}^t$. (More precisely, multiplying scalars to each row to make entries of d are same. Then multiplying scalars to columns a, b, c, d to make all non-zero entries are 1.)
2. Since $\{a, b, e\}$, $\{b, c, e\}$ are independent, and $\{c, a, e\}$ is dependent, $e = \begin{pmatrix} * & 0 & * \end{pmatrix}^t$. By row operations and multiplying a scalar to a column, WMA $e = \begin{pmatrix} 1 & 0 & x \end{pmatrix}^t$ where x is non-zero.
3. Similarly, WMA $f = \begin{pmatrix} 1 & y & 0 \end{pmatrix}^t$, $g = \begin{pmatrix} 0 & 1 & z \end{pmatrix}^t$, and $h = \begin{pmatrix} 1 & s & t \end{pmatrix}^t$ where y, z, s, t are non-zero unknowns.

$$\begin{array}{cccccccc} & a & b & c & d & e & f & g & h \\ \left(\begin{array}{ccc|cccc} 1 & 0 & 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & y & 1 & s \\ 0 & 0 & 1 & 1 & x & 0 & z & t \end{array} \right) \end{array}$$

4. Since $\{b, d, e\}$ is a circuit,

$$0 = \det \begin{pmatrix} & b & d & e \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & x & x \end{pmatrix} = x - 1.$$

So $x = 1$.

5. Since $\{d, g, f\}$ is a circuit,

$$0 = \det \begin{pmatrix} & d & g & f \\ 1 & 0 & 1 & 1 \\ 1 & 1 & y & y \\ 1 & z & 0 & 0 \end{pmatrix} = -(y - 1)z - 1.$$

So $y \neq 1$ and $z = (1 - y)^{-1}$.

6. Since $\{c, d, h\}$ is a circuit,

$$0 = \det \begin{pmatrix} c & d & h \\ 0 & 1 & 1 \\ 0 & 1 & s \\ 1 & 1 & t \end{pmatrix} = s - 1.$$

So $s = 1$.

7. Since $\{a, g, h\}$ is a circuit,

$$0 = \det \begin{pmatrix} a & g & h \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 0 & z & t \end{pmatrix} = t - z.$$

So $t = z$.

8. Since $\{e, f, h\}$ is a circuit,

$$0 = \det \begin{pmatrix} e & f & h \\ 1 & 1 & 1 \\ 0 & y & 1 \\ 1 & 0 & z \end{pmatrix} = yz + 1 - y.$$

Since $yz - z + 1 = 0$ by 5, we deduce that $z = y$ and $y^2 - y + 1 = 0$.

Therefore, if the matroid $M_{\sqrt{-3}}$ has a matroid representation over a field $\mathbb{F}$, then its form is

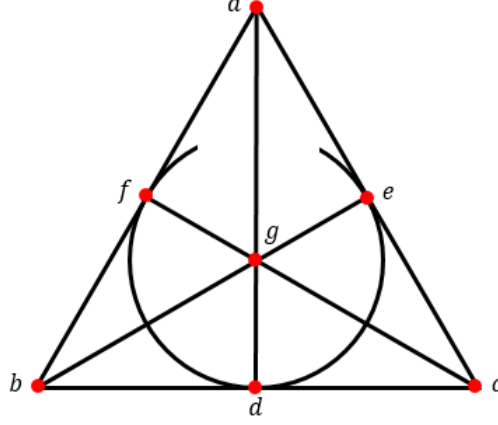
$$\begin{pmatrix} a & b & c & d & e & f & g & h \\ 1 & 0 & 0 & 1 & 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 & 0 & y & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 & 0 & y & y \end{pmatrix}$$

by suitable row operations and multiplying scalars for columns, and $y^2 - y + 1 = 0$ over $\mathbb{F}$. (Obviously, $y \neq 0, 1$.) Hence

$M_{\sqrt{-3}}$ is $\mathbb{F}$ -representable if and only if $\mathbb{F}$ has a solution of $y^2 - y + 1 = 0$.

(Here $y = \frac{1 \pm \sqrt{-3}}{2}$, and this is a reason why we name this matroid $M_{\sqrt{-3}}$.)

Example 19.2 (Fano and non-Fano matroids). *Fano matroid* F_7 is a matroid with a geometric representation below.



$\{a, b, c\}$ is a base of F_7 . Now we start to find a matrix representation of F_7 .

$$\begin{pmatrix} a & b & c & g & f & e & d \\ \begin{array}{ccc|ccc} 1 & 0 & 0 & & & & \\ 0 & 1 & 0 & & & & \\ 0 & 0 & 1 & & & & \end{array} \end{pmatrix}$$

Similar with the previous example, we can consider the dependency and independency of $\{a, b, *\}$, $\{b, c, *\}$, $\{c, a, *\}$ for $* = g, f, e, d$. Then we deduce that

$$\begin{pmatrix} a & b & c & g & f & e & d \\ \begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & x & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & y & z \end{array} \end{pmatrix}$$

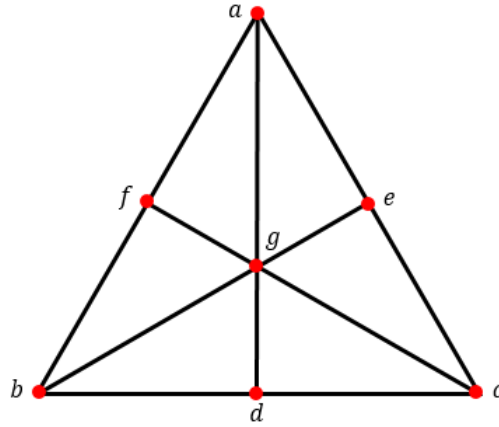
where x, y, z are non-zero unknowns. Since $\{a, d, g\}$ is a circuit, $z = 1$. Since $\{b, e, g\}$ is a circuit, $y = 1$. Since $\{c, f, g\}$ is a circuit, $x = 1$. The only remaining circuit to check is $\{d, e, f\}$.

$$0 = \det \begin{pmatrix} d & e & f \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} = 2.$$

Hence

Fano matroid F_7 is representable over a field $\mathbb{F}$ if and only if $\text{char}\mathbb{F} = 2$.

Non-Fano matroid F_7^- is a matroid with a geomtric configuration below.



Only difference with Fano matroid is that $\{d, e, f\}$ is a circuit in F_7 , but it is an independent set in F_7^- . So

$$0 \neq \det \begin{pmatrix} d & e & f \\ 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix} = 2.$$

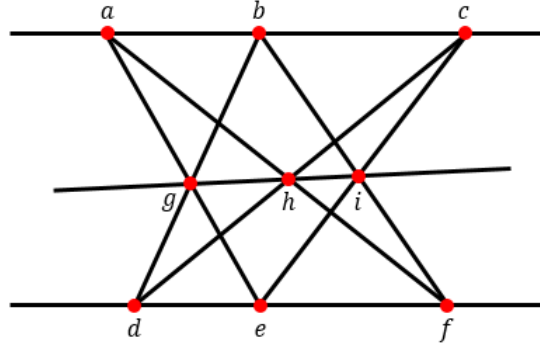
Hence

Non-Fano matroid F_7^- is representable over a field $\mathbb{F}$ if and only if $\text{char}\mathbb{F} \neq 2$.

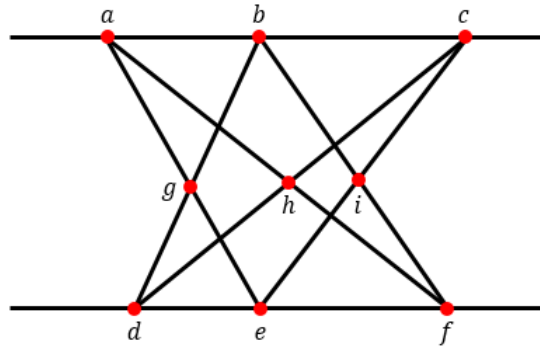
20 Week14-1, 2019.12.02. Mon

One basic question in matroid theory: Is there a matroid not representable over any fields? The answer is yes. Actually, almost all matroids are not representable over any fields.

Example 20.1 (Pappus and non-Pappus matroids). One of the most famous theorem in Euclidean geometry is Pappus theorem: When we choose colinear triples (a, b, c) and (d, e, f) of points in $\mathbb{R}^2$, and get a new triple (g, h, i) as like a below configuration, the new triple (g, h, i) is colinear.



We call this Pappus configuration. Actually, it deduce a matroid of rank 3, called *Pappus matroid*. Similarly, we can consider a below configuration named non-Pappus configuration. Its corresponding matroid is called *non-Pappus matroid*.



Theorem 20.1. *The non-Pappus matroid is not representable over any field.*

Proof. The non-Pappus matroid is of rank 3. From the above figure, $\{a, c, d\}$ is a base of the non-Pappus matroid. Let us construct its matroid representation from a below matrix.

$$\begin{pmatrix} a & c & d & f & h & b & e & g & i \\ \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} & & & & & & & & \end{pmatrix}$$

1. Since $\{a, c, f\}$, $\{a, d, f\}$, $\{c, d, f\}$ are independent, WMA $f = (1 \ 1 \ 1)^t$.
2. Since c, d spans h , i.e., $\{c, d, h\}$ is a circuit, $h = (0 \ * \ *)^t$. Since a, f spans h , WMA $h = (0 \ 1 \ 1)^t$.
3. Since a, c spans b , WMA $b = (1 \ x \ 0)^t$ where x is non-zero.

4. Since d, f spans e , WMA $e = (1 \ 1 \ y)^t$ where y is non-zero.
5. Since b, d spans g , WMA $g = (1 \ x \ z)^t$ where z is non-zero. Since a, e spans g , $z = xy$.
6. Since c, e spans i , WMA $i = (1 \ w \ y)^t$ where w is non-zero.

$$\begin{array}{cccc|cccc} a & c & d & f & h & b & e & g & i \\ \hline 1 & 0 & 0 & 1 & 0 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & x & 1 & x & w \\ 0 & 0 & 1 & 1 & 1 & 0 & y & xy & y \end{array}$$

7. $\{b, f, i\}$ is a circuit, so

$$0 = \det \begin{array}{ccc} b & f & i \\ \hline 1 & 1 & 1 \\ 1 & x & w \\ 1 & 0 & y \end{array} = w - x + (x - w)y.$$

We deduce $w = x + y - xy$.

8. Finally, since $\{g, h, i\}$ is independent,

$$0 \neq \det \begin{array}{ccc} g & h & i \\ \hline 1 & 0 & 1 \\ x & 1 & x + y - xy \\ xy & 1 & y \end{array} = y - (x + y - xy) + x - xy = 0.$$

It is impossible.

Hence we can conclude that

Non-Pappus matroid is not representable over any fields.

□

In Pappus matroid, $\{g, h, i\}$ is a circuit. We think that by suitable choice of x and y in some field $\mathbb{F}$, Pappus matroid is $\mathbb{F}$ -representable. Be careful that $x = y = 1$ is a bad choice since $e = g = i = (1 \ 1 \ 1)^t$ in this case, so any two of them consist a circuit of size 2. What are fields that Pappus matroid is representable?

Sub-chapter. Binary matroids

Theorem 20.2. *Let M be a matroid. The following are equivalent:*

- (1) M is binary.

- (2) $|C \cap D|$ is even for all circuits C and cocircuits D .
- (3) If C_1, C_2 are circuits, then $C_1 \Delta C_2$ is a disjoint union of circuits.
- (4) If $C_1, C_2, \dots, C_k$ are circuits, then $C_1 \Delta C_2 \Delta \dots \Delta C_k$ is a disjoint union of circuits.
- (5) If B is a base and C is a circuit, then $C = \Delta_{e \in C-B} C(e, B)$ where $C(e, B)$ is the fundamental circuit of e with respect to B .
- (6) There is a base B such that for ever circuit C , $C = \Delta_{e \in C-B} C(e, B)$.

In a sense of graphs, every conditions in the above is quite obvious. Consider (2). We know that an intersection of a cycle and a edge-cut is of size even. Consider (3) and (4). A symmetric difference of cycles is Eulerian, i.e., the degree of each vertex is even, so we can decompose it into edge-disjoint cycles. How about (5) and (6) for a graph???

Proof. We will show $(1) \Rightarrow (3) \Rightarrow (2) \Rightarrow (4) \Rightarrow (5) \Rightarrow (6) \Rightarrow (1)$.

[(1) $\Rightarrow$ (3)] Remind that the base field is binary. Identify each element of M as a column vector of its matrix representation. Then $0 = \sum_{e \in C_1} e + \sum_{f \in C_1} = \sum_{e \in C_1 \Delta C_2} e$. So $C_1 \Delta C_2$ contains a circuit C'_1 , and $\sum_{e \in (C_1 \Delta C_2) - C'_1} e = 0$. Still we can find a circuit C'_2 contained in $(C_1 \Delta C_2) - C'_1$ unless it is empty. Repeating this process, we obtain a disjoint circuits of which union is $C_1 \Delta C_2$.

[(3) $\Rightarrow$ (2)] Suppose not, i.e., we can choose a circuit C and a cocircuit D so that $|C \cap D|$ is odd and is minimized. Since $|C \cap D| \neq 1$ for any matroid, $|C \cap D| \geq 3$. Let x, y, z be distinct elements of $C \cap D$. Note that $E(M) - D$ is a hyperplane not containing x, y . The rank of $E(M) - D$ is $r(M) - 1$. $E - D + x$, $E - D + y$ and $E - D + \{x, y\}$ have the full rank. It implies that $E - D + \{x, y\}$ contains a circuit C' containing both x, y . Then $z \in C \Delta C' \subseteq (C \cup (E(M) - D)) - \{x, y\}$. By (3), $C \Delta C'$ is a disjoint union of circuits $C_1, \dots, C_k$ with $k \geq 1$. Since $C_i \cap D \subseteq (C \cap D) - \{x, y\}$, $|C_i \cap D|$ is even for all i by our choice of C and D . However, $\sqcup(C_i \cap D) = (C \cap D) - \{x, y\}$ has odd number of elements. It is a contradiction.

[(2) $\Rightarrow$ (4)] ETS $C_1 \Delta \dots \Delta C_k$ contains a circuit. The reason why it is enough is an induction on $|C_1 \Delta \dots \Delta C_k|$: If we can find a circuit C_{k+1} contained the symmetric difference of circuits, then $C_1 \Delta \dots \Delta C_k \Delta C_{k+1}$ is a disjoint union of circuits by the induction hypothesis. So we conclude that $C_1 \Delta \dots \Delta C_k$ is a disjoint union of circuits. The basic case is when $C_1 \Delta \dots \Delta C_k$ is empty. We can regard it as the disjoint union of zero circuits.

Suppose not, i.e., $C_1 \Delta \dots \Delta C_k$ is independent. WMA $C_1 \Delta \dots \Delta C_k \neq \emptyset$ since the empty set is the disjoint union of zero circuits. Let $e \in C_1 \Delta \dots \Delta C_k$. There is a base B containing $(C_1 \Delta \dots \Delta C_k) - e$. $\mathcal{cl}(B - e)$ is a hyperplane, and its complement D is a cocircuit such that $D \cap (C_1 \Delta \dots \Delta C_k) = \{e\}$. However, $D \cap (C_1 \Delta \dots \Delta C_k) = (D \cap C_1) \Delta \dots \Delta (D \cap C_k)$ is of size even since every $|D \cap C_i|$ is even for each i by (2). It is a contradiction.

[(4) $\Rightarrow$ (5)] It is obvious that $(\Delta_{e \in C-B} C(e, B)) \Delta C \subseteq B$. Combining this fact with (4), we deduce that $(\Delta_{e \in C-B} C(e, B)) \Delta C = \emptyset$. Hence $\Delta_{e \in C-B} C(e, B) = C$.

[(5) $\Rightarrow$ (6)] It is trivial since every matroid has a base.

[(6) $\Rightarrow$ (1)] We construct a binary matroid $M' = M(A)$ with

$$A = \begin{array}{c|c} B & E(M) - B \\ \hline I_r & * \end{array}$$

For $f \in B$ and $e \in E(M) - B$,

$$A_{f,e} = \begin{cases} 1 & \text{if } f \in C(e, B), \\ 0 & \text{otherwise.} \end{cases}$$

Claim: $M = M'$.

Let C be a circuit of M . By (6), $C = \Delta_{e \in C-B} C(e, B)$. Note that $C(e, B)$ is a circuit of both M and M' by our construction of A , i.e., $\sum_{f \in C(e, B)} f = 0$ when we regard f as a column vector of A . It implies $\sum_{f \in C} f = \sum_{e \in C-B} \sum_{f \in C(e, B)} f = 0$, i.e., C is dependent in M' , since the base field is binary. So C contains a circuit of M' . $\dots (*)$

Now WTS a circuit C of M' contains a circuit of M . Let $e \in C - B$. $C - e$ is independent in M' . There is a base B' of M' extending $C - e$. B' is independent in M' , so it is independent in M by $(*)$. $C_M(e, B') = \Delta_{e' \in C_M(e, B') - B} C_M(e', B)$ by (6). Here the subscript means that we get the fundamental circuit on the given matroid. Remind that $C_M(e', B)$ is also a circuit of M' by our construction of A . Hence $C_M(e, B')$ is a symmetric difference of some circuits of M' . It implies that $C_M(e, B')$ is the disjoint union of circuits of M' since M' is binary (we proved that (1) $\Rightarrow$ (4)). Since C is the unique circuit of M' contained in $B' + e$, and $\emptyset \neq C_M(e, B') \subseteq B' + e$ is a disjoint union of circuits of M' , we deduce that $C = C_M(e, B')$. So C is a circuit of M .

By Lemma 9.7, we can conclude that $M = M'$. $\square$

21 Week14-2, 2019.12.04. Wed

Proposition 21.1. *Let $\mathbb{F}$ be a field. $U_{2,k}$ is representable over $\mathbb{F}$ if and only if $|\mathbb{F}| \geq k - 1$.*

Proof. If $U_{2,k} = M(A)$ over a field $\mathbb{F}$, then we consider A as a standard form,

$$\begin{array}{c|ccccc} e_1 & e_2 & e_3 & e_4 & \dots & e_k \\ \hline 1 & 0 & 1 & 1 & \dots & 1 \\ 0 & 1 & * & * & \dots & * \end{array}$$

Here $k - 2$ $*$'s are non-zero and all distinct. It implies that $|\mathbb{F}^\times| \geq k - 2$, i.e., $|\mathbb{F}| \geq k - 1$.

Conversely, if $|\mathbb{F}| \geq k - 1$, then we can take the above standard form A . Hence $U_{2,k} = M(A)$. $\square$

The above proposition implies that $U_{2,k}$ is binary iff $k \leq 3$.

Theorem 21.2. *A matroid M is binary if and only if M has no minor isomorphic to $U_{2,4}$.*

Proof. ($\Rightarrow$) Remind Proposition 10.5: If a matroid is representable over a field $\mathbb{F}$, then its minors are also representable over $\mathbb{F}$. Hence by the previous observation, M has no minor isomorphic to $U_{2,4}$.

($\Leftarrow$) Suppose M is not binary. WMA M is minor-minimal, which is not binary, i.e., M is not binary, but all its proper minors are binary. M has no loops and coloops. (Let e be a loop of M . $M \setminus e$ is binary, and equals to $M(A)$ where A is a binary matrix. Then $M = M(A')$ where A' is a matrix from A by adding a zero column. For a coloop e of M , consider a same process for M^* .) M has no circuits and cocircuits of size 2. (Let $\{e, f\}$ be a circuit of size 2 of M . $M \setminus e$ is binary, and equals to $M(A)$. Then $M = M(A')$ where A' is a matrix from A by adding a column same with the column corresponding to f . For a cocircuit $\{e, f\}$ of M , consider a same process for M^* .)

Let $x, y \in E(M)$ be distinct. Since $\{x, y\}$ is coindependent,

$$r(M \setminus \{x, y\}) = |E(M) - \{x, y\}| - r(M^*) + r_{M^*}(\{x, y\}) = |E(M)| - r(M^*) = r(M).$$

By our choice of M , $M \setminus \{x, y\}$ is binary. Let $(I_r \ D)$ be a binary representation of $M \setminus \{x, y\}$, where $r = r(M)$. We can also check that $r(M \setminus x) = r(M \setminus y) = r(M)$. Two matroids are also binary, and we can represent them as $(I_r \ D \ v_y)$ and $(I_r \ D \ v_x)$, where ordering of element corresponding to first $|E(M) - \{x, y\}|$ column vectors are same with the binary representation of $M \setminus \{x, y\}$. v_x and v_y are column vectors corresponding to x and y respectively.

Can we guarantee that binary representations of $M \setminus x$ and $M \setminus y$ are represented to $(I_r \ D \ v_y)$ and $(I_r \ D \ v_x)$ by sequences of suitable row operations and column changing? Yes, since we do it over $GF(2)$. More precisely, we pick a base B of M corresponding to I_r . It is also the base of $M \setminus x$ and $M \setminus y$. Let $(I_r \ D' \ v_y)$ and $(I_r \ D'' \ v_x)$ be binary representations of $M \setminus x$ and $M \setminus y$ respectively. We can make these forms by suitable row operations, multiplying scalars for columns and interchanging columns. Here I_r corresponds to B . Then by considering the fundamental circuits of $e \notin B$ with respect to B , remaining columns (not corresponding to an element of B) are uniquely determined since we do it over binary. So it must be $D = D' = D''$. (Unfortunately, this method does not hold for a big field. If size of a field is large, then D', D'' is not determined uniquely, i.e., $D \neq D', D \neq D''$ possibly.)

Let M' be a binary matroid represented by $(I_r \ D \ v_x \ v_y)$. (We can give the natural bijection between column vectors of the above matrix representation and $E(M)$.) Then $r(M') = r$, $M' \setminus x = M \setminus x$ and $M' \setminus y = M \setminus y$. Since $M' \neq M$,

there is a minimal set Z that is independent in one of M' or M , and dependent in the other (actually, is a circuit by the minimality of Z). Z is a circuit in one of M or M' , say M_c , and an independent set in the other, say M_i . By the earlier observation that deleting x or y in M and M' are isomorphic, we deduce that $\{x, y\} \subseteq Z$.

Claim: If I is independent in M_i and $I \supseteq Z$, then $I = \{x, y\}$.

The proof of claim is following: Suppose $J = I - \{x, y\} \neq \emptyset$. J is independent in M_i and not containing $\{x, y\}$, so J is independent in M_c . Now both M_c/J and M_i/J are binary. Note that $\{x, y\}$ is still dependent in M_c/J , and independent in M_i/J by Proposition 9.5. In addition, by Proposition 9.4 and $|J| = r_{M_i}(J) = r_{M_c}(J)$,

$$r(M_c/J) = r(M_c) - |J| = r(M_i) - |J| = r(M_i/J).$$

There is a base B of $M_c/J \setminus \{x, y\}$. Remind that $M_c/J \setminus \{x, y\} = M_i/J \setminus \{x, y\}$, so B is a base of $M/J \setminus \{x, y\}$. Suppose $r(M/J \setminus \{x, y\}) < r(M/J)$, then

$$r_{M^* \setminus J}(\{x, y\}) = |\{x, y\}| - r(M/J) + r(M/J \setminus \{x, y\}) < 2.$$

It contradicts that $\{x, y\}$ is coindependent in M . Since $r(M_c/J) = r(M_i/J)$, we deduce that B is also the base of M_c/J and M_i/J .

For $e \notin B \cup J$, the fundamental circuit of e with respect to B in M_i/J is also the fundamental circuit of e with respect to B in M_c/J since such fundamental circuit does not contain at least one of x, y (remind $M \setminus x = M' \setminus x$ and $M \setminus y = M' \setminus y$). So M_i/J and M_c/J have the same binary representation. However, remind that $\{x, y\}$ is independent in M_i/J , and dependent in M_c/J . It is a contradiction.

Now come back to the original proof. By the claim $Z = \{x, y\}$. M_i has a base extending Z , and it is actually Z by the claim.

$$2 = r(M_i) = r(M_c).$$

So $r(M) = 2$. Remind that $r(M) = r(M \setminus \{x, y\})$. So $|E(M)| \geq 4$. Pick any set S of four elements of M . Let N be a minor obtained from M by deleting $E(M) - S$. Then any set of size 2 in N is independent since we showed that M has no circuit of size ≤ 2 , and any set of size 3 is dependent since $r(M) = 2$. It implies that N is equivalent to $U_{2,4}$. Therefore, we get $U_{2,4}$ as a minor of M . $\square$

Theorem 21.3 (Seymour). *If an algorithm decides whether a matroid given by independent orcale, is binary, then it has to query $\geq 2^{\frac{n}{2}-1}$ sets to the independent orcale.*

Proof. Note *Circuit-hyperplane relaxation*: If M is a matroid, and C is a circuit as well as a hyperplane, then there is a matroid M' such that $\mathcal{B}(M') = \mathcal{B}(M) \cup \{C\}$. (We proved it in the midterm exam.)

Let us consider a matroid M on a ground set of size $2n$ with a binary representation

$$A = (I_n \quad J_n - I_n) = \left(\begin{array}{cccc|cccc} x_1 & x_2 & \dots & x_n & y_1 & y_2 & \dots & y_n \\ 1 & 0 & \dots & 0 & 0 & 1 & \dots & 1 \\ 0 & 1 & \dots & 0 & 1 & 0 & \dots & 1 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 & 1 & 1 & \dots & 0 \end{array} \right)$$

Here J_n is an $n \times n$ matrix whose all entries are 1. Then a circuit of M has two types:

1. $\{x_i, y_i, x_j, y_j\}$ for $i \neq j$, and
2. $\{d_1, d_2, \dots, d_n\}$ for some $d_i \in \{x_i, y_i\}$, where $|\{d_1, \dots, d_n\} \cap \{y_1, \dots, y_n\}|$ is odd. (?)

The second type is also a hyperplane. (?) So M has 2^{n-1} circuits that are hyperplanes. (Remind that $\binom{n}{1} + \binom{n}{3} + \binom{n}{5} + \dots = 2^{n-1}$.) Let M' be a matroid obtained from M by relaxing one circuit-hyperplane into a base.

Claim: M' is not a binary. Suppose M' is binary. Let C be a second type circuit of M such that $\mathcal{B}(M') = \mathcal{B}(M) \cup \{C\}$. Note that

$$\triangle_D: \text{a second type circuit of } M \quad D = \emptyset$$

since each x_i, y_i appear in exactly 2^{n-2} times. (?) It implies that

$$\triangle_D: \text{a second type circuit of } M, \text{ not } C \quad D = C.$$

Note that circuits of M except C is still circuits of M' . (?) So C is a disjoint union of circuits of M' . By Theorem 20.2, C cannot be an independent set of M' . It is a contradiction.

What is the good explanation of this result, to conclude the theorem??? $\square$

21.1 Midterm Exam 4, Circuit-hyperplane relaxation

Proposition 21.4. *Let M be a matroid on E . We write $\mathcal{B}(M)$ to denote the set of all bases of a matroid M . Let B be a subset of $E(M)$ not in $\mathcal{B}(M)$.*

There is a matroid M' on E such that $\mathcal{B}(M') = \mathcal{B}(M) \cup \{B\}$ if B is a circuit as well as a hyperplane in M . However, the converse does not holds.

Proof. Note that for any $e \in B$, $r_M(B - e) = r_M(B) = r(M) - 1$ since B is a circuit as well as a hyperplane in M . In addition, $|B - e| = r(M) - 1$ (since $B - e$ is independent), so $|B| = r(M)$.

Let us check the base axioms. Obviously, $\mathcal{B}(M) \cup \{B\}$ satisfies (B1). Now WTS $\mathcal{B}(M) \cup \{B\}$ satisfies (B2). ETS for any $B' \in \mathcal{B}(M)$,

1. $e \in B' - B \Rightarrow \exists f \in B - B'$ s.t. $B' - e + f \in \mathcal{B}(M) \cup \{B\}$, and

2. $e \in B - B' \Rightarrow \exists f \in B' - B$ s.t. $B - e + f \in \mathcal{B}(M)$.

Let $e \in B' - B$. Then $r_M(B' - e) = r(M) - 1$. If there is $f \in B - B'$ such that $r_M(B' - e + f) = r(M)$, then $B' - e + f \in \mathcal{B}(M)$. So WMA for all $f \in B - B'$, $r_M(B' - e + f) = r(M) - 1$. It implies that $B \subseteq cl(B' - e)$. Since B is a hyperplane, $B = cl(B' - e)$. It implies that $B \supseteq B' - e$, so $B' - B = \{e\}$. We already showed that $|B| = r(M)$ ($= |B'|$), so $B - B' = \{g\}$. Hence $B' - e + g = B$.

Let $e \in B$. Suppose $r_M(B - e + f) = r(M) - 1$ for all $f \in B' - B$. It implies that $B' \subseteq cl_M(B - e)$, so $r(M) = r_M(B') \leq r_M(B - e) = r(M) - 1$, which is a contradiction. Hence there is $f \in B' - B$ such that $r_M(B - e + f) = r(M)$, so $B \in \mathcal{B}(M)$ (since $|B - e + f| = |B| = r(M)$).

To disprove the converse, consider this case: Let M and M' be matroids on [4] such that $\mathcal{B}(M) = \{\{1, 2, 3\}, \{1, 2, 4\}\}$ and $\mathcal{B}(M') = \{\{1, 2, 3\}, \{1, 2, 4\}, \{1, 3, 4\}\}$. (We can easily check that both $\mathcal{B}(M)$ and $\mathcal{B}(M')$ satisfies the base axioms.) Set $B = \{1, 3, 4\}$. Then B is a hyperplane of M , but not a circuit of M ($\{3, 4\}$ is a circuit of M properly contained in B). $\square$

22 Week15-1, 2019.12.09. Mon

Conjecture 22.1 (Rota's conjecture). *For every finite field $\mathbb{F}$, there is a finite list of matroids $M_1, M_2, \dots, M_l$ such that*

M is representable over $\mathbb{F}$ if and only if no M_i 's are minor of M .

We say them as finitely many forbidden (excluded) minors.

It is not true if $\mathbb{F}$ is infinite. Moreover, for any infinite fields, there are infinitely many forbidden minors. (How???)

Remark (Oxley, Semple, Vertigan). If $\mathbb{F}$ is finite with $|\mathbb{F}| = q$, then $l \geq 2^{q-4}$.

Theorem 22.1 (Robertson and Seymour). *Every infinite sequence of graphs admits a pair G_i, G_j such that G_i is isomorphic to a minor of G_j . In other words, the graph class with the minor relation forms a well-quasi-ordering (WQO).*

Note that a relation is WQO if it is a reflexive and transitive binary relation with no infinite antichain.

Corollary 22.1.1. *Any minor-closed graph class $\mathcal{P}$ has finitely many forbidden graph minors.*

Proof. Let $\mathcal{C}$ be the class of graphs such that $G \in \mathcal{C}$ if $G \notin \mathcal{P}$ and its every proper minor is in $\mathcal{P}$. In other words, $\mathcal{C}$ is the class of forbidden graph minors with respect to $\mathcal{P}$. It is obviously an antichain under the graph minor relation. By Robertson-Seymour theorem, it is finite. $\square$

Theorem 22.2 (Matroid minors theorem by Geelen, Gerards and Whittle). *Every infinite sequence of matroids representable over $\mathbb{F}$ admits a pair M_i, M_j*

such that M_i is isomorphic to a minor of M_j , if $\mathbb{F}$ is finite. In other words, for a finite field $\mathbb{F}$, the class of $\mathbb{F}$ -representable matroids with the matroid minor relation forms WQO.

In 2014, Geelen, Gerards and Whittle announced that they proved it, but a paper is not published yet.

Unfortunately, the matroid minors theorem does not imply Rota's conjecture. The reason why is that minimal excluded (forbidden) matroids with respect to a field $\mathbb{F}$ are not representable over $\mathbb{F}$, so we cannot apply the theorem. There is an infinite antichain of matroids under the matroid minor relation, so the class of matroids is not WQO. (As we mentioned earlier, moreover, we can find an infinite antichain of matroids representable over a fixed infinite field.)

Sub-chapter. Regular matroids

Remind the definition: A matroid is *regular* if it is representable by a totally unimodular matrix (of which every square submatrix has determinant 0 or ± 1).

Proposition 22.3. *The following hold:*

1. *Every graphic matroid is regular.*
2. *A is totally unimodular if and only if $(I_r \ A)$ is also totally unimodular.*
3. *$(I_r \ A)$ is totally unimodular if and only if every $r \times r$ submatrix has determinant 0 or ± 1 .*
4. *If $(I_r \ A)$ is totally unimodular, $\det B = \pm 1$, and $B(I_r \ A)$ has an $r \times r$ identity submatrix, then $B(I_r \ A)$ is totally unimodular.*
Actually, if A is an $r \times n$ totally unimodular matrix with $r \leq n$, $\det B = \pm 1$, and BA has an $r \times r$ identity submatrix, then BA is totally unimodular.
5. *A is an invertible square matrix which is totally unimodular. Then A^{-1} is also totally unimodular.*
6. *If M is regular, then so is M^* .*

Proof. 1. It is proved by Theorem 16.2 and Example 16.1.

2. Let A be TU. Let B be a square submatrix of $(I_r \ A)$. Let us divide B into B' and B'' , which the first one is in I_r part, and the second is in A part. WMA B' has columns, and every column of B' contains 1. Let C be a square submatrix of B'' , which deleting rows having 1 in B' . Then $\det B' = \pm \det C$. Since C is the square submatrix of A , its determinant is 0 or ± 1 .

The converse is obvious.

3. The forward direction is obvious.

Now let us prove the reverse direction. Let B be a $m \times m$ submatrix of $(I_r \ A)$. (Of course, $m \leq r$.) We can take $r - m$ columns of I_r part. In addition, we can take $r - m$ rows of A , which have 1's of the earlier choice of

columns. By adding these, we can extend B to an $r \times r$ submatrix C such that $\det B = \det C$. By the given assumption, the determinant of C is 0 or ± 1 .

4. Let C be an $r \times r$ submatrix of $B(I_r \ A)$. Then $B^{-1}C$ is a square submatrix of $(I_r \ A)$. It has determinant 0 or ± 1 , so $\det C = \det B \cdot \det(B^{-1}C)$ is 0 or ± 1 . Remind the condition that $B(I_r \ A)$ has an $r \times r$ identity submatrix. By 3, we can conclude that $B(I_r \ A)$ is TU.

Note that the condition having an $r \times r$ submatrix is critical. Consider a below case: Take $r = 2$ and

$$B = \begin{pmatrix} 2 & 0 \\ 0 & 1/2 \end{pmatrix}.$$

In this case, $B(I_r \ A) = (B \ BA)$ cannot be TU since it has an entry with a value $1/2$. (Note that for an integral square matrix B , its inverse is integral iff $\det B = \pm 1$, over $\mathbb{R}$.)

It is enough to prove the second statement replacing $(I_r \ A)$, in the proof of the first statement, to A .

5. Of course $\det A = \pm 1$. Consider an augmented matrix $(I_r \ A)$. It is TU, and $A^{-1}(I_r \ A) = (A^{-1} \ I_r)$. So by 4, it is TU. Hence A^{-1} is TU.

6. Let S be a base of M . Let A be a totally unimodular matrix with $M(A) = M$. It is the $r \times n$ matrix with $r(M) = r \leq n$. Let B be an $r \times r$ submatrix of A , of which columns corresponding to elements of S . Since S is the base, B is invertible. $\det B = \pm 1$. Of course, $B^{-1}A$ has an $r \times r$ identity submatrix. By 4, $B^{-1}A$ is TU. WMA $B^{-1}A = (I_r \ D)$ by interchanging columns. (The set of columns of I_r part corresponds to S .) D is TU. Without doubt, D^t is TU. Hence $(D^t \ I_{n-r})$ is TU. Remind the proof of Theorem 8.4. $(D^t \ I_{n-r})$ is a matrix representation of M^* . Therefore, M^* is regular. $\square$

Theorem 22.4. [Tutte] TFAE:

- (1) M is regular.
- (2) M is representable over every field.
- (3) M is binary, and M is representable over some field $\mathbb{F}$ with $\text{char}\mathbb{F} \neq 2$.

Now some people research what matroids are representable over $GF(3)$ and $GF(5)$.

Remind that a good point of binary matroids: Let M be a binary matroid. Let B be a base of M . Then it determines a unique binary representation $(I_{r(M)} \ D)$ where the identity part corresponds to B , up to interchanging rows and columns. Each column e of D is determined precisely by the fundamental circuit of e with respect to B .

Proof. (1) $\Rightarrow$ (2) $\Rightarrow$ (3) is trivial. The precise proof of (1) $\Rightarrow$ (2) is in Remark before Definition 16.3.

Now we only focus on verifying (3) $\Rightarrow$ (1). Let $E = E(M)$ and $r = r(M)$. Let B be a base of M . Let A be a $\mathbb{F}$ -representation of M . WMA

$$A = \begin{array}{c|c} B & E - B \\ \hline I_r & D \end{array}$$

by suitable row operations, multiplying scalars to columns and interchanging columns.

Let us define the fundamental graph G_B with respect to B . Its vertex-set is E , and $e \in B$ is adjacent to $f \in E - B$ if and only if the fundamental circuit of f with respect to B contains e if and only if the (e, f) -entry of D is non-zero. Obviously, G_B is the bipartite graph with a bipartition $(B, E - B)$. We can identify an edge between $e \in B$ and $f \in E - B$ of G_B as a non-zero entry (e, f) of D .

(Sometimes, we denote G_B as $G(D^*)$. Here D^* is a 0-1 matrix such that $D^*(e, f) = 1$ if and only if $D(e, f) \neq 0$. $G(D')$ for a 0-1 matrix D is a bipartite graph with a bipartition R, C , of which adjacency matrix with row R and column C is D' .)

Let T be a spanning forest of G_B .

Claim 1: WMA each edge of T has value 1 in D .

Claim 2: WMA all non-zero entries of D are ± 1 .

Claim 3: $A = \begin{pmatrix} I_r & D \end{pmatrix}$ is TU.

(See Chapter 6.6: Regular matroids in Matroid Theory by J. Oxley. More precisely, Lemma 6.6.2, Theorem 6.6.3 and Lemma 6.6.4. Additionally, see p. 81 to confirm the definition of *pivoting*.) $\square$

(The remaining proof of Theorem 22.4 was done in December 11th.)

23 Week15-2, 2019.12.11. Wed

Sub-chapter. Regular matroids

Theorem 23.1 (Tutte, 1958). *A matroid M is regular if and only if M has no minor isomorphic to $U_{2,4}$, F_7 and F_7^* .*

Remaining questions are following. How to decide whether a matroid is regular? How to decide whether a matrix is totally unimodular? Note that a totally unimodular matrix is closely related to integer programming:

Objective: $\max c^T x$,

Restrictions: $Ax \leq b$, $x \geq 0$ and $x \in \mathbb{Z}^n$.

If A is TU, then solutions of IP is also solutions of its LP relaxation (change a condition $x \in \mathbb{Z}^n$ to $x \in \mathbb{R}^n$).

Theorem 23.2 (Regular matroid decomposition, Seymour, 1980). *A matroid M is regular if and only if M is graphic, cographic, R_{10} , or 1-sum/2-sum/3-sum of two smaller regular matroids.*

Here $R_{10} = M(A)$ where

$$A = \left(\begin{array}{ccccc|ccccc} 1 & 0 & 0 & 0 & 0 & -1 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & -1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & -1 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & -1 \end{array} \right)$$

Consider a K_5 -minor-free graph. It is planar, having $K_{3,3}$ -minor, or some exceptional cases. In addition, some operations on graphs preserve a structure (in this case, K_5 -minor-free). In 1937, Wagner proved that every K_5 -minor-free graph is a planar graph, the Wagner graph (the Möbius graph of order 8) or a clique-sums of planar graphs and copies of the Wagner graph. (Note that $K_{3,3}$ is a 3-clique-sum of three disjoint tetrahedrons.) The regular matroid decomposition theorem proved by Seymour is analogous to above Wagner's theorem or other structure theorems in graph theory.

An algorithm recognizing TU matrix is similar with the above structure theorem.

One open problem in this area is that:

For a given skew-symmetric matrix A , can we decide whether $\det A[X, X]$ (determinant of principal submatrix) is equal to 0 or ± 1 for all X ?

If A satisfies the condition, then we say that A is *principally unimodular (PU)*. It is related to regular delta-matroids.

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